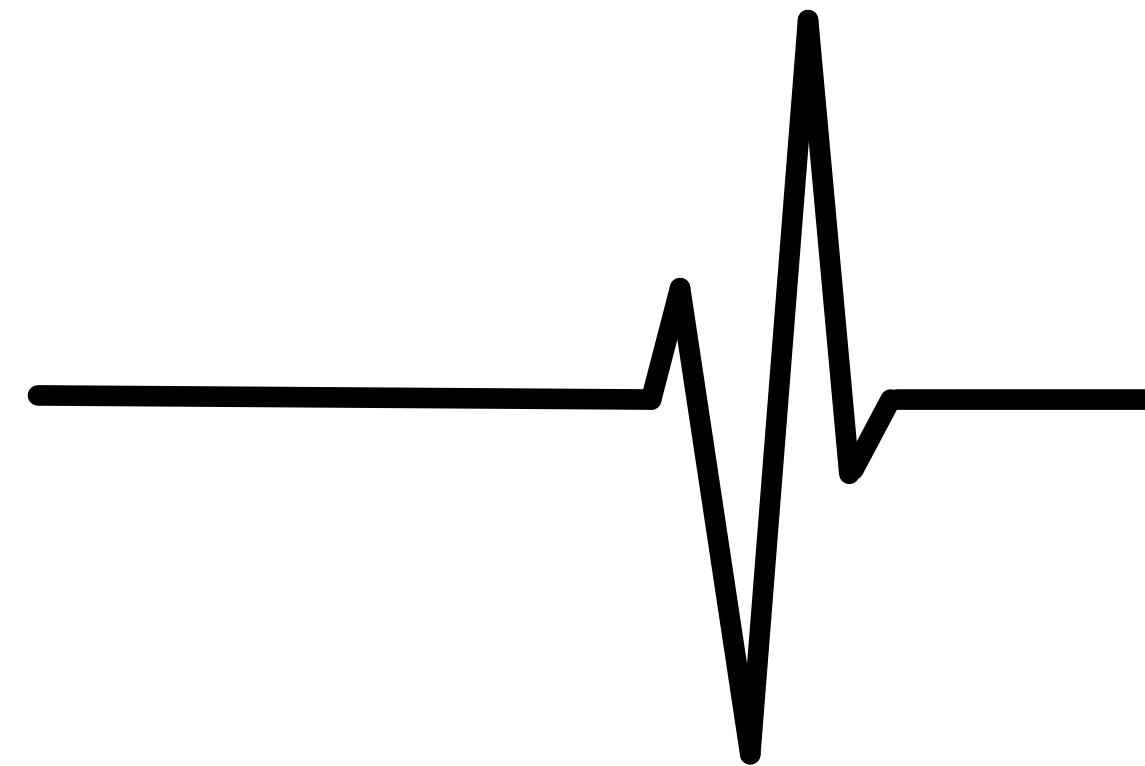


Team E

Dongyun Nam, Eunsu Park, Yerin Park, Julia Irsalina

A Dual-branch VAE-LSTM with Multi-head Attention

for Real-time Stress Detection



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01

Introduction

Motivation: Preventive Medicine & Stress detection



Preventive Medicine

Definition: A paradigm shift from the traditional medical model focused on post-occurrence **"Cure,"**

→ Emphasizing instead the prevention of disease onset or early detection for proactive **"Care."**

Paradigm Shift in Healthcare: From Cure to Care

Motivation: Preventive Medicine & Stress detection



Why Preventive Medicine?

1. Rising Healthcare Costs and Chronic Diseases
2. Out-of-Hospital Risks
3. Data-Driven Early Intervention

Motivation: Preventive Medicine & Stress detection

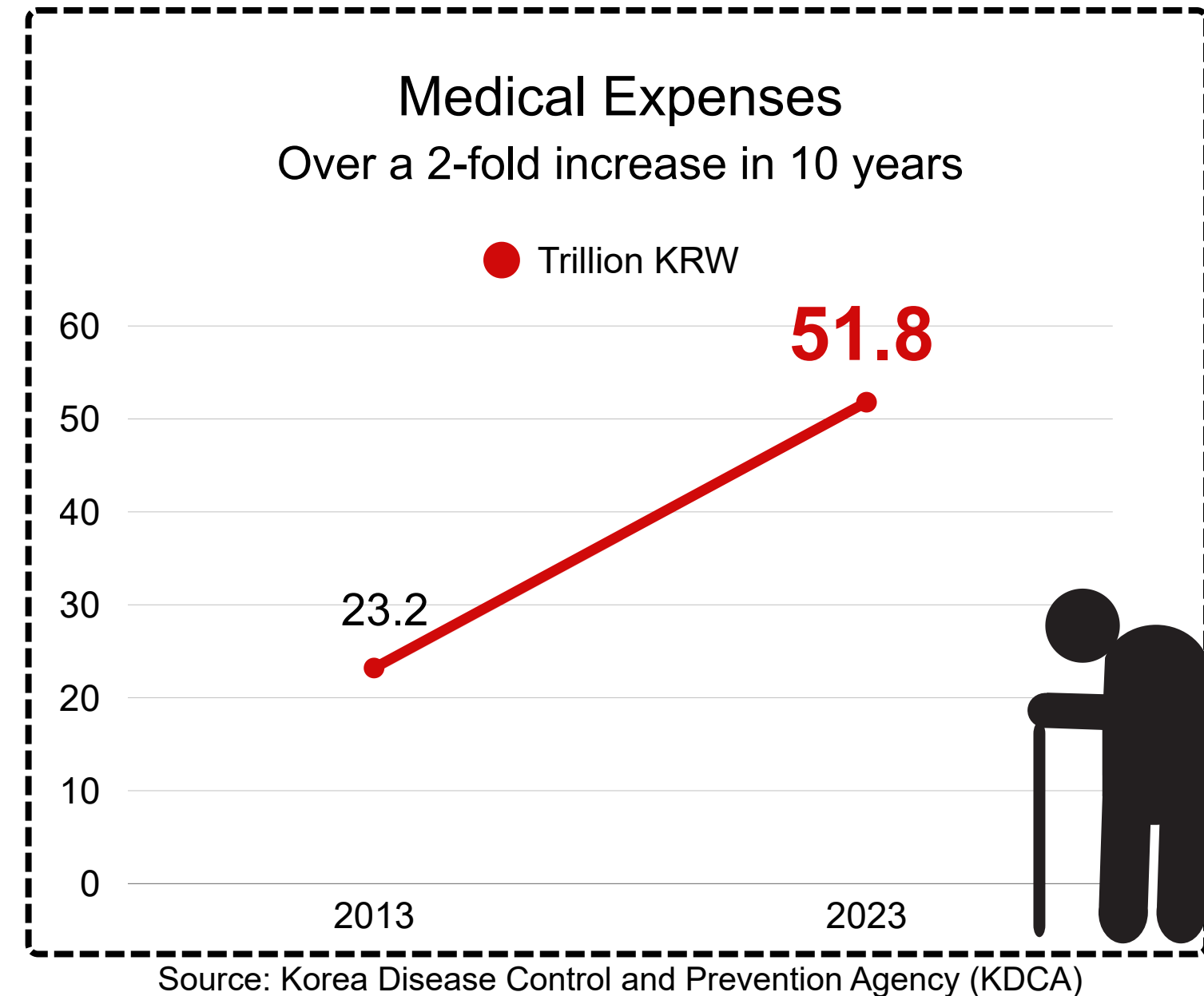


Why Preventive Medicine?

1. Rising Healthcare Costs and Chronic Diseases

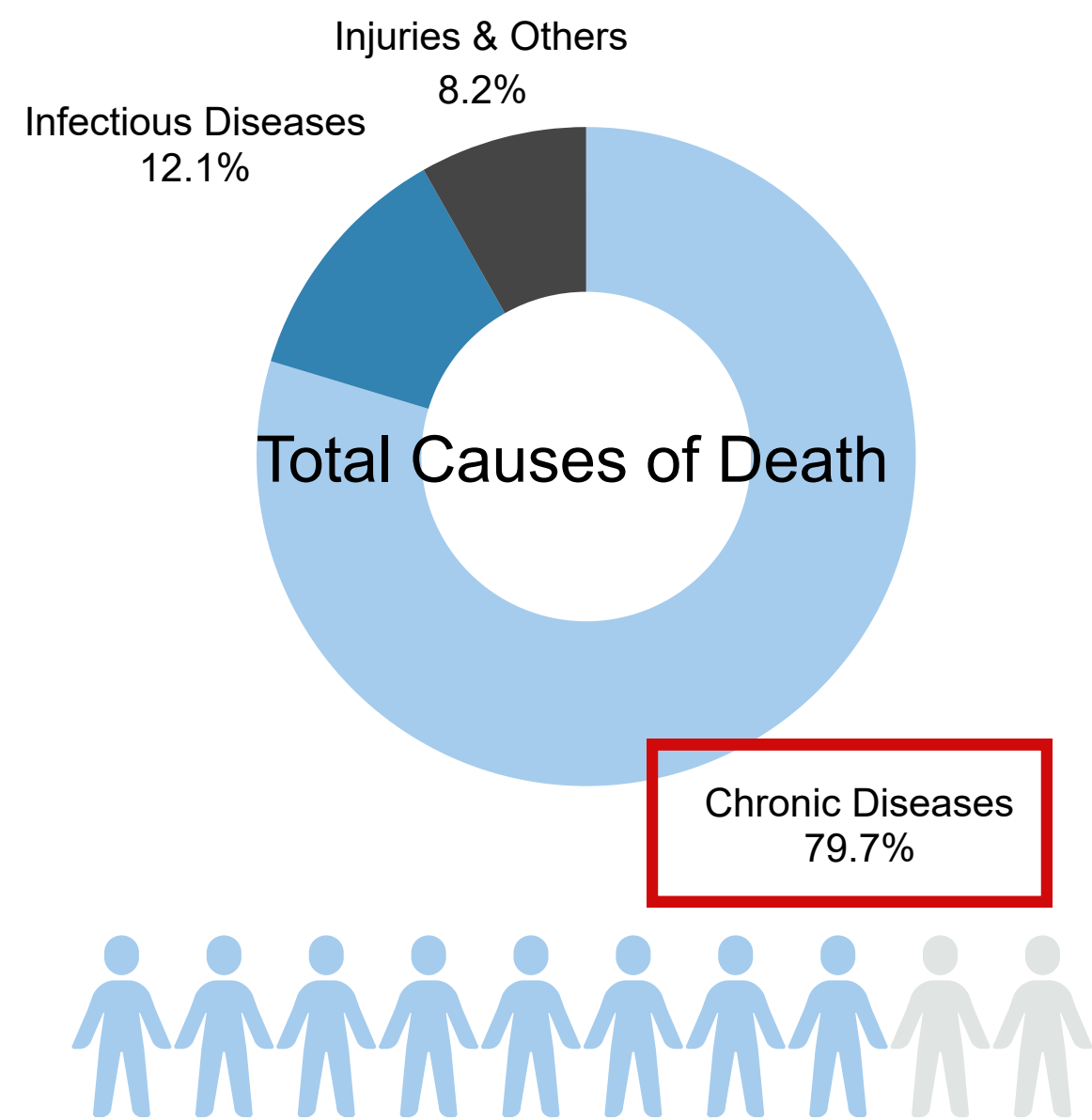
2. Out-of-Hospital Risks

3. Data-Driven Early Intervention

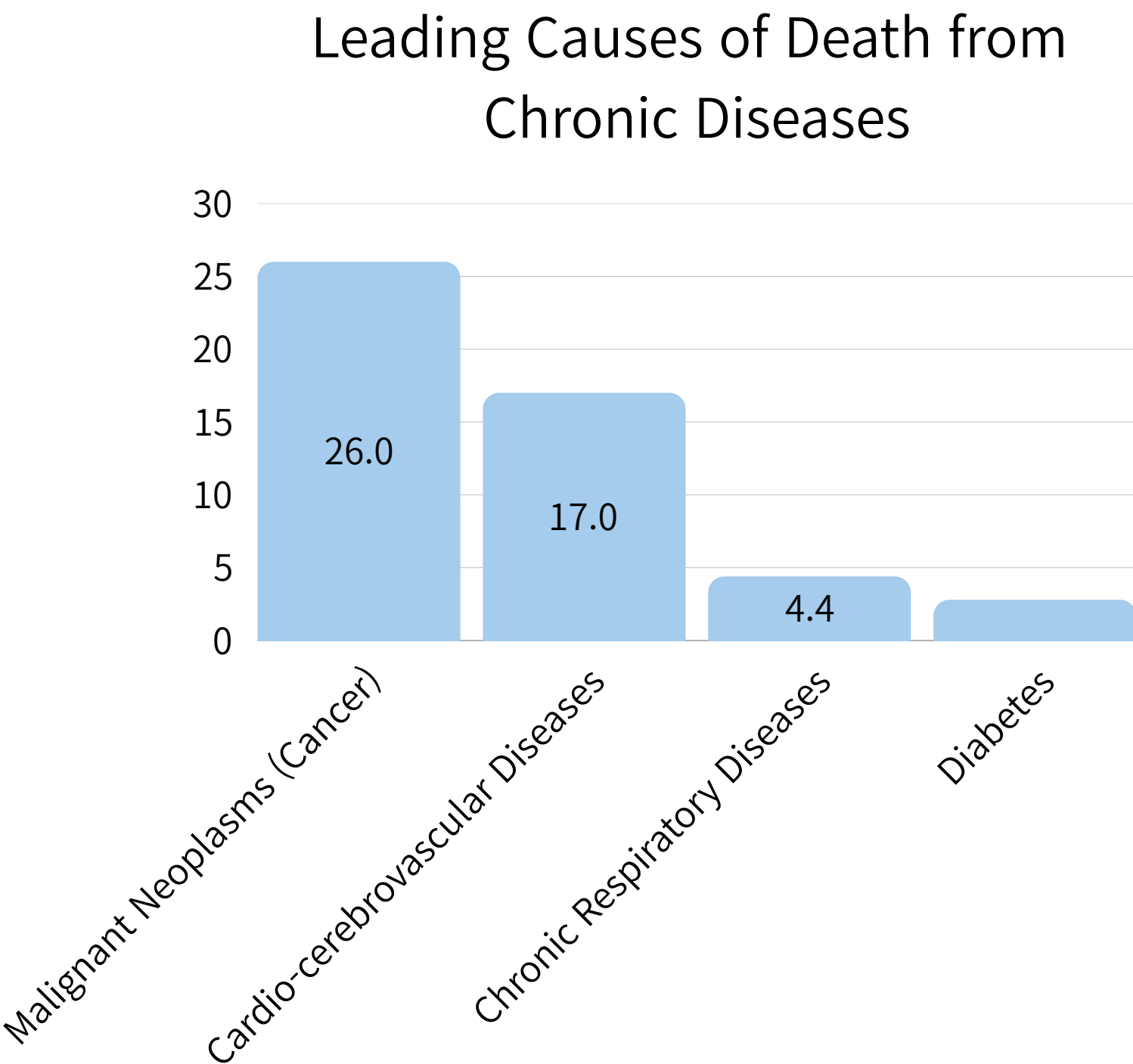


Motivation: Preventive Medicine & Stress detection

Managing chronic conditions for prevention



Source: Korea Disease Control and Prevention Agency (KDCA)

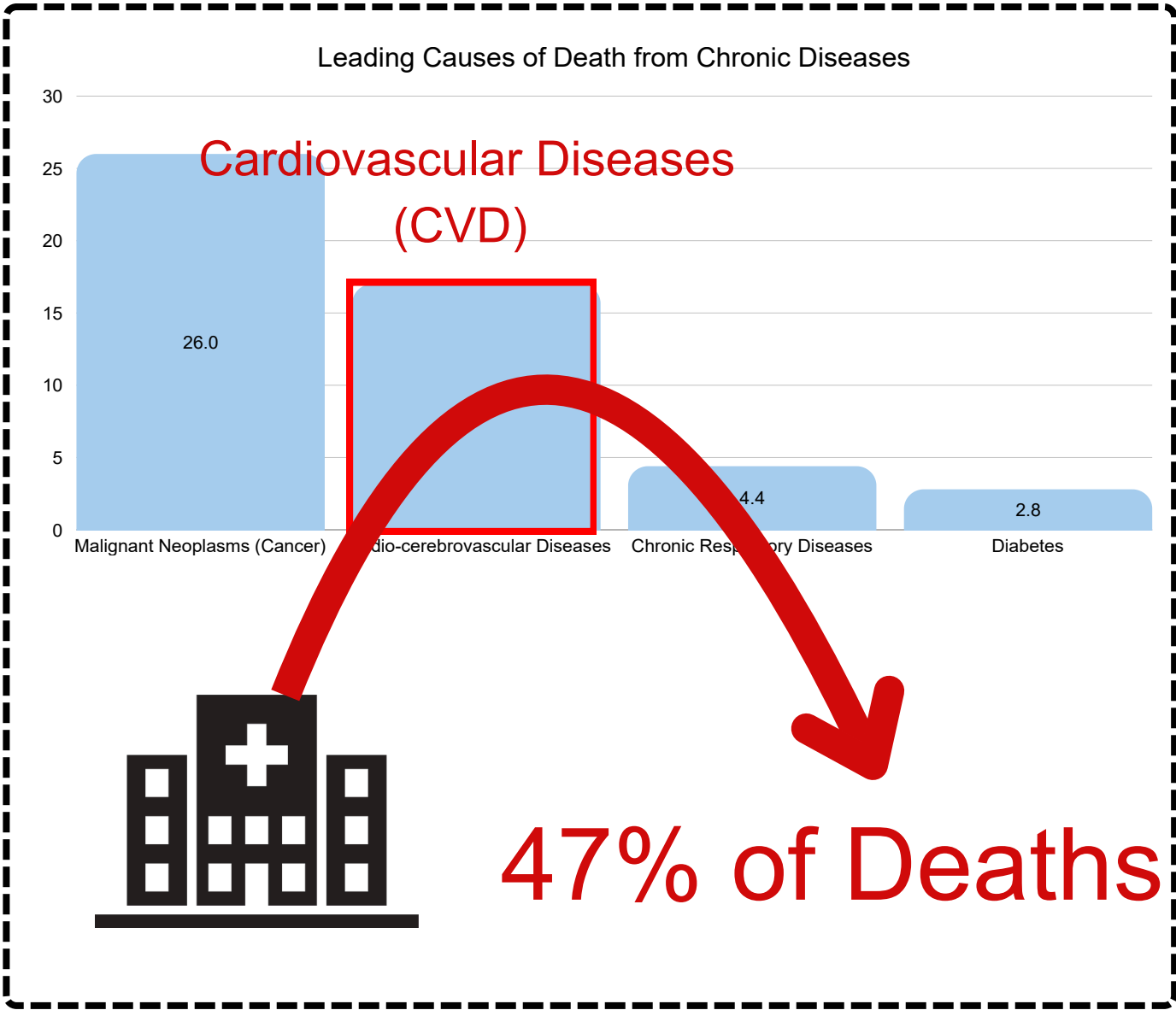


Motivation: Preventive Medicine & Stress detection



Why Preventive Medicine?

- 1. Rising Healthcare Costs and Chronic Diseases
- 2. Out-of-Hospital Risks
- 3. Data-Driven Early Intervention



Motivation: Preventive Medicine & Stress detection



Why Preventive Medicine?

1. Rising Healthcare Costs and Chronic Diseases
2. Out-of-Hospital Risks
3. **Data-Driven Early Intervention**



Advancements of **Wearable Sensors**

Motivation: Preventive Medicine & Stress detection



Why Preventive Medicine?

Cure



Care

Motivation: Preventive Medicine & Stress detection



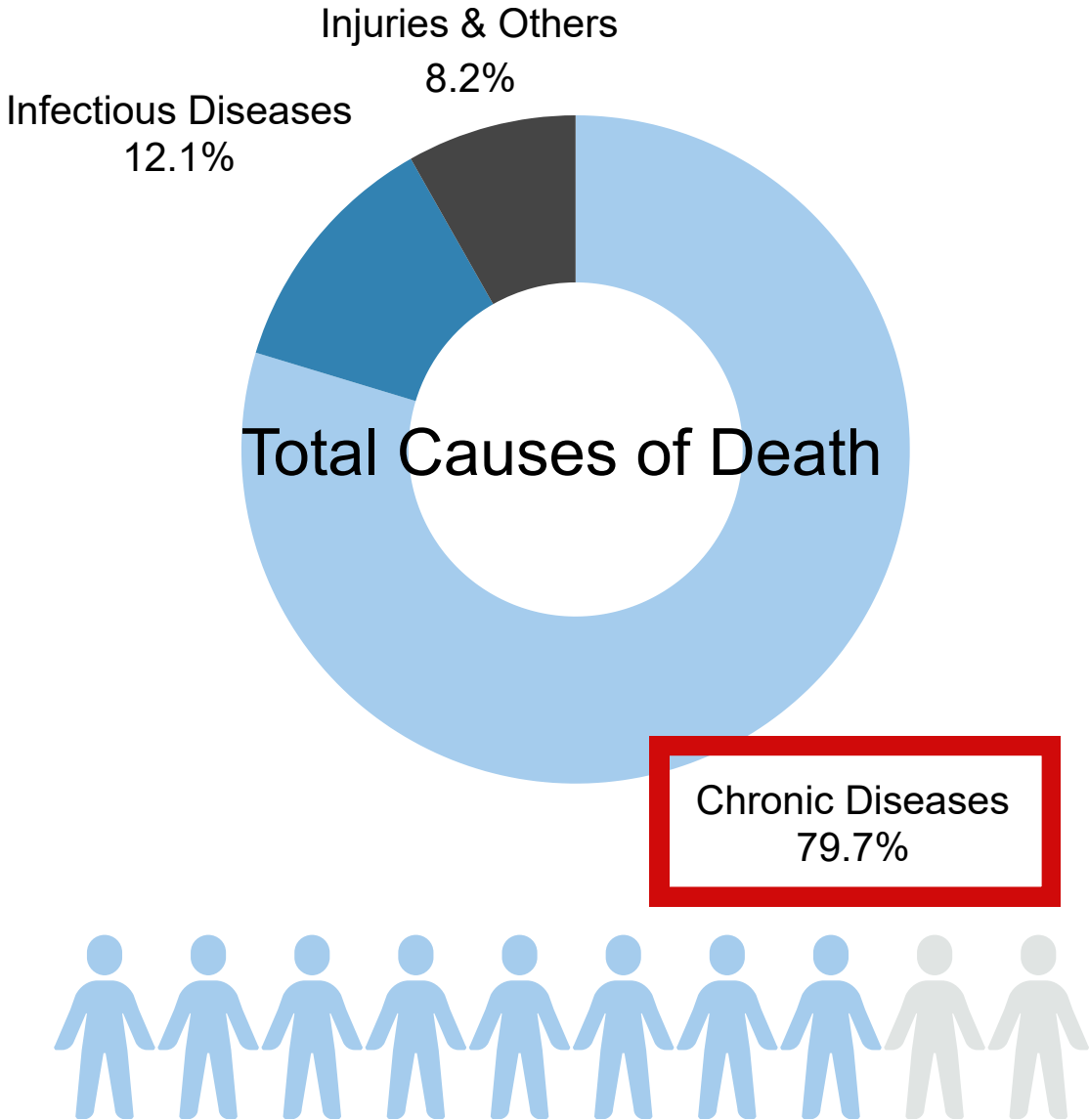
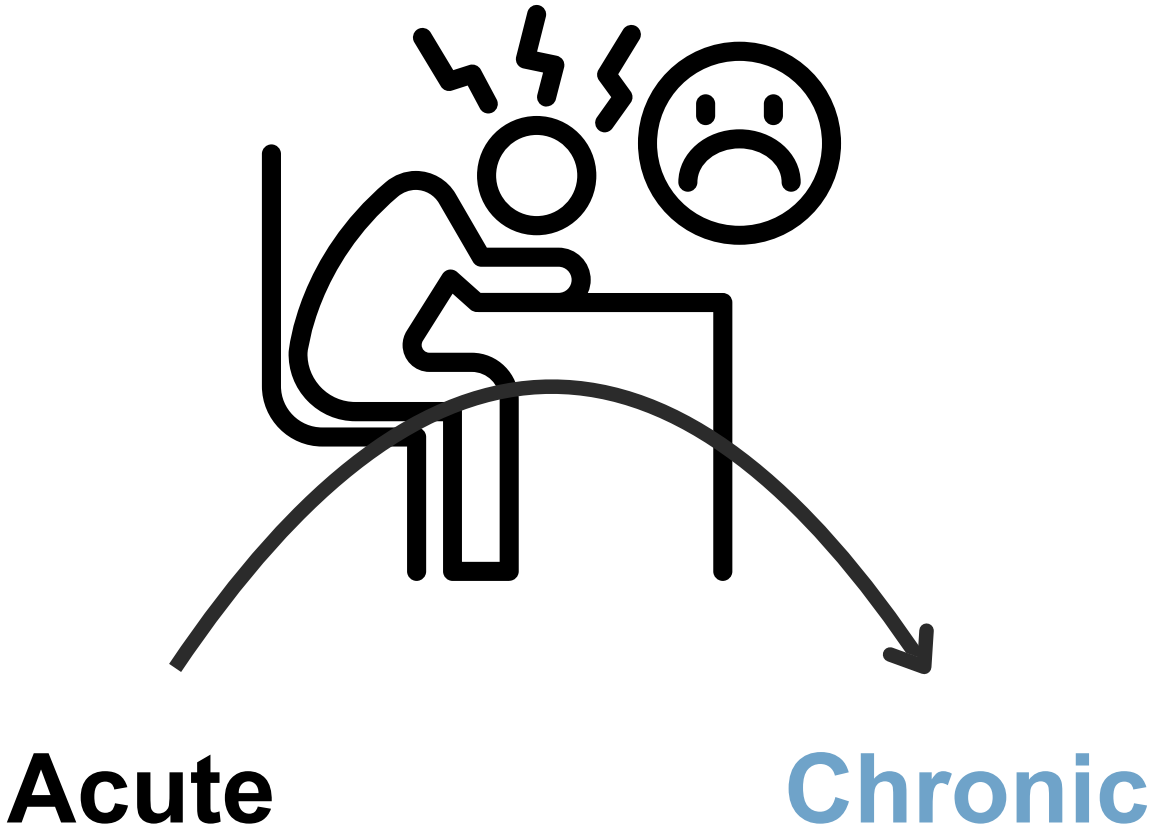
Why do we select 'Stress'?

Stress Detection in Preventive Medicine

Motivation: Preventive Medicine & Stress detection

Why do we select 'Stress'?

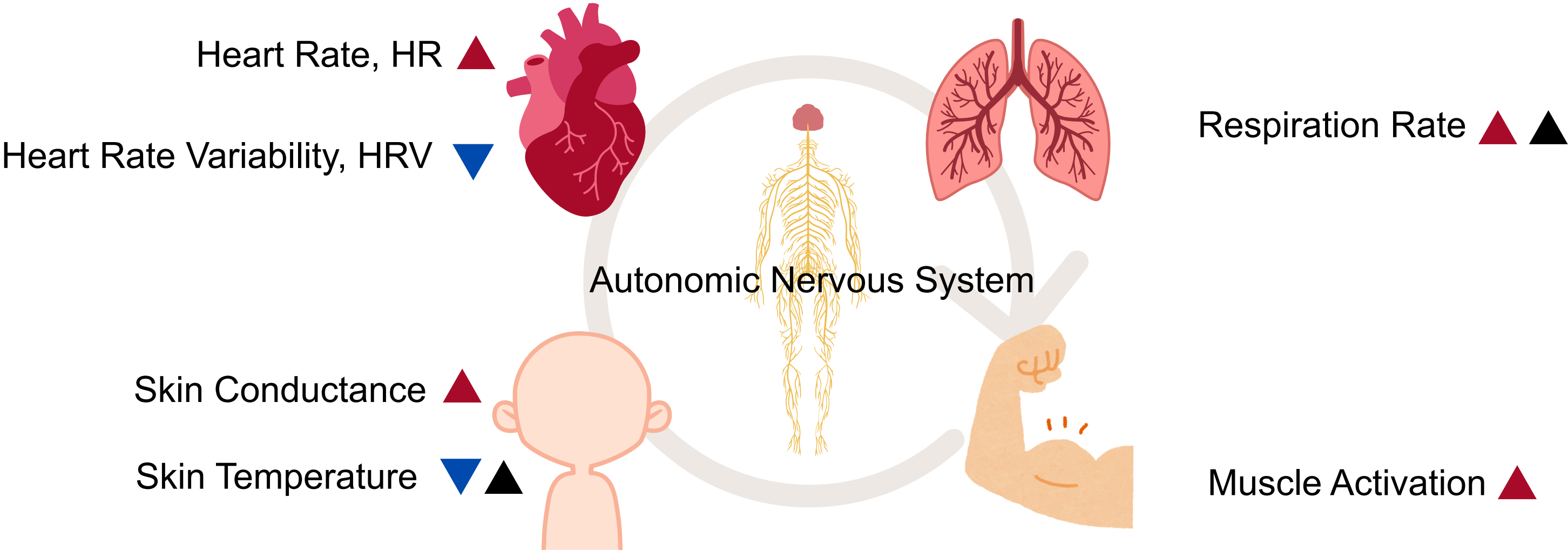
The root of all diseases



Source: Korea Disease Control and Prevention Agency (KDCA)

Motivation: Preventive Medicine & Stress detection

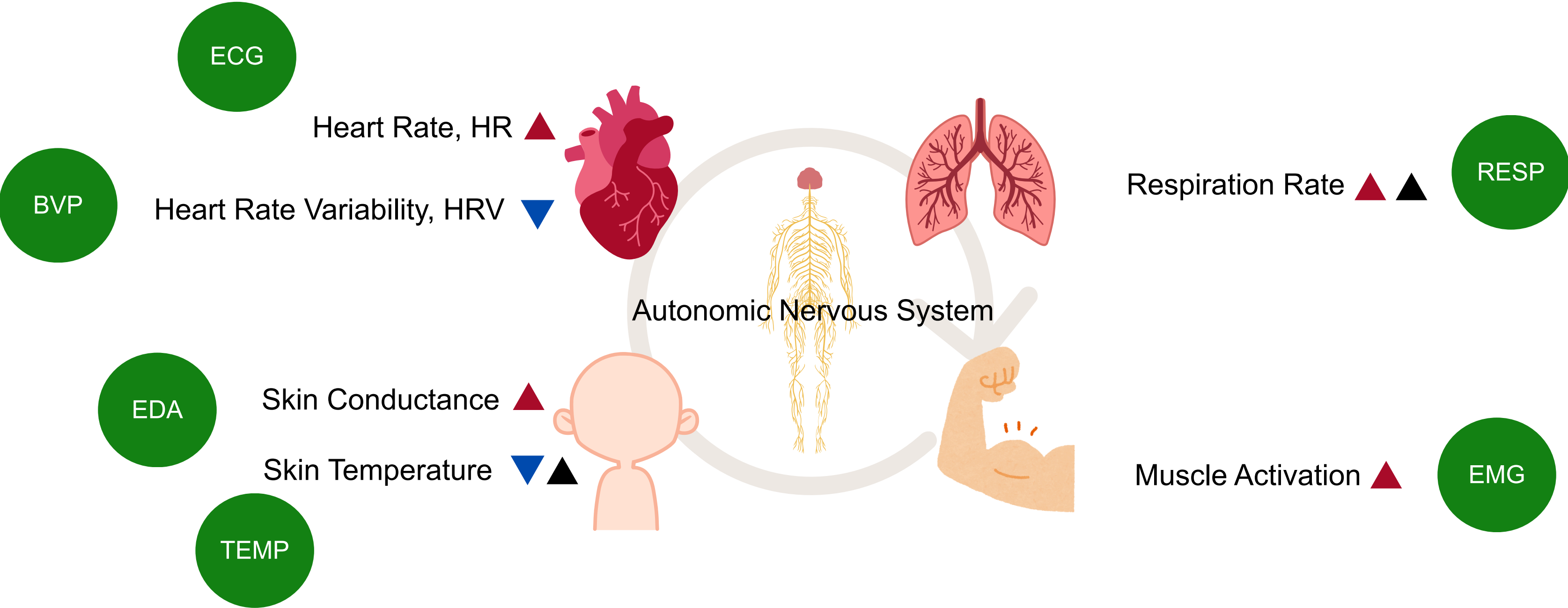
What is 'Stress'?



Fight-or-Flight Response: A survival mechanism driven by the Sympathetic Nervous System (SNS)

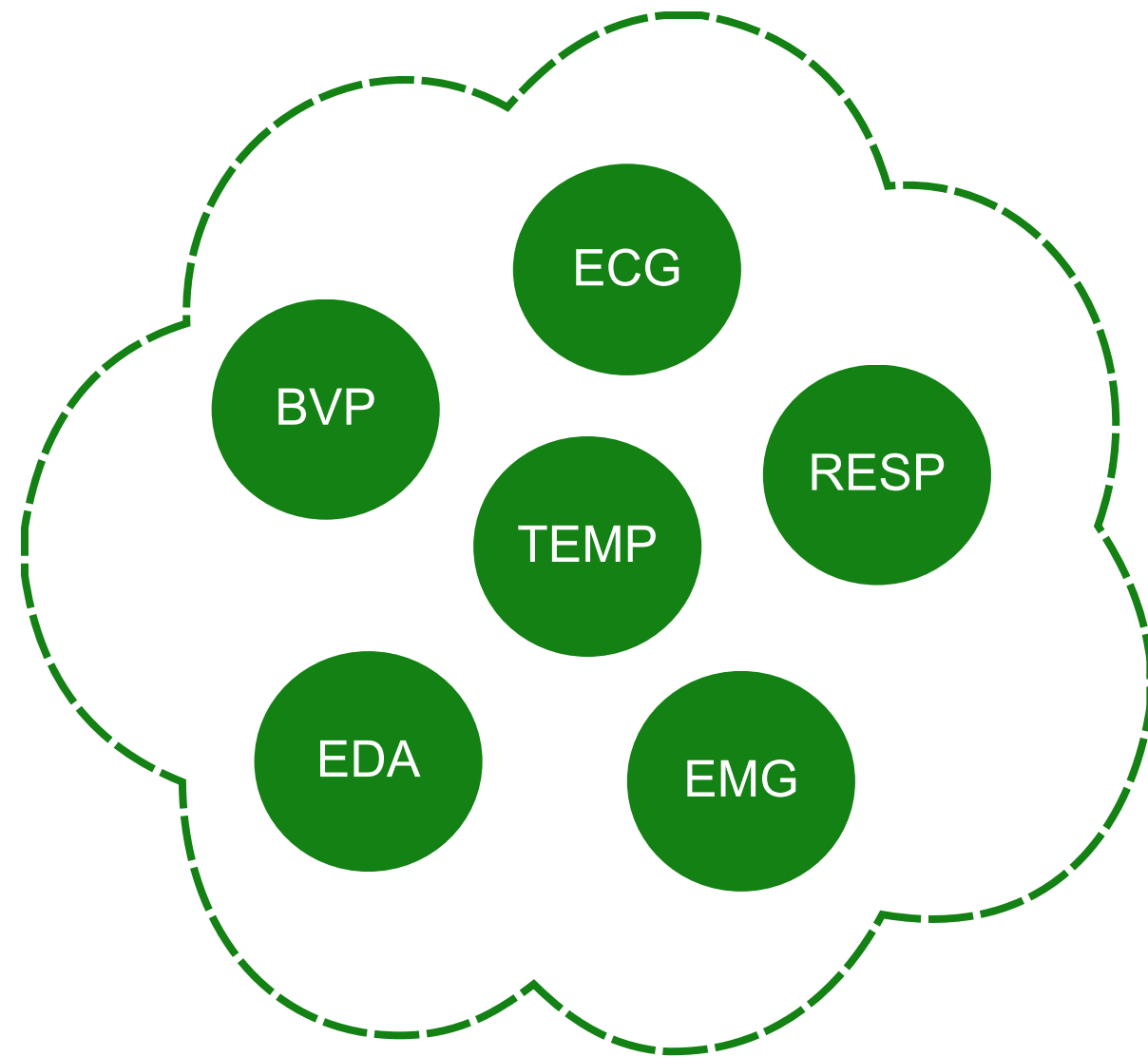
Motivation: Preventive Medicine & Stress detection

How do we detect 'Stress'?



Motivation: Preventive Medicine & Stress detection

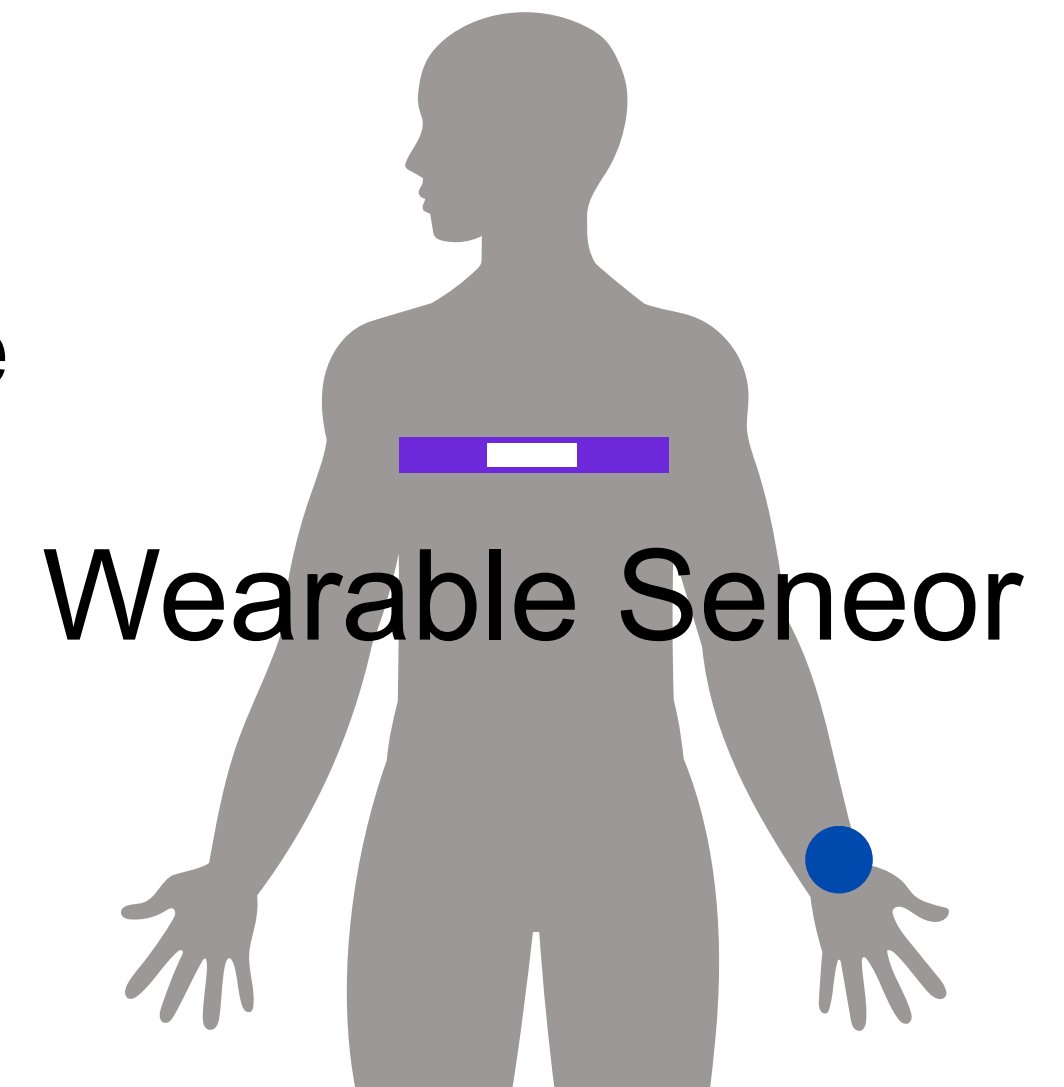
How do we detect 'Stress'?



Simultaneously change



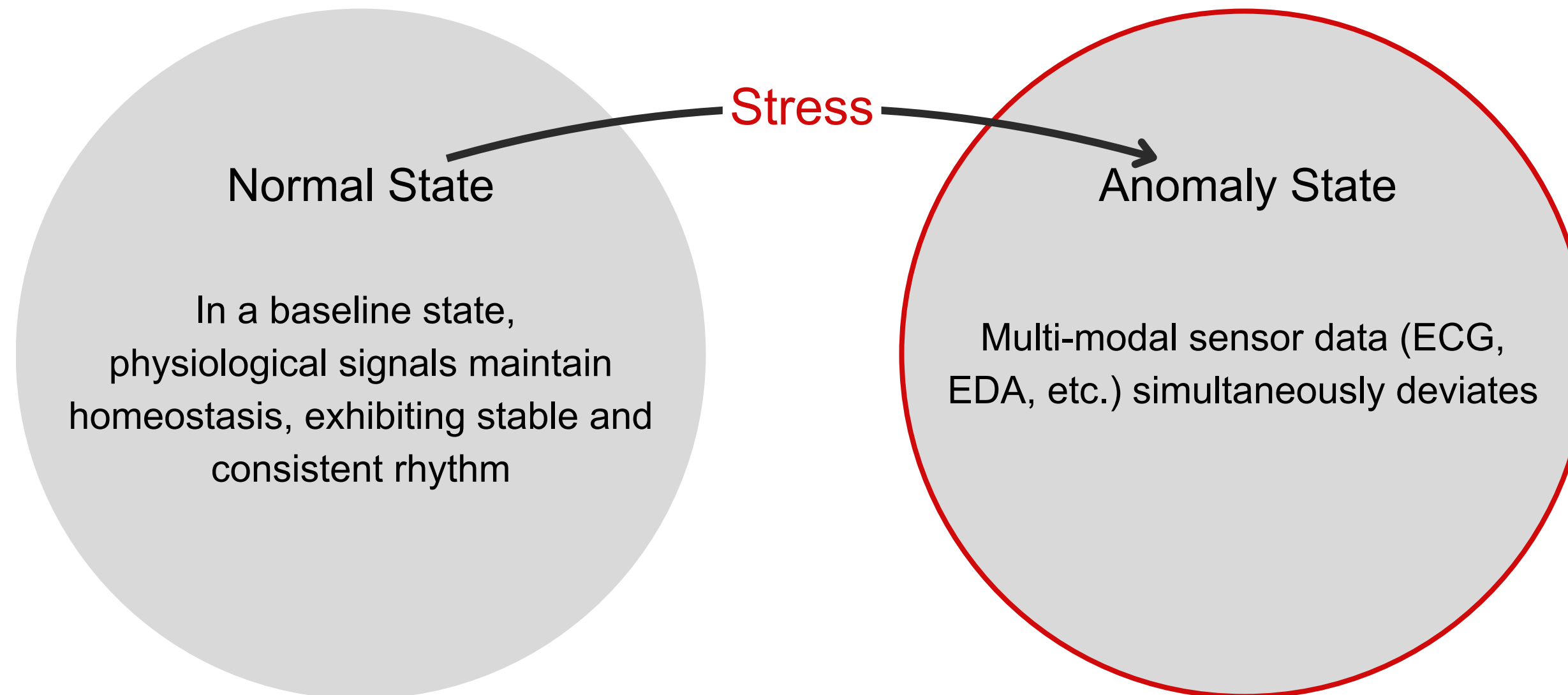
Multi-modal data



Wearable Seneor

Motivation: Preventive Medicine & Stress detection

What is 'Stress'?

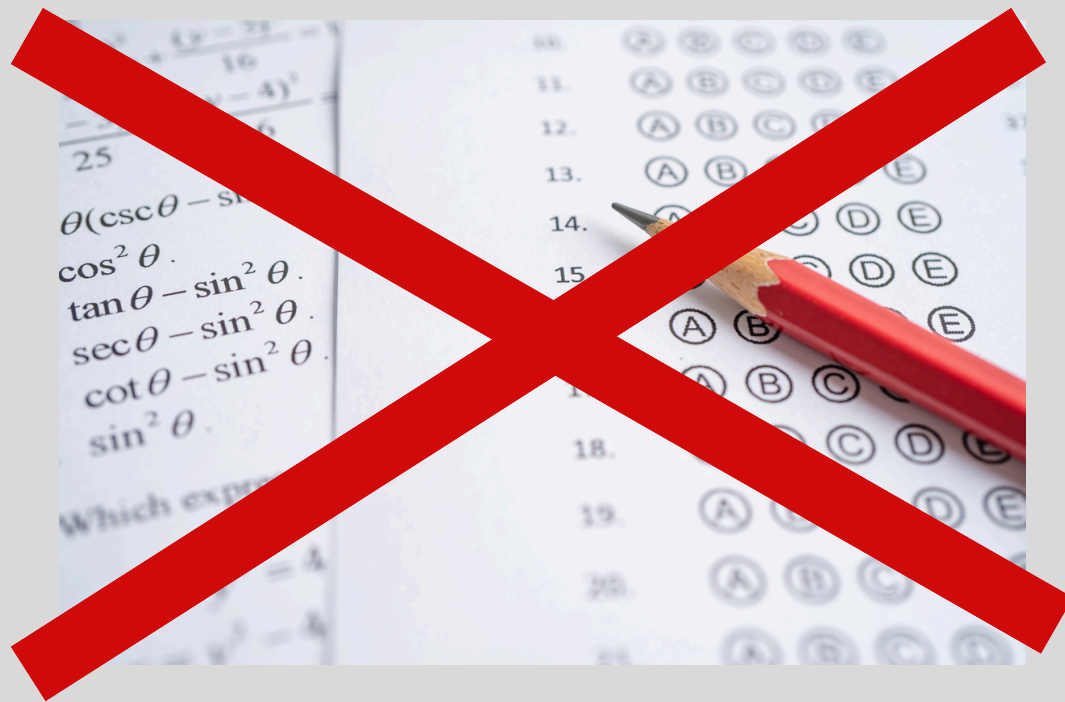


- **Stress:** Multi-modal Co-occurring **Anomaly**
- Distinct physiological signals simultaneously deviate from their learned normal distributions, driven by the Autonomic Nervous System

Limitation of Previous Research

2-step problem: Data accessibility + Model Level

Limitations of Self-Reports



- Reliability Issues
- Disruptive
- Discontinuous

Advantages of Wearable Sensing

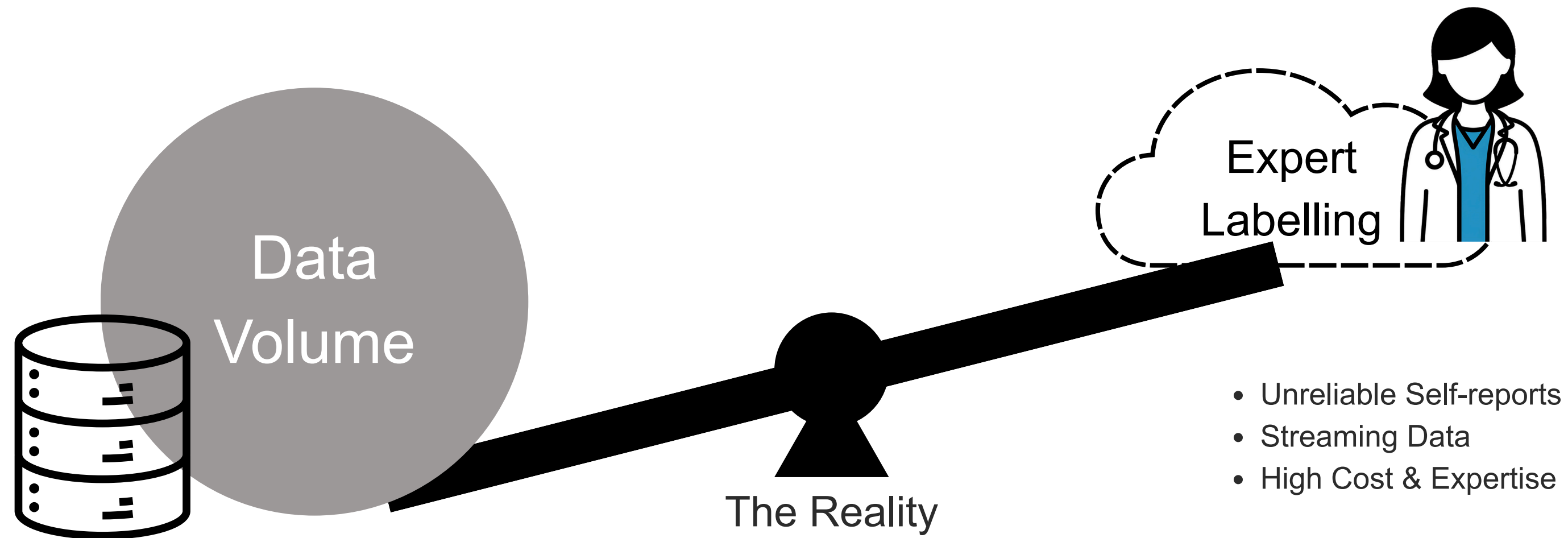


Empatica E4

- Non-invasive & Continuous: 24/7 monitoring
- Objective

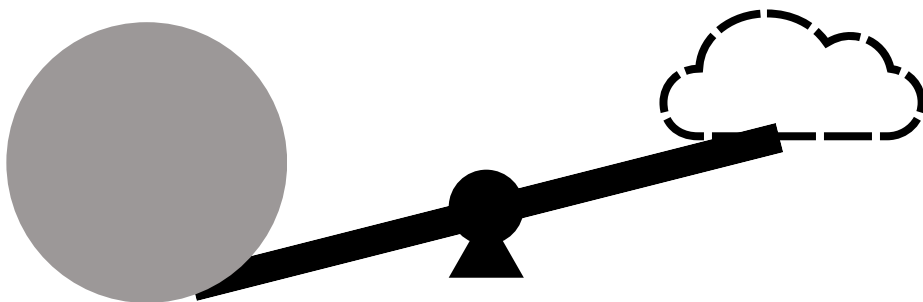
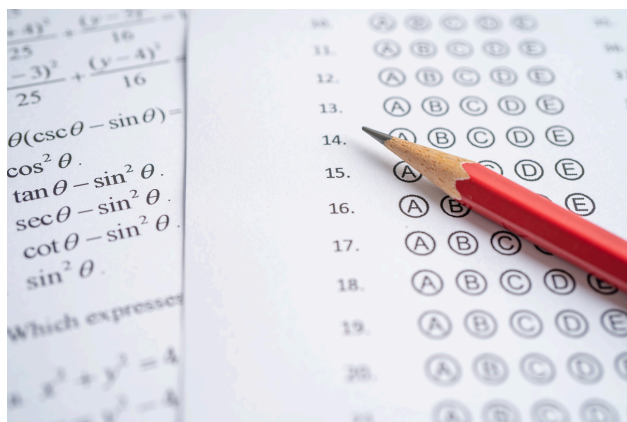
Limitation of Previous Research

2-step problem: Data accessibility + Model Level

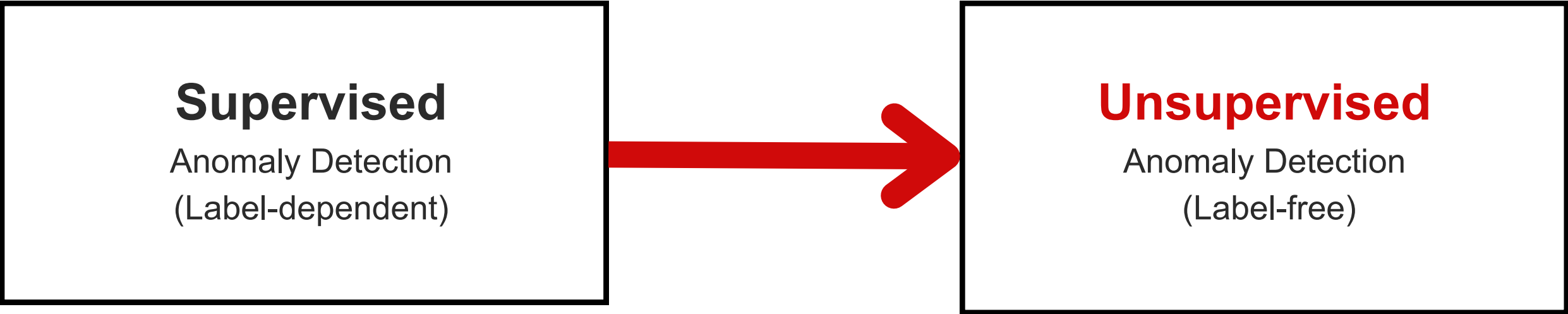


Limitation of Previous Research

2-step problem: Data accessibility + Model Level



Label Scarcity

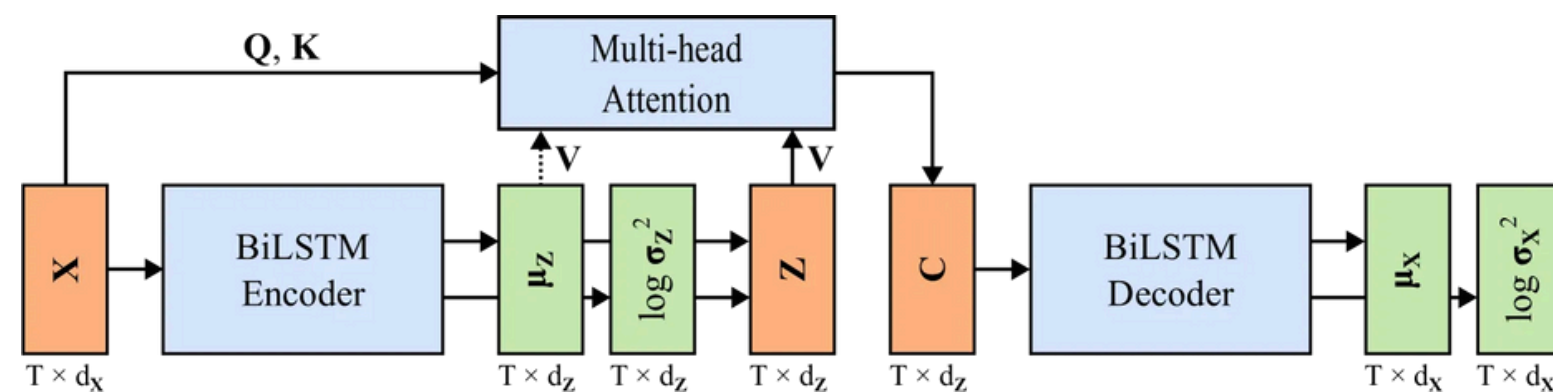


Limitation of Previous Research

2-step problem: Data accessibility + Model Level

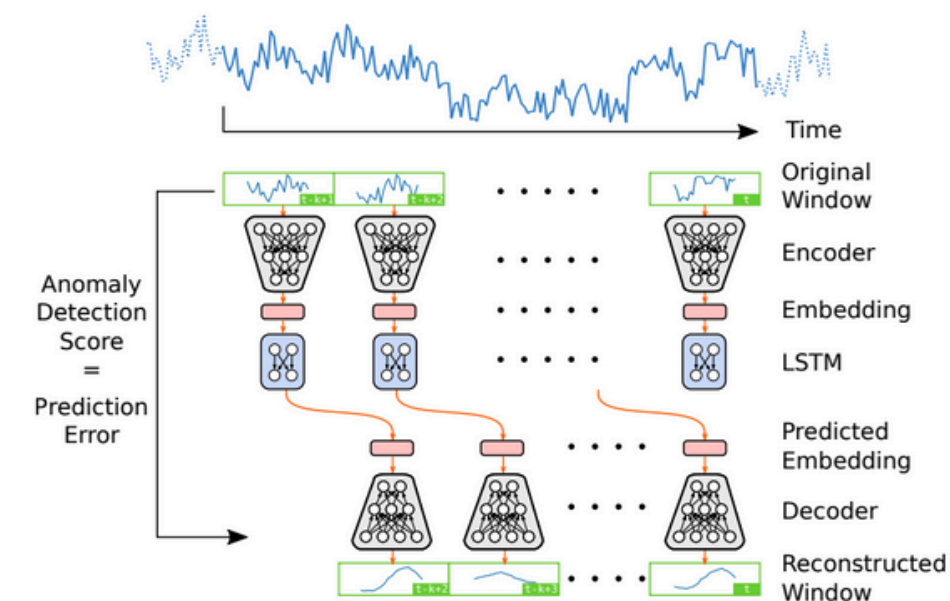
MA-VAE (Reconstruction-based)

- Unsupervised approach
- Variational Autoencoder (VAE)
+ MA (Multi-head Attention)
- Reconstructs input data using VAE with MA
- The model structure is simple.



VAE-LSTM (Prediction-based)

- Unsupervised approach
- VAE $\rightarrow$ Local structure extraction
- LSTM $\rightarrow$ Predicts latent representations
 $\rightarrow$ Reconstructs the input



Limitation of Previous Research

Limitation of 2 model in Anomaly detection

MA-VAE (Reconstruction-based)

Pros

- The simple model structure
- Learns stable global patterns

Cons

- Over-sensitive to Noise
- Misses Temporal Shifts
- Normal and anomaly scores overlap → Unstable Boundaries
- very slow inference speed ← BiLSTM

VAE-LSTM (Prediction-based)

Pros

- Faster than MA-VAE
- Learns Temporal Patterns well
- Good at Detecting True Anomalies

Cons

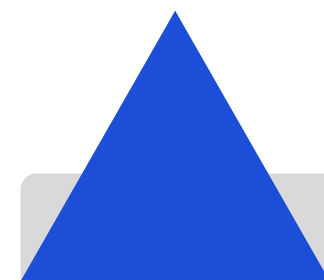
- Overreacts to small fluctuations
- Focused on Local Context
- Lack of global context information

Problem Definition

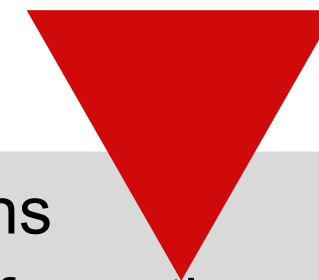
Label Scarcity & Subjectivity

Unsupervised Anomaly
Detection

This motivates our Global–Local Fusion design !!



Pros
inference speed



Cons
lost global informations

Inspired by MA-VAE's attention mechanism (which fused raw input X with latent Z),
we instead fuse Global Latent (long-range) + Local Latent (short-term) to recover both temporal and global information.

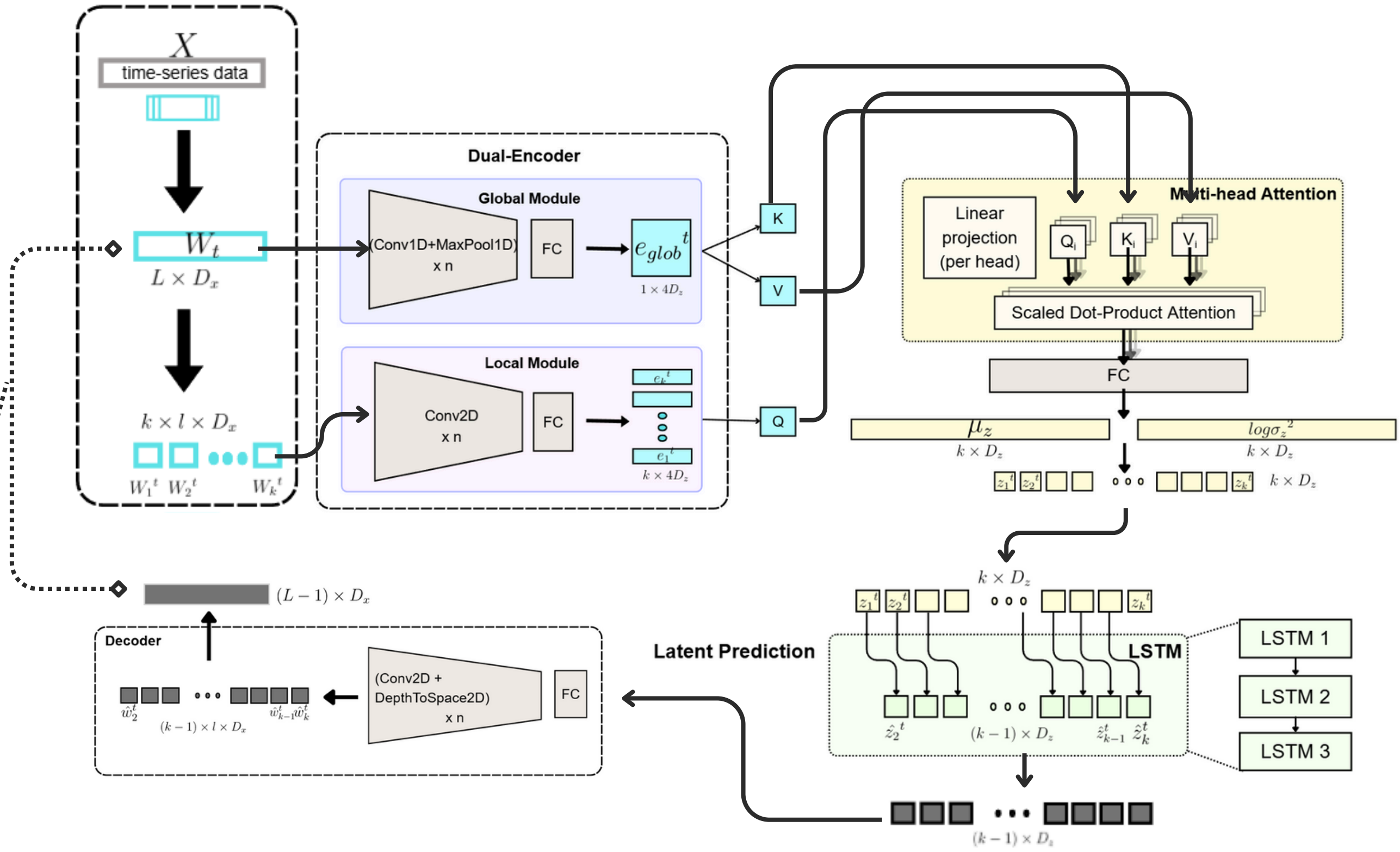
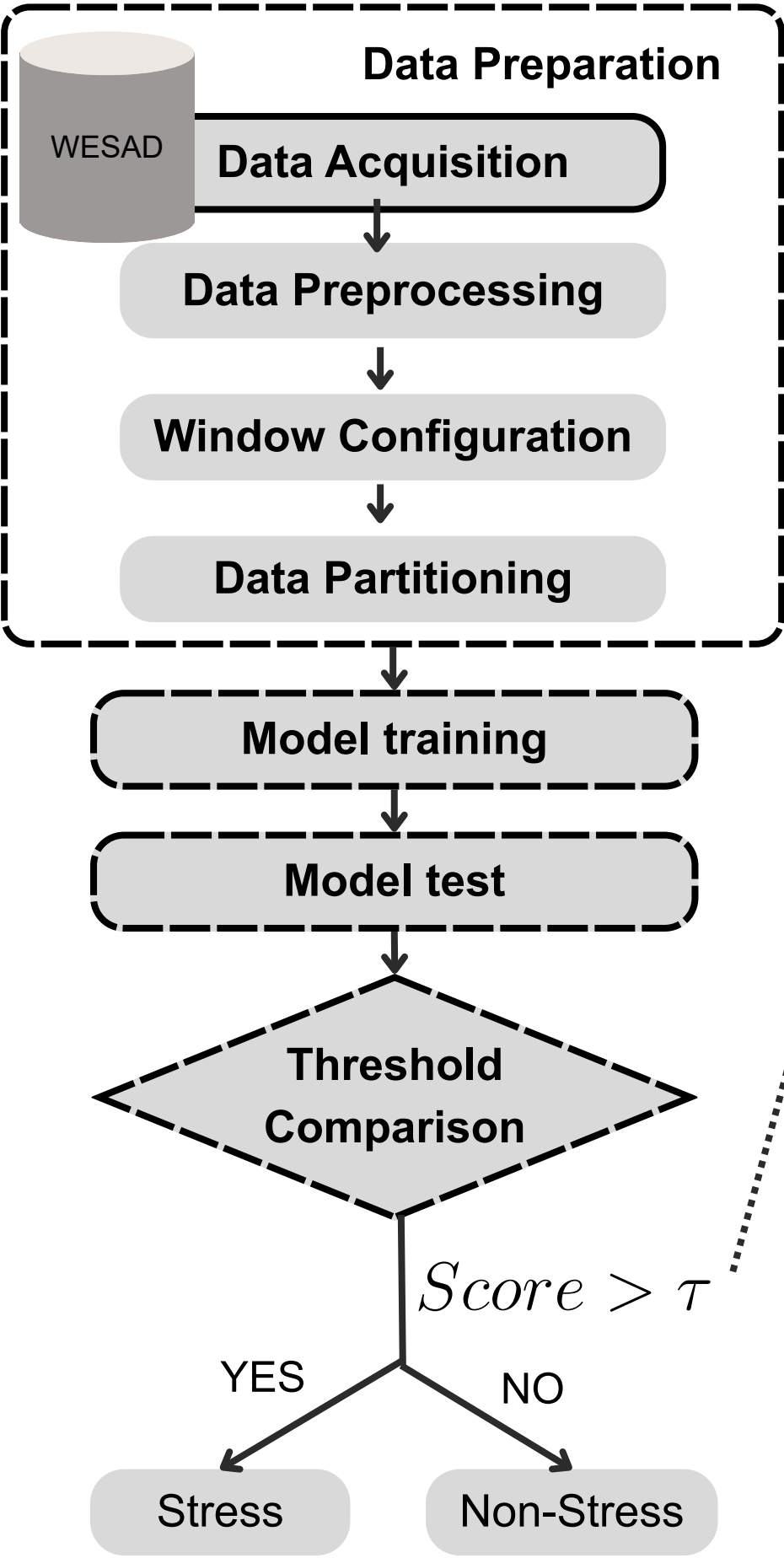
Research Hypothesis & Objectives

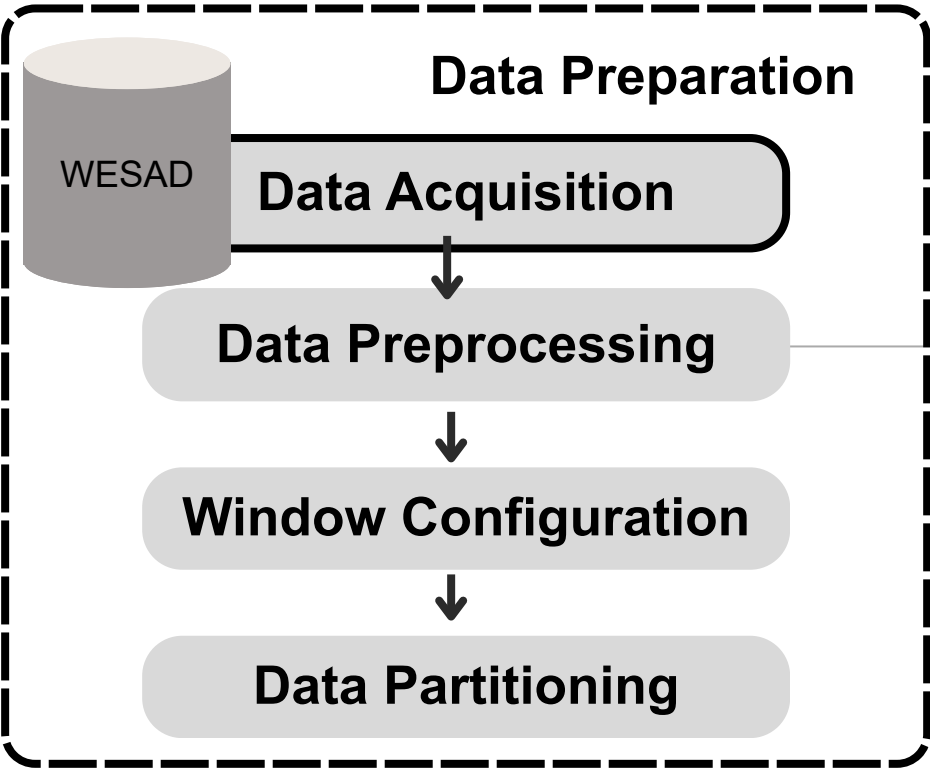
Dual-branch Architecture Design

Performance	F1 Score 0.883, AUROC 0.959
Efficiency	Fast inference time (less than 50ms) Lightweight model architecture (2M Parameters)
Robustness	Stable and consistent performance across varying window sizes

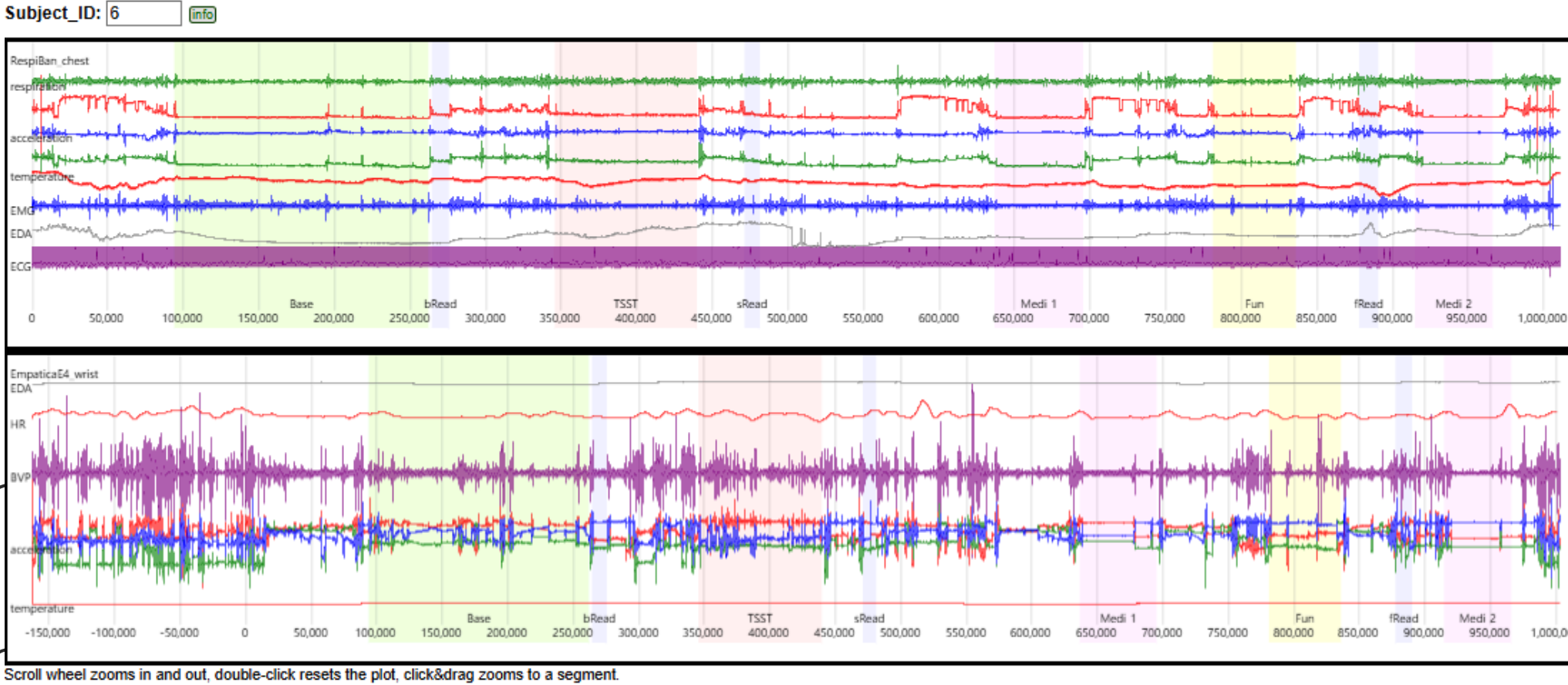
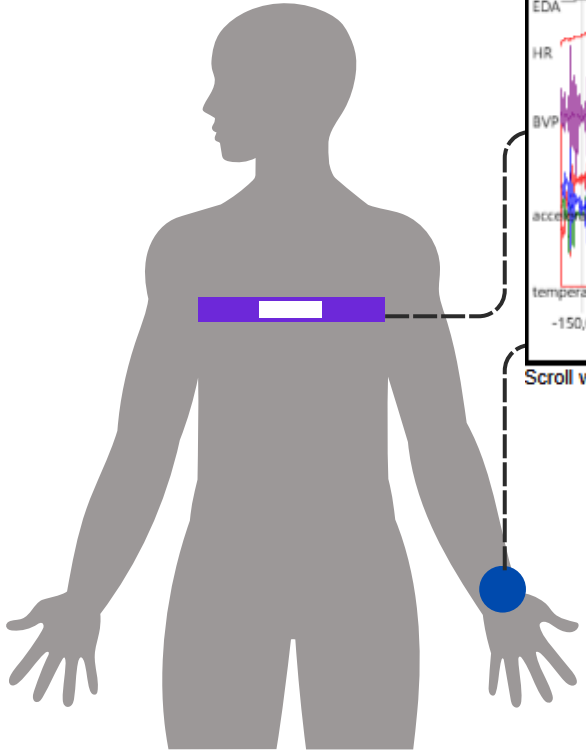
02

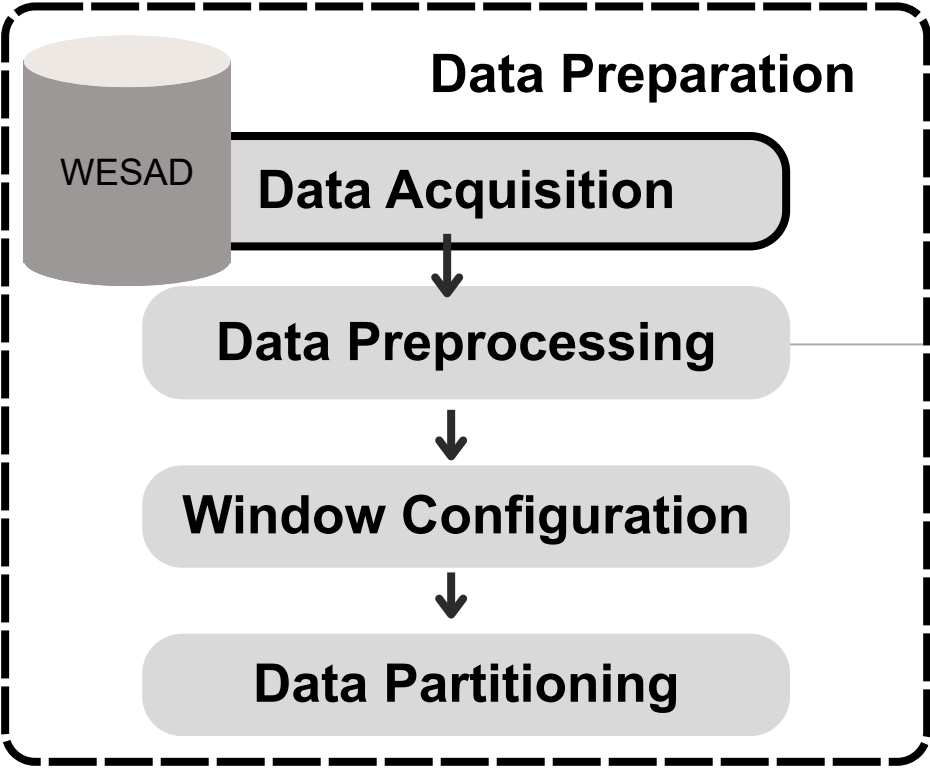
Dataset





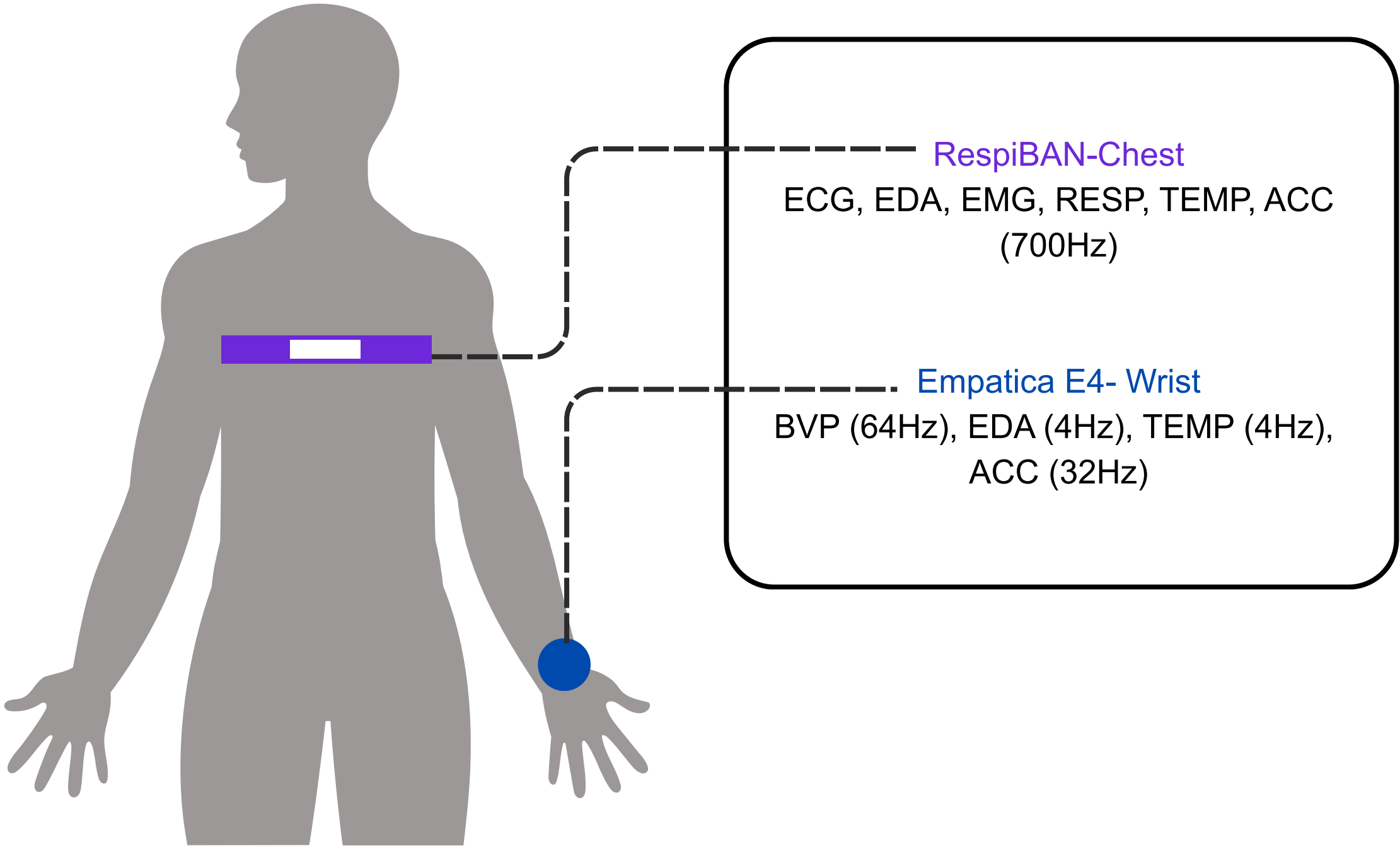
WESAD = Multimodal Dataset (Wearable Stress and Affect Detection)

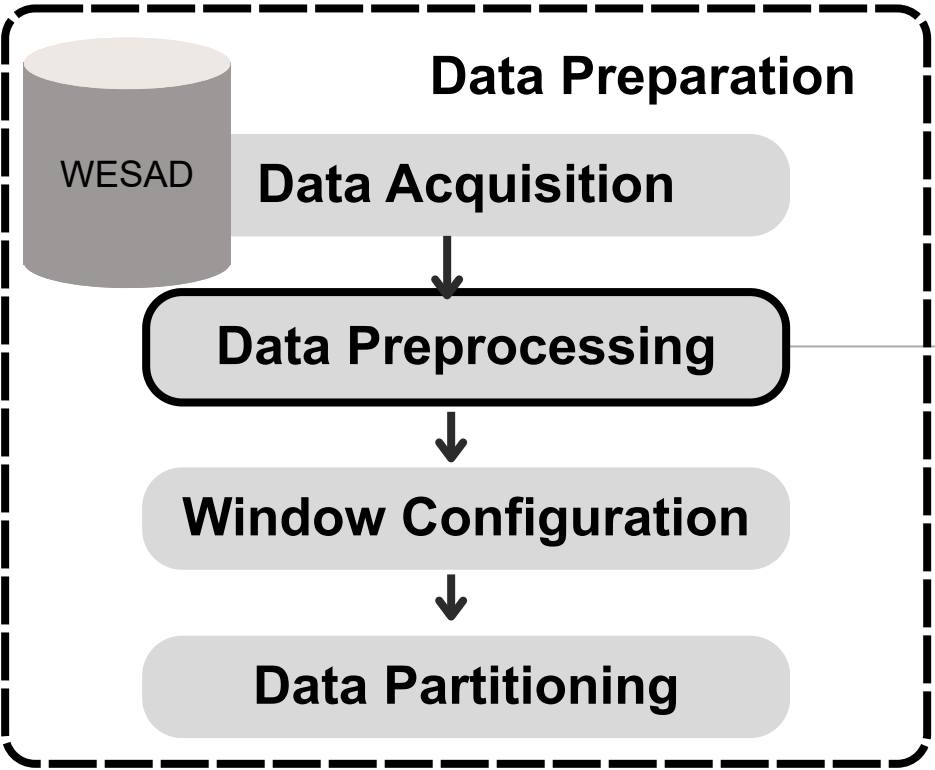




WESAD = Multimodal Dataset
(Wearable Stress and Affect Detection)

15 Participants





DATA Preparation

Data Preprocessing

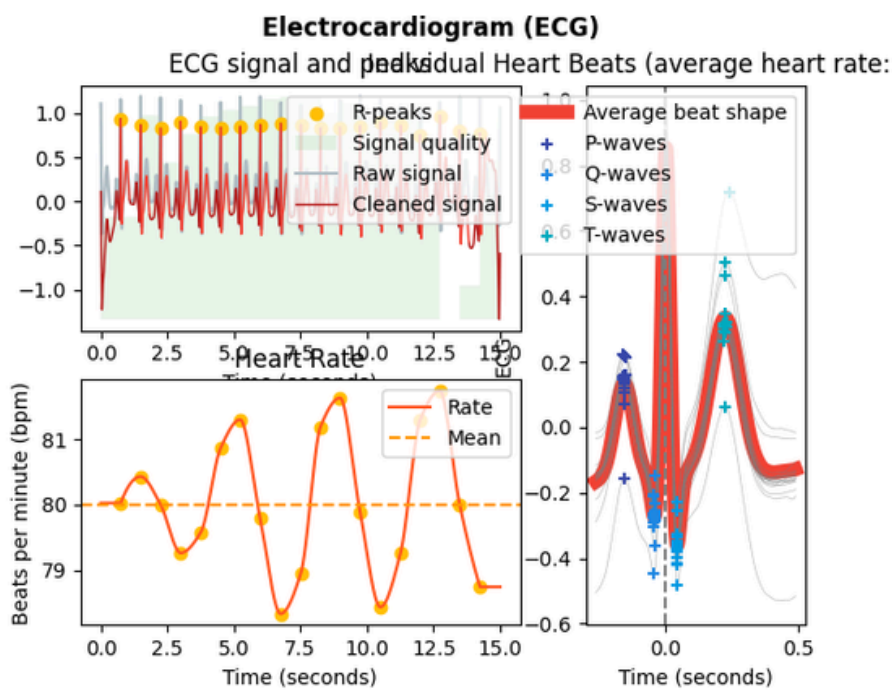
- Resampling
 - 4 Hz
- Filtering & Denoising & Feature Extraction
 - NeuroKit2 Library
 - Extracted 55 time-domain features
 - Ex) Peaks, Heart Rate, Amplitude, ...
- Normalization
 - Standardization (z-score)

RespiBAN-Chest

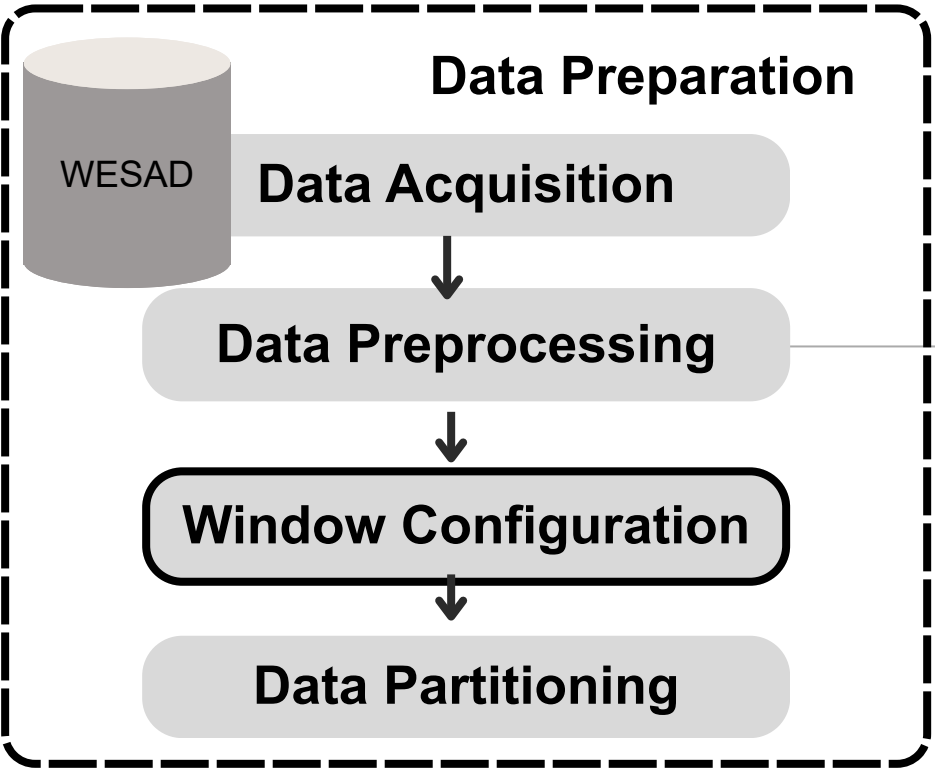
ECG, EDA, EMG, RESP, TEMP, ACC
(700Hz)

Empatica E4- Wrist

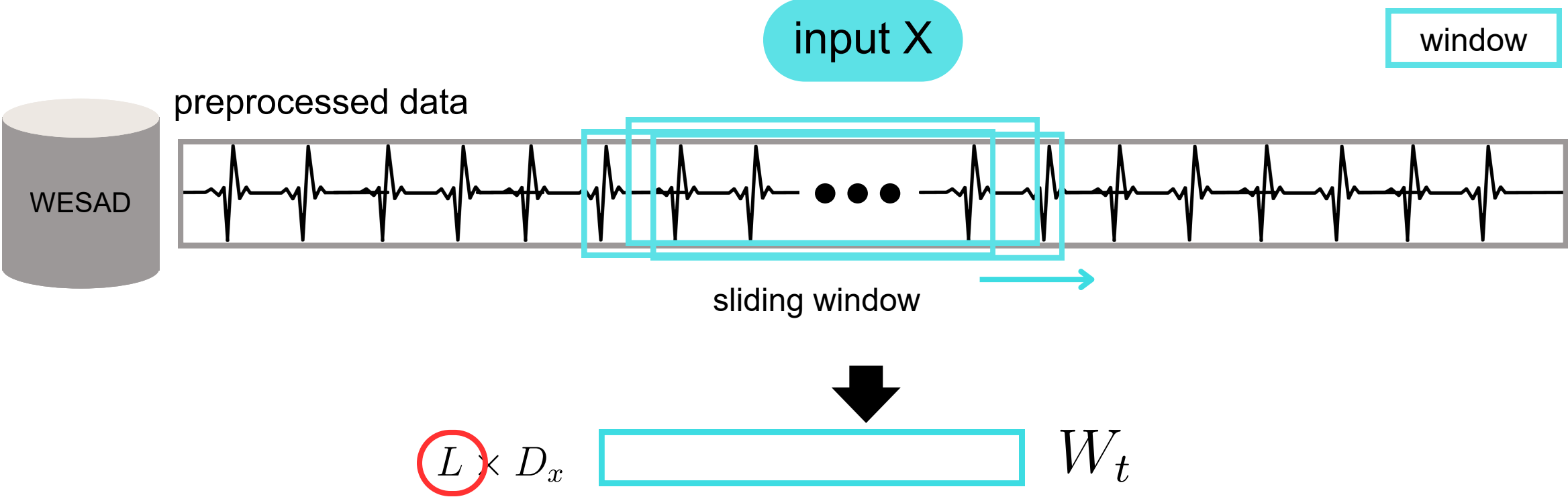
BVP (64Hz), EDA (4Hz), TEMP (4Hz),
ACC (32Hz)



Source: NeuroKit2 Documentation

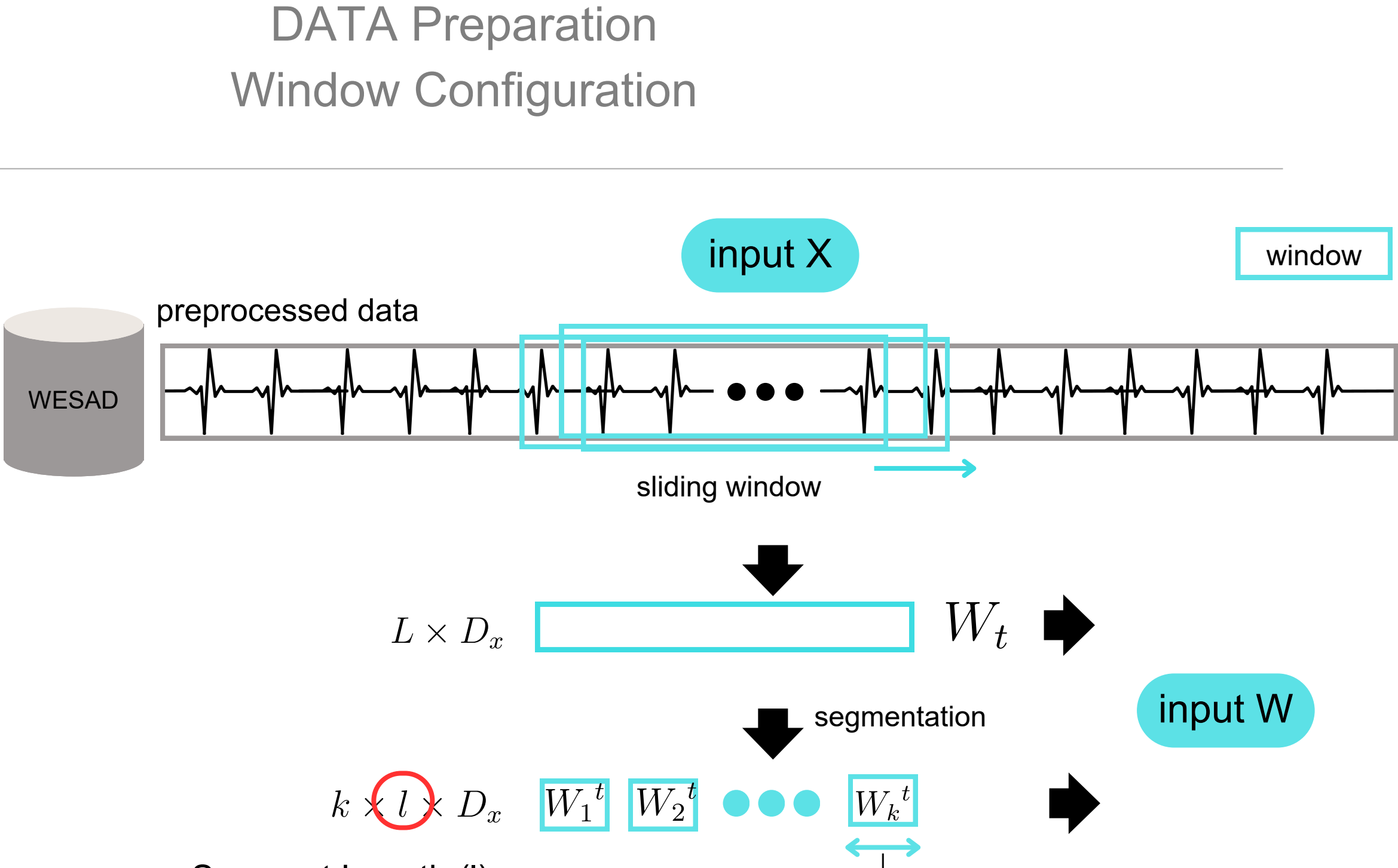
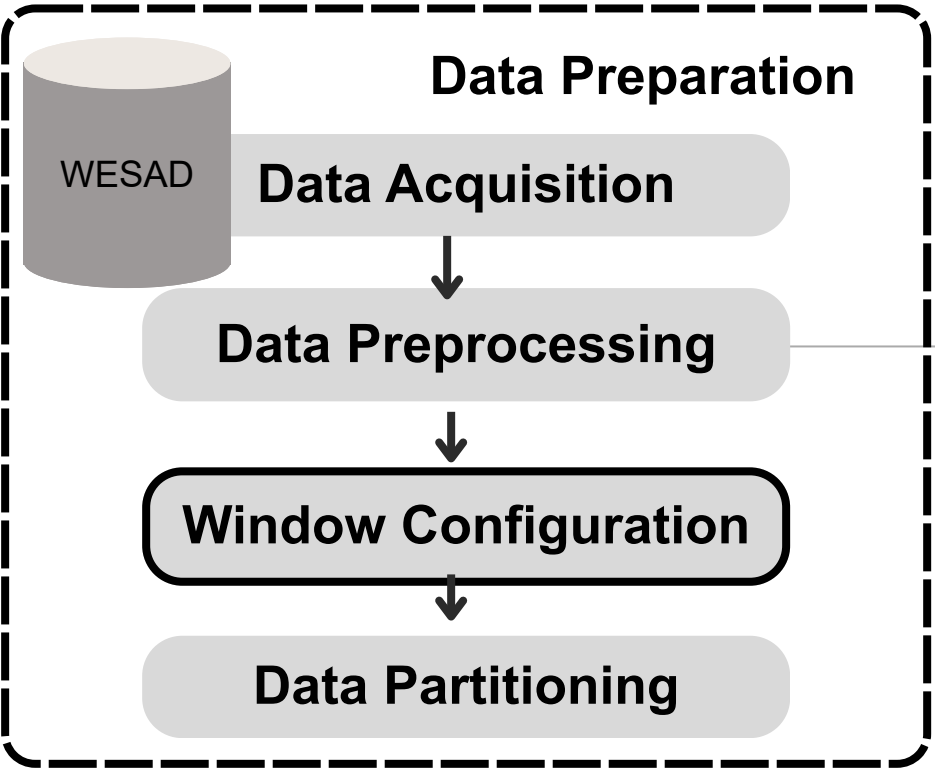


DATA Preparation Window Configuration



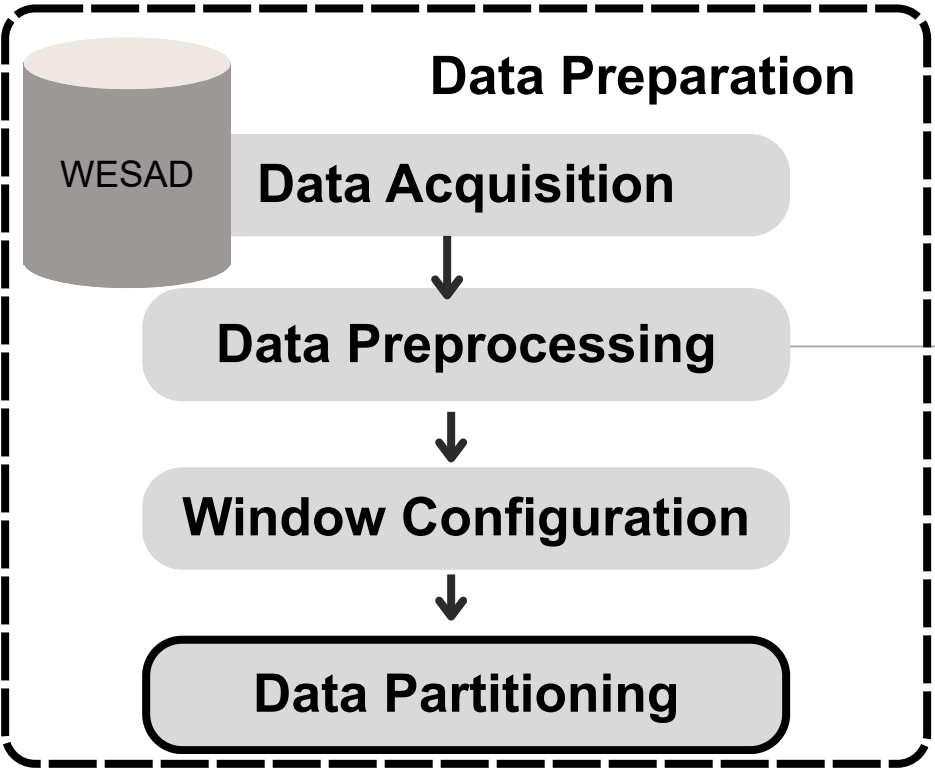
Window Length (L)

- Entire length = Window Size
- For Global Module
- Capture the long-term context and overall trends of the data



Segment Length (l)

- The length of the smaller sub-segments derived by dividing the entire window
- For Local Module
- Capture fine-grained and rapid patterns within each local segment



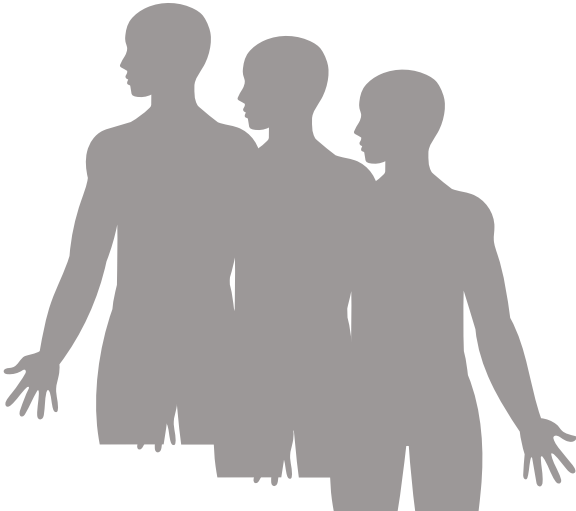
DATA Preparation

Data Partitioning - 3-Fold Cross-Validation

The Number of Subjects: 15

Data Split

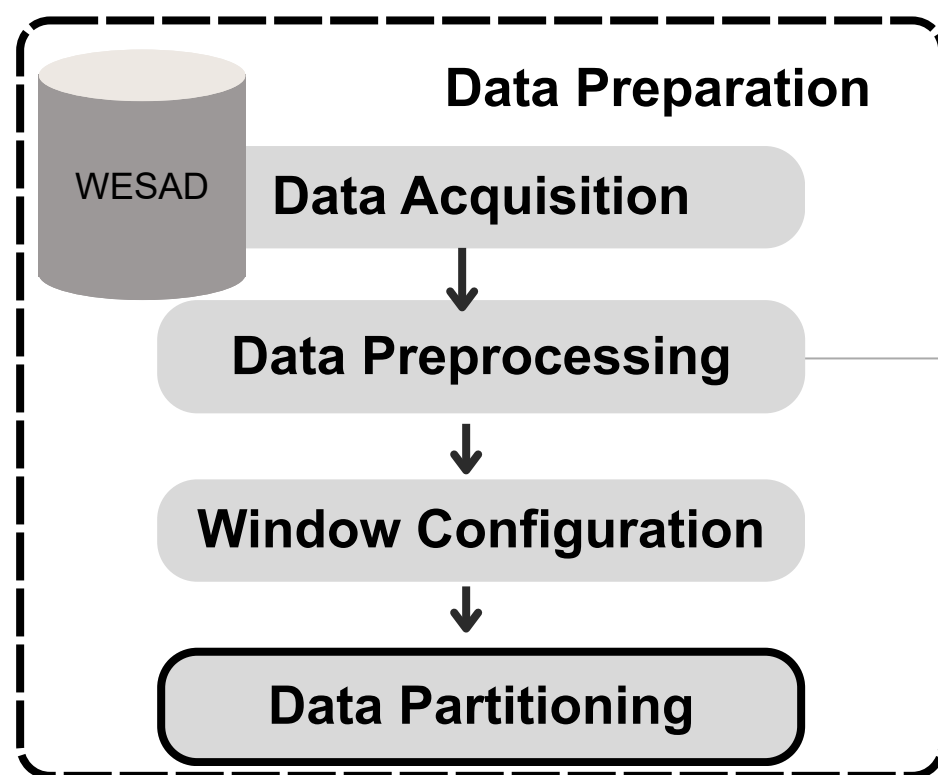
- Train (10 subjects)
- Test (5 subjects)



All Data = 15 Participants			
	Fold1	Fold2	Fold3
Split 1	Test	Train	Train
Split 2	Train	Test	Train
Split 3	Train	Train	Test

3-Fold Subject-Independent Cross-Validation

- For Generalization Performance
- Not Random suffling, But Subject-Independent Split



DATA Preparation

Data Partitioning - 3-Fold Cross-Validation

Training Dataset

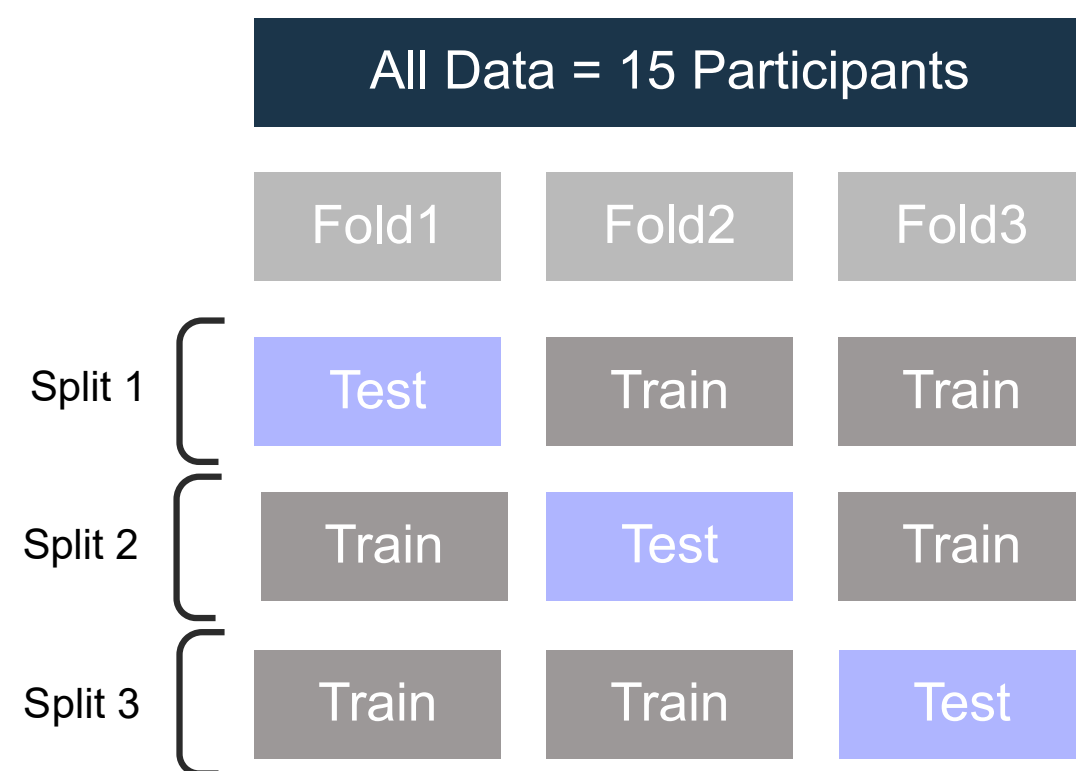
- Used for training
- Only Non-stress Data

Test Dataset

- Used for evaluating
- Both Non-stress + Stress (Anomaly)
- Binary Classification: Non-Stress vs Stress

Validation Dataset

- Used for early stopping while training
- After training data is segmented into windows, randomly sampled 10% is used as validation dataset.

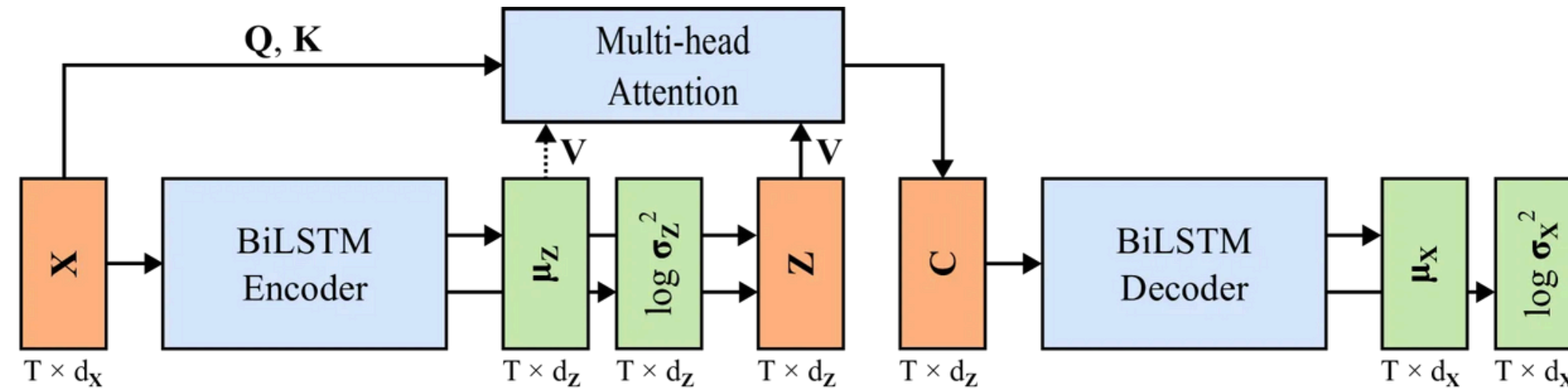


03

Model Structure

Baseline Models

Baseline Models: MA-VAE



$$\mathbf{Q}_i = \mathbf{X}\mathbf{W}_i^Q, \quad \mathbf{K}_i = \mathbf{X}\mathbf{W}_i^K, \quad \mathbf{V}_i = \mathbf{Z}\mathbf{W}_i^V.$$

$$\mathbf{C}_i = \text{Softmax} \left(\frac{\mathbf{Q}_i \mathbf{K}_i^T}{\sqrt{d_K}} \right) \mathbf{V}_i.$$

- Each attention head **focuses on a different time step**
- Learns to **prioritize relevant features while suppressing noise**
- Multi-head attention effectively integrates **contextual information**

Baseline Models: MA-VAE

Loss Function

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{rec}}}_{\text{Reconstruction Loss}} + \beta \underbrace{\mathcal{L}_{\text{KL}}}_{\text{Latent Regularization}}$$

Reconstruction Loss

$$\mathcal{L}_{\text{rec}} = \|x - \hat{x}\|^2$$

KL Divergence

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2} \sum_{j=1}^d (1 + \log(\sigma_j^2) - \mu_j^2 - \sigma_j^2)$$

Why KL Vanishing Happens in VAE?

- In early training, the KL term **dominates the loss function**.
- The **encoder ignores the latent space** → latent collapse.
- Model behaves like a **normal autoencoder** → **poor anomaly separation**.

→ **MA-VAE uses a Cyclical KL Annealing schedule**

Baseline Models: MA-VAE

MA-VAE uses a Cyclical KL Annealing schedule

1. Grace Period (Epoch 0–25)

- β increases **very slowly**: $\beta = \text{linear_cycle}(0 \rightarrow 10^{-4})$
- Ensures **pure reconstruction learning first** $\rightarrow$ **stable encoder**.

2. Cyclical Phase (After Epoch 25)

Every 25 epochs: $\beta = \text{linear_cycle}(10^{-4} \rightarrow 10^{-1})$

- Each cycle **re-introduces KL pressure gradually**
- **Prevents KL from exploding**
- Avoids latent collapse while keeping training stable

“Linear cycle”

Linear cycle simply means:

- **β increases linearly** within each cycle
- It resets back to the starting β at the beginning of the next cycle

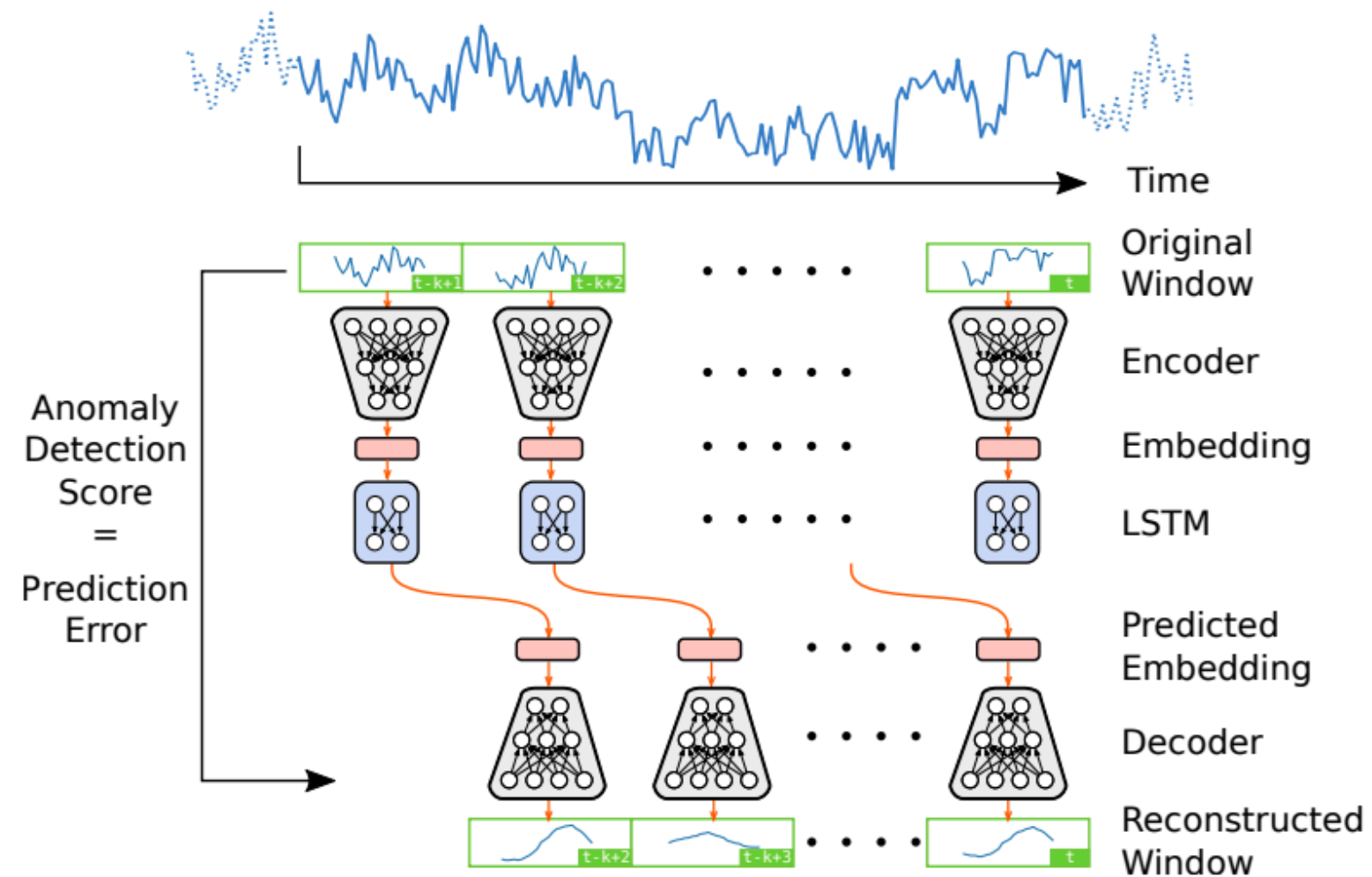
Mathematically:

$$\beta_t = \beta_{\min} + (\beta_{\max} - \beta_{\min}) \cdot \frac{(t \bmod T_{\text{cycle}})}{T_{\text{cycle}}}$$

Where,

- T_{cycle} = cycle length (25 epochs)
- $\beta_{\min} = 10^{-4}$
- $\beta_{\max} = 10^{-1}$

Baseline Models: VAE-LSTM



Core Idea

VAE-LSTM combines:

- **VAE** → **Local structure extraction:** Learns robust representations from short windows, captures local regularity (normal behavior patterns)
- **LSTM** → **Temporal dependency modeling:** Predicts how embeddings evolve over time, captures sequential trends.

Process: Predicts the next window's embedding

→ Reconstructs window

→ Uses prediction error as the anomaly score

Baseline Models: VAE-LSTM

Loss Function

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{rec}}}_{\text{Reconstruction Loss}} + \beta \underbrace{\mathcal{L}_{\text{KL}}}_{\text{Latent Regularization}}$$

Reconstruction Loss

$$\mathcal{L}_{\text{rec}} = \frac{1}{(k-1)} \sum_{i=2}^k \|w_i - \hat{w}_i\|_2^2$$

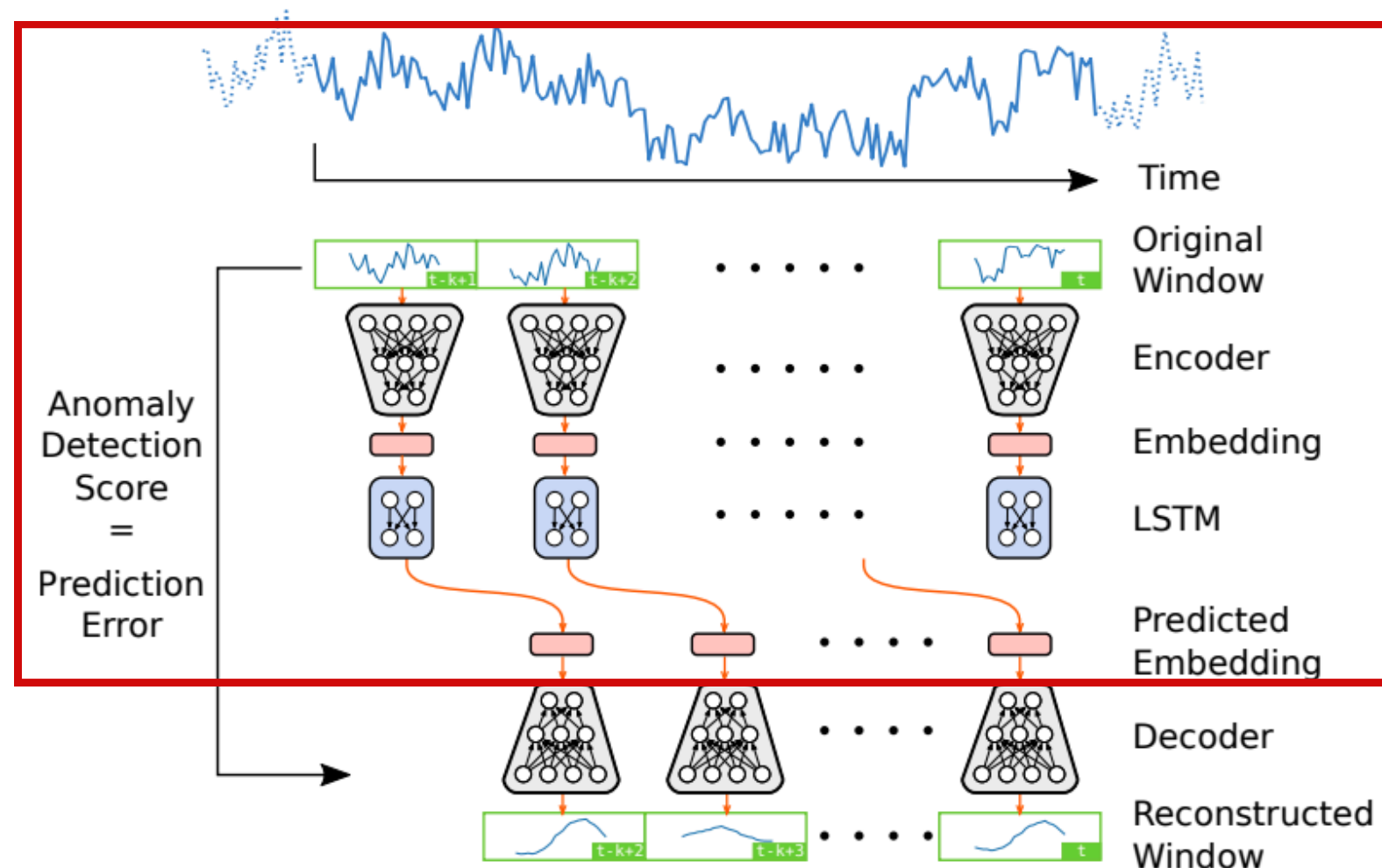
KL Divergence

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2} \sum_{j=1}^d (1 + \log(\sigma_j^2) - \mu_j^2 - \sigma_j^2)$$

- k : Total number of segments
- $w_{b,i}$: The i -th ground truth segment
- $\hat{w}_{b,i}$: The i -th reconstructed segment via LSTM prediction

Baseline Models: VAE-LSTM

VAE-LSTM is a hybrid model that processes each time-series window in two stages:



1. VAE → Local Structure Extraction

- The **original window** is segmented into multiple shorter sub-windows (e.g., 576 → 12 segments of length 48).
- Each segment is fed independently into the **VAE encoder**.
- This means the **VAE only “sees” local patterns (segment), not the entire global structure.**

2. LSTM → Temporal Dependency Modeling

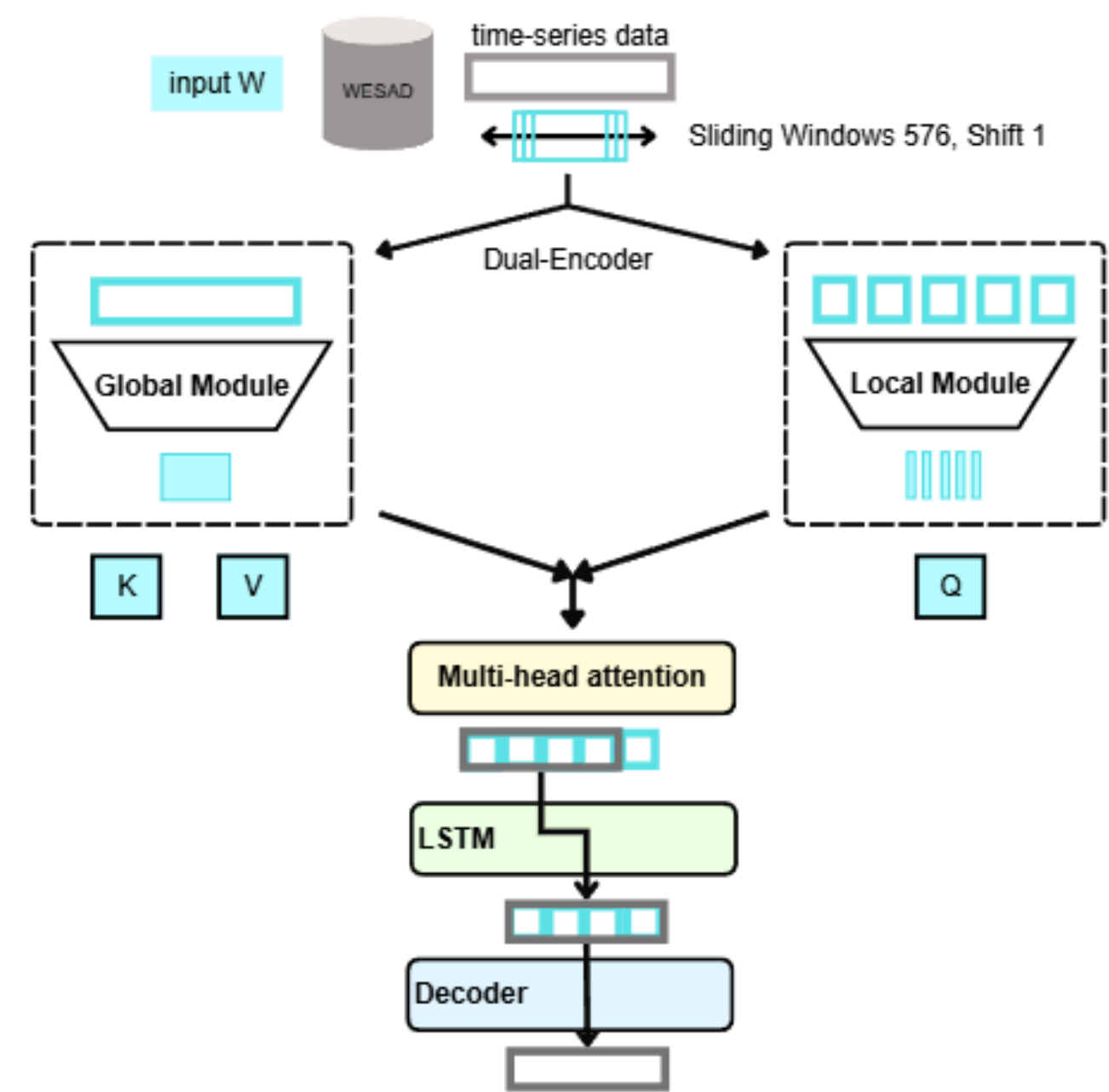
- The sequence of latent vectors from VAE is passed to the **LSTM**.
- LSTM predicts the next latent vector, capturing **temporal evolution** and sequential trends.

Lin, S., Clark, R., Birke, R., Schönborn, S., Trigoni, N., and Roberts, S. Anomaly detection for time series using VAE-LSTM hybrid model. In Proceedings of the 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 4322–4326. IEEE, 2020. doi: 10.1109/ICASSP40776.2020.9053558.

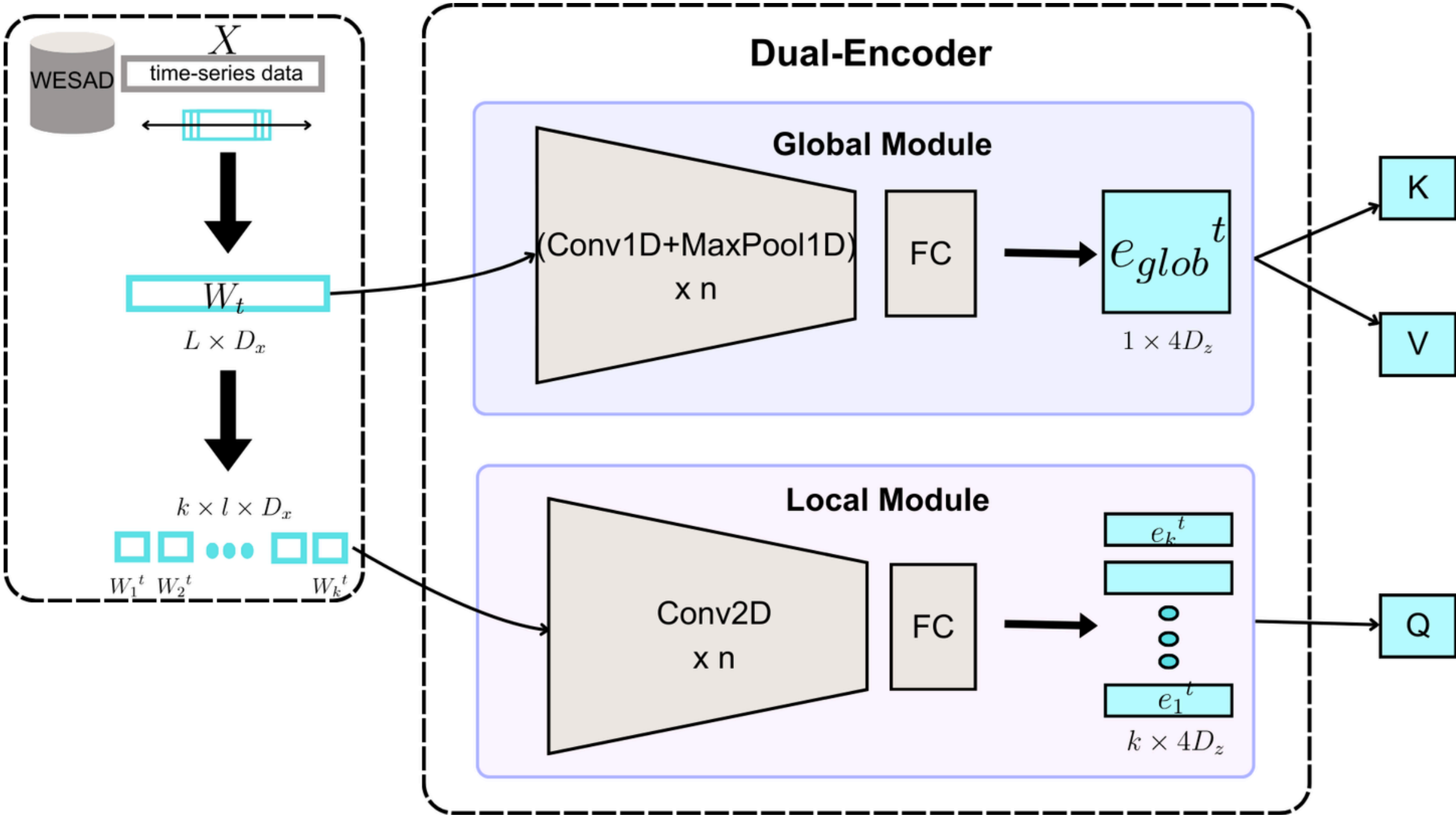
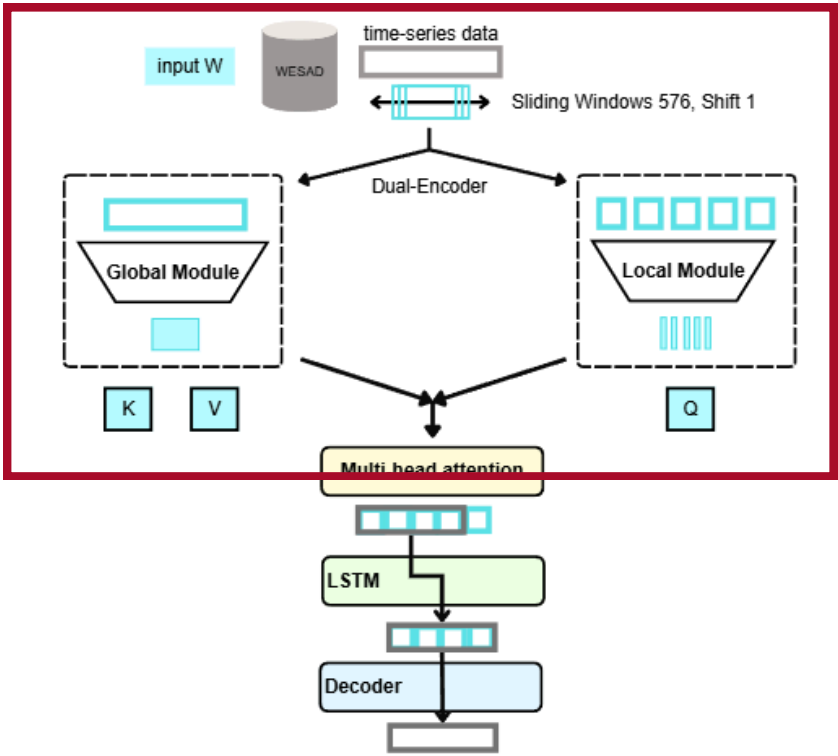
VAE-LSTM is a Prediction Based Model

Proposed Model

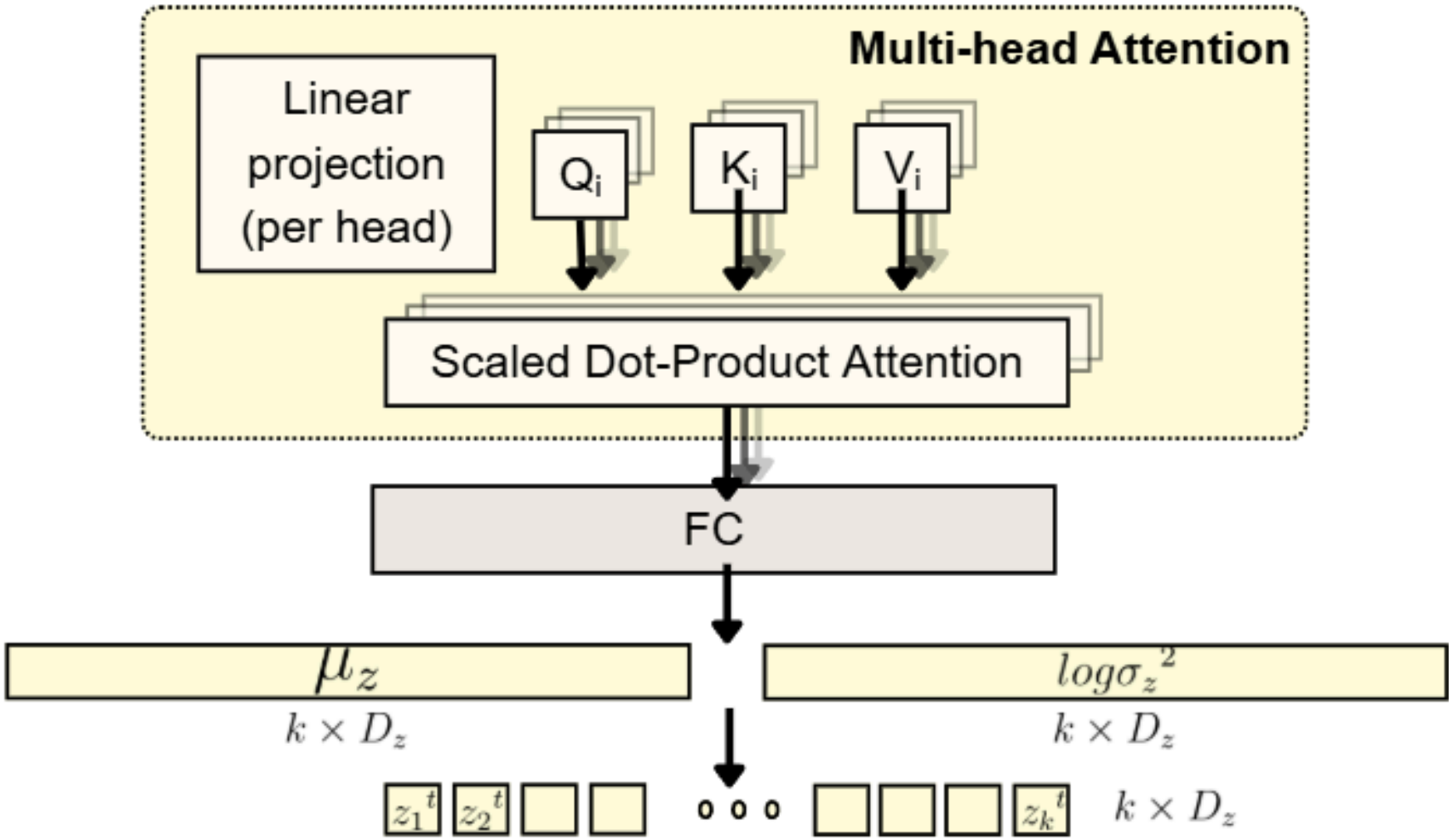
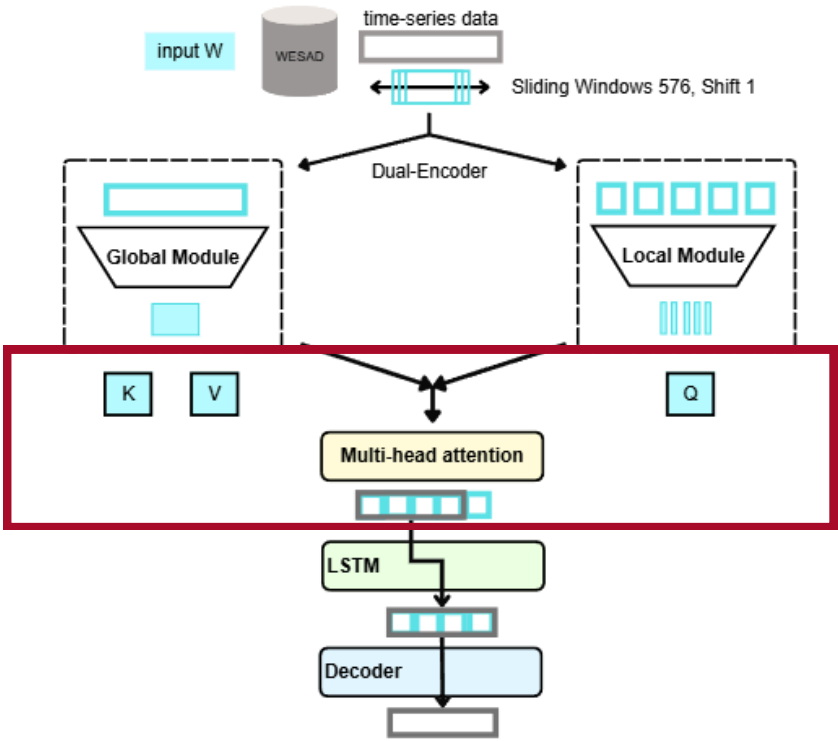
Proposed Dual-Branch VAE-LSTM Framework



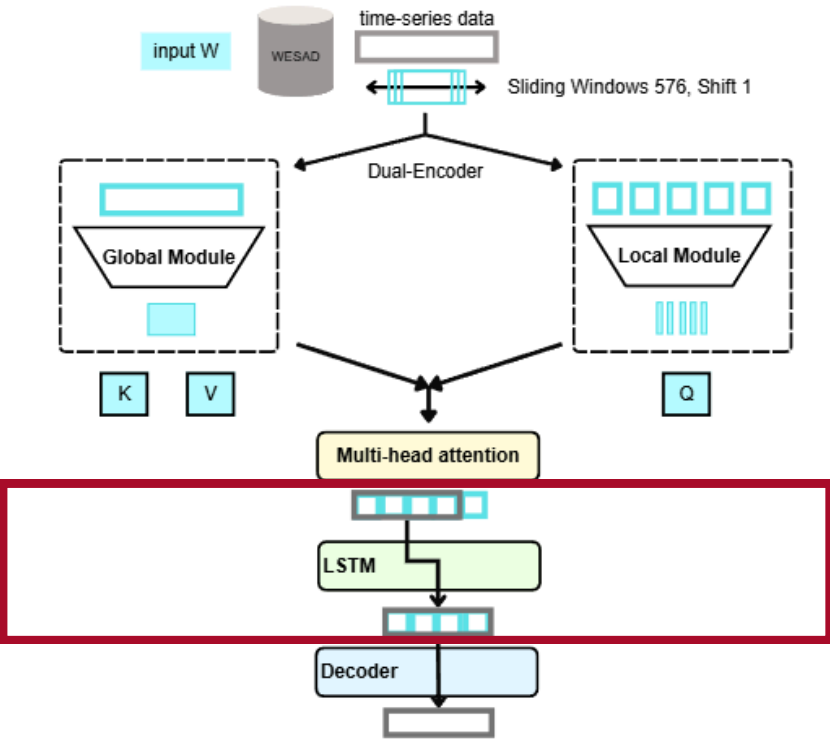
Proposed Dual-Branch VAE-LSTM Framework



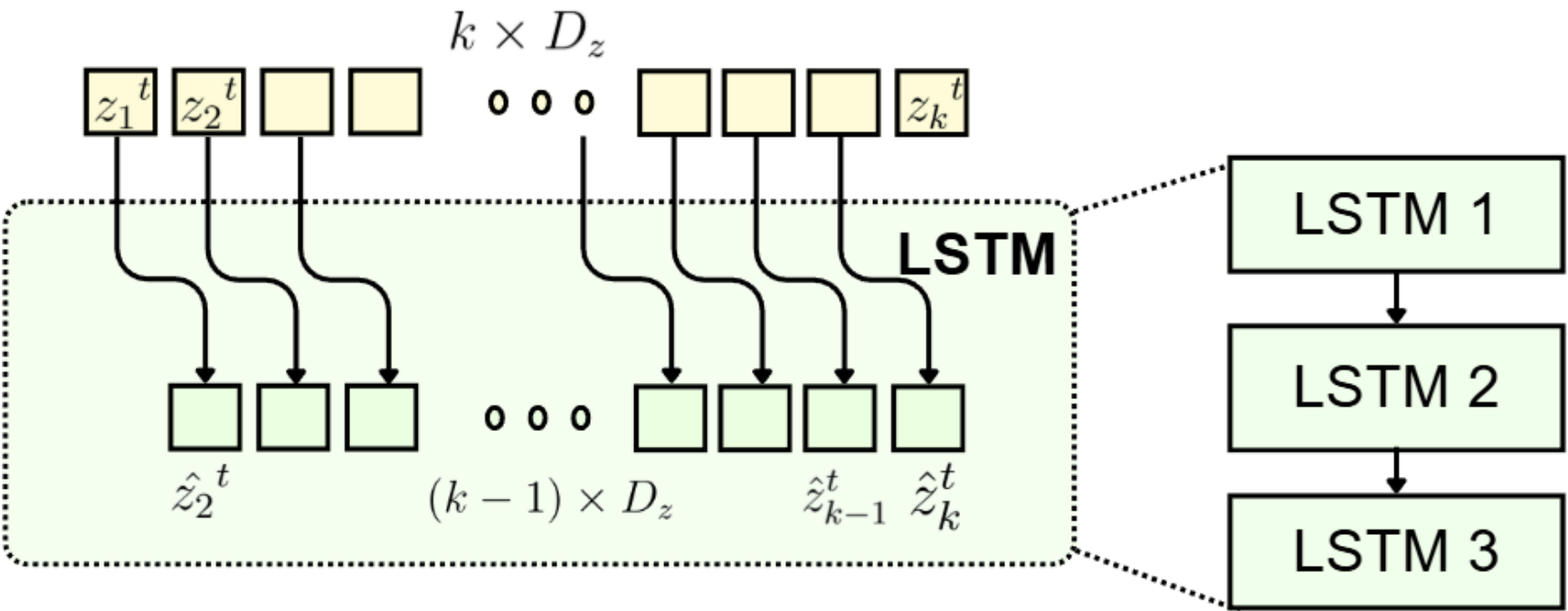
Proposed Dual-Branch VAE-LSTM Framework



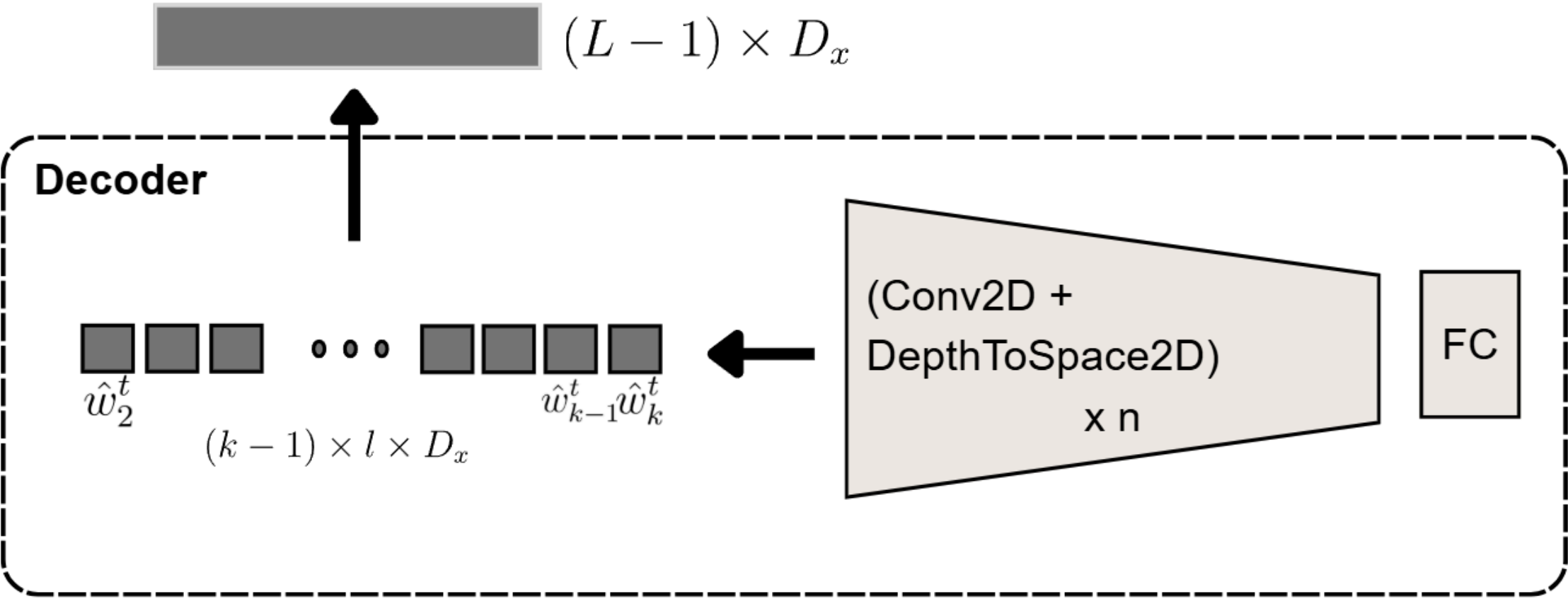
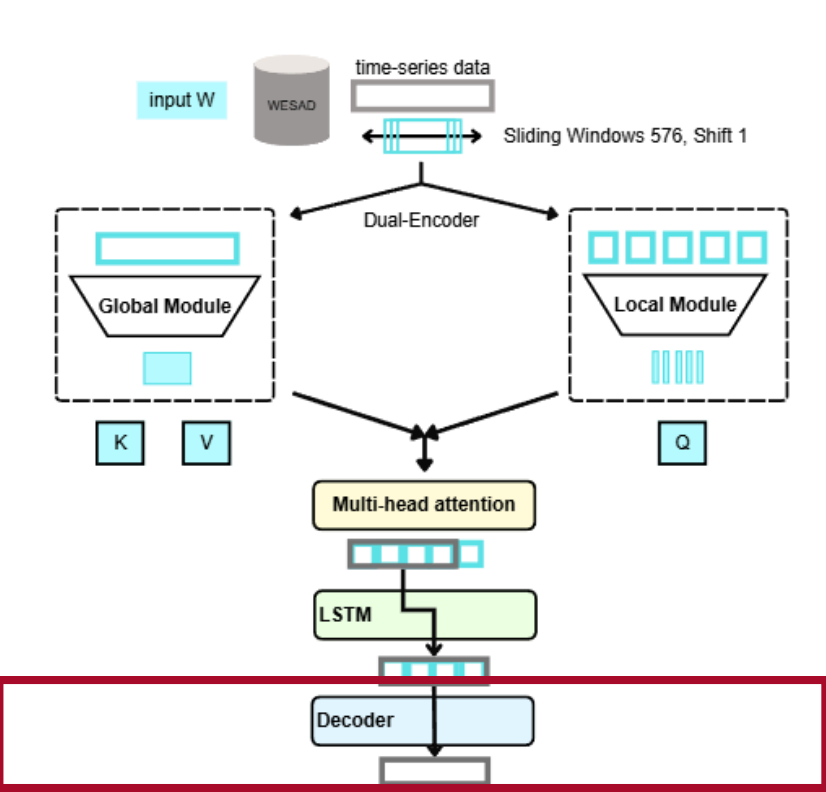
Proposed Dual-Branch VAE-LSTM Framework



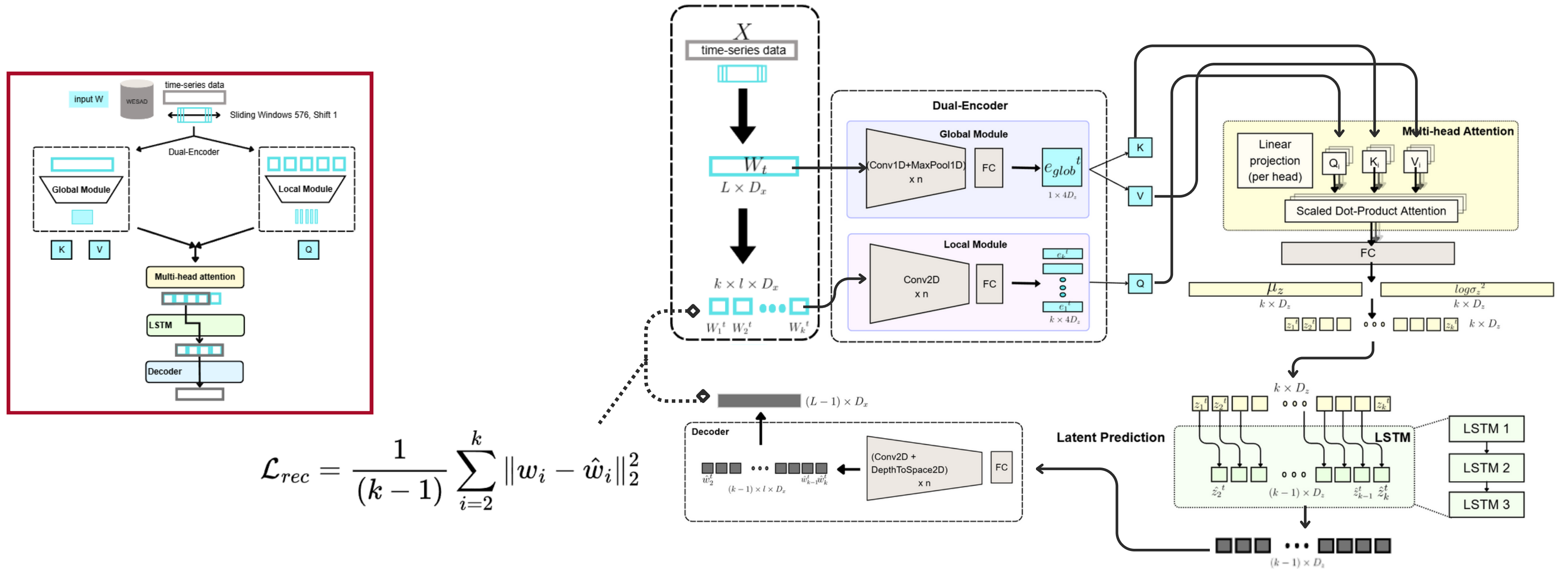
Latent Prediction



Proposed Dual-Branch VAE-LSTM Framework



Proposed Dual-Branch VAE-LSTM Framework



04

Experiment

Experimental Setup

Models Evaluated

- MA-VAE
- VAE-LSTM, Local module only
- Dual-branch VAE-LSTM (Linear Concat)
- Dual-branch VAE-LSTM (Multi-head Attention) ★

Training Configuration

- Optimizer: Adam ($LR = 4 \times 10^{-4}$, $\beta=(0.9, 0.95)$)
- Early stopping = 10 epochs, max epochs = 100
- ELBO loss, Cyclical KL annealing

Window Parameters

- VAE-LSTM: $L=576, l=48, shift=1$
- MA-VAE: $L=576, shift=48$

Evaluation Metrics: Accuracy, Precision, Recall, F1, AUROC

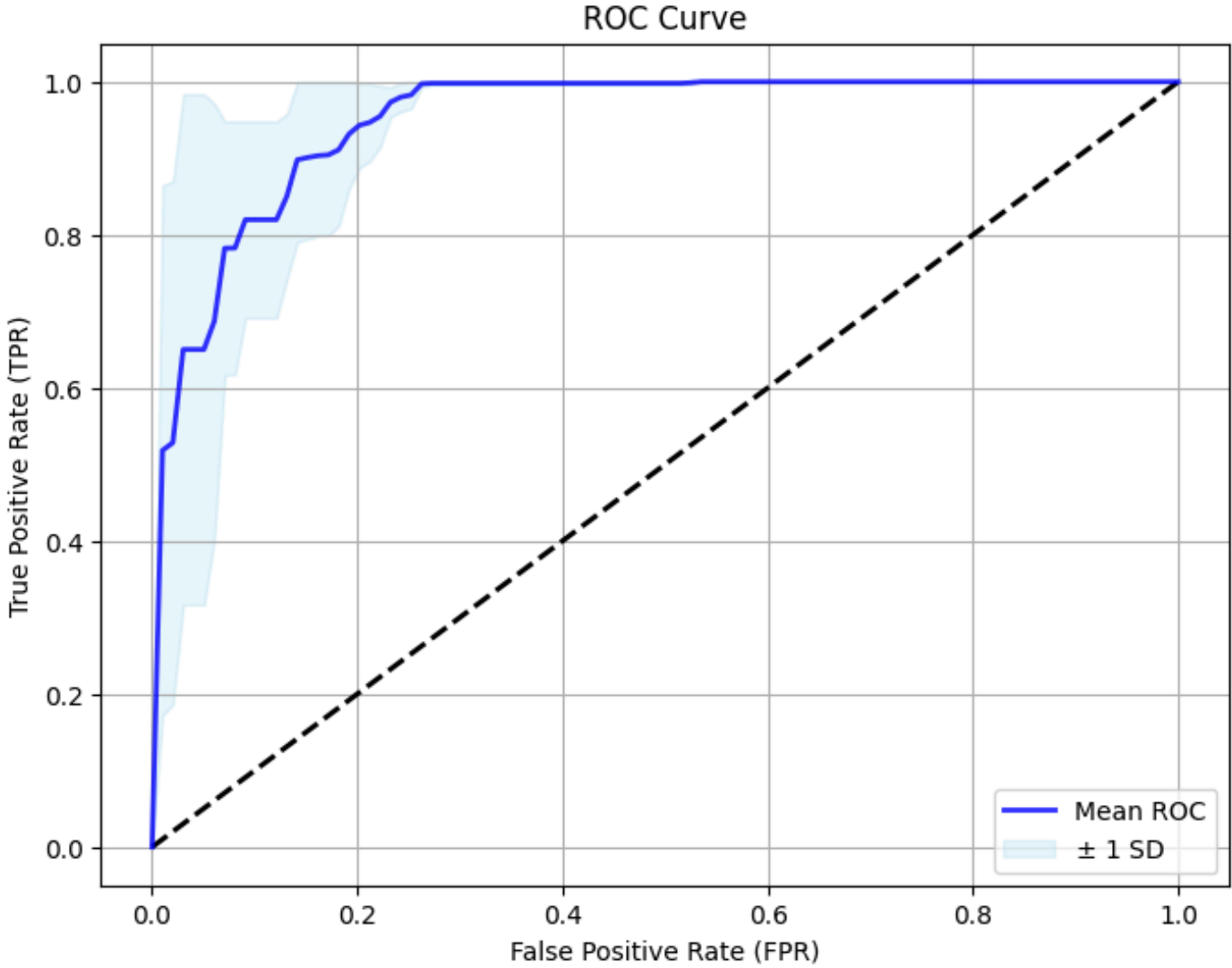
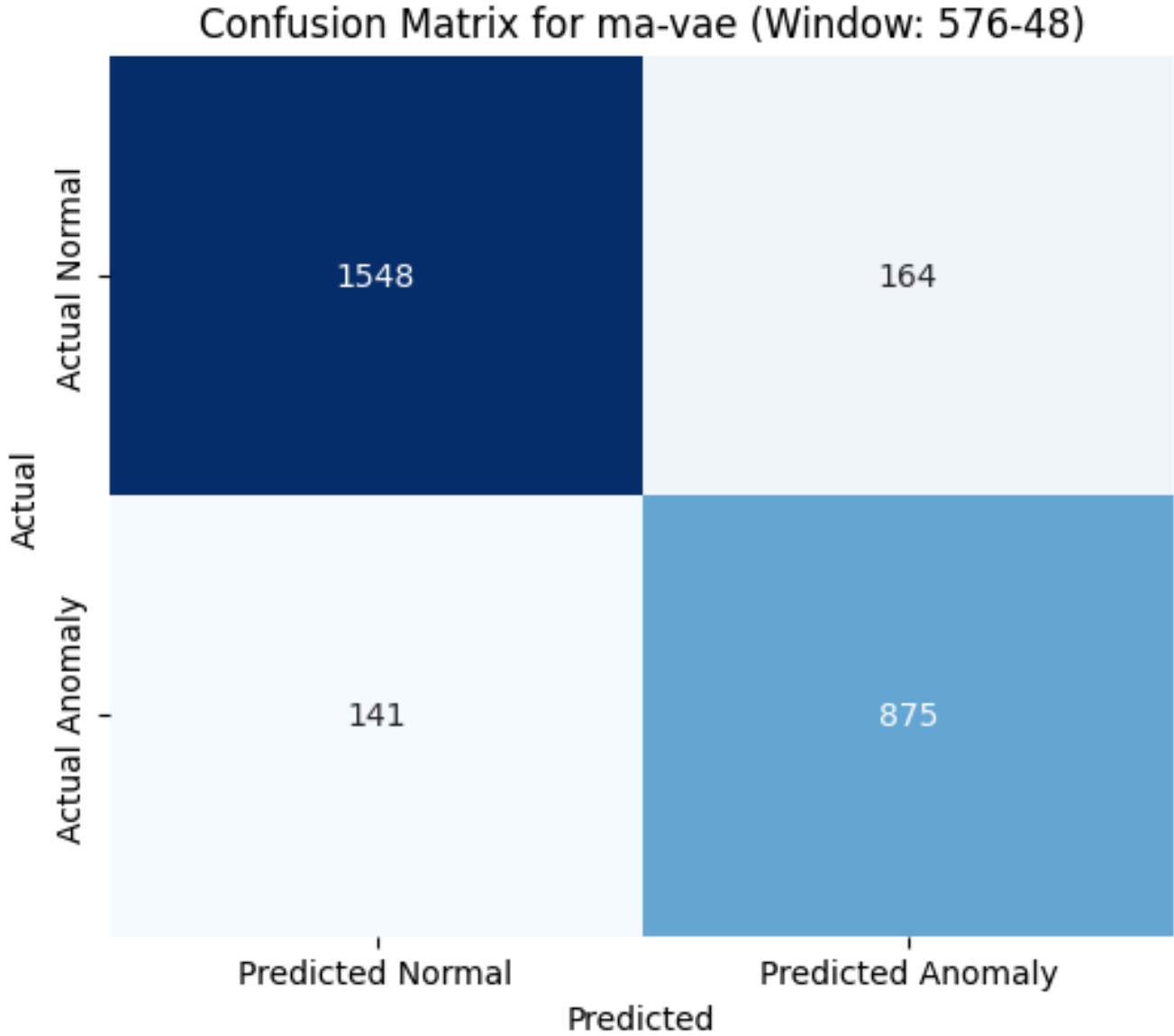
Why Cyclical KL annealing ?

- VAEs suffer from KL-vanishing → latent space collapse.
- Cyclical KL annealing gradually increases KL weight so the model learns meaningful latent features before being regularized

MA-VAE

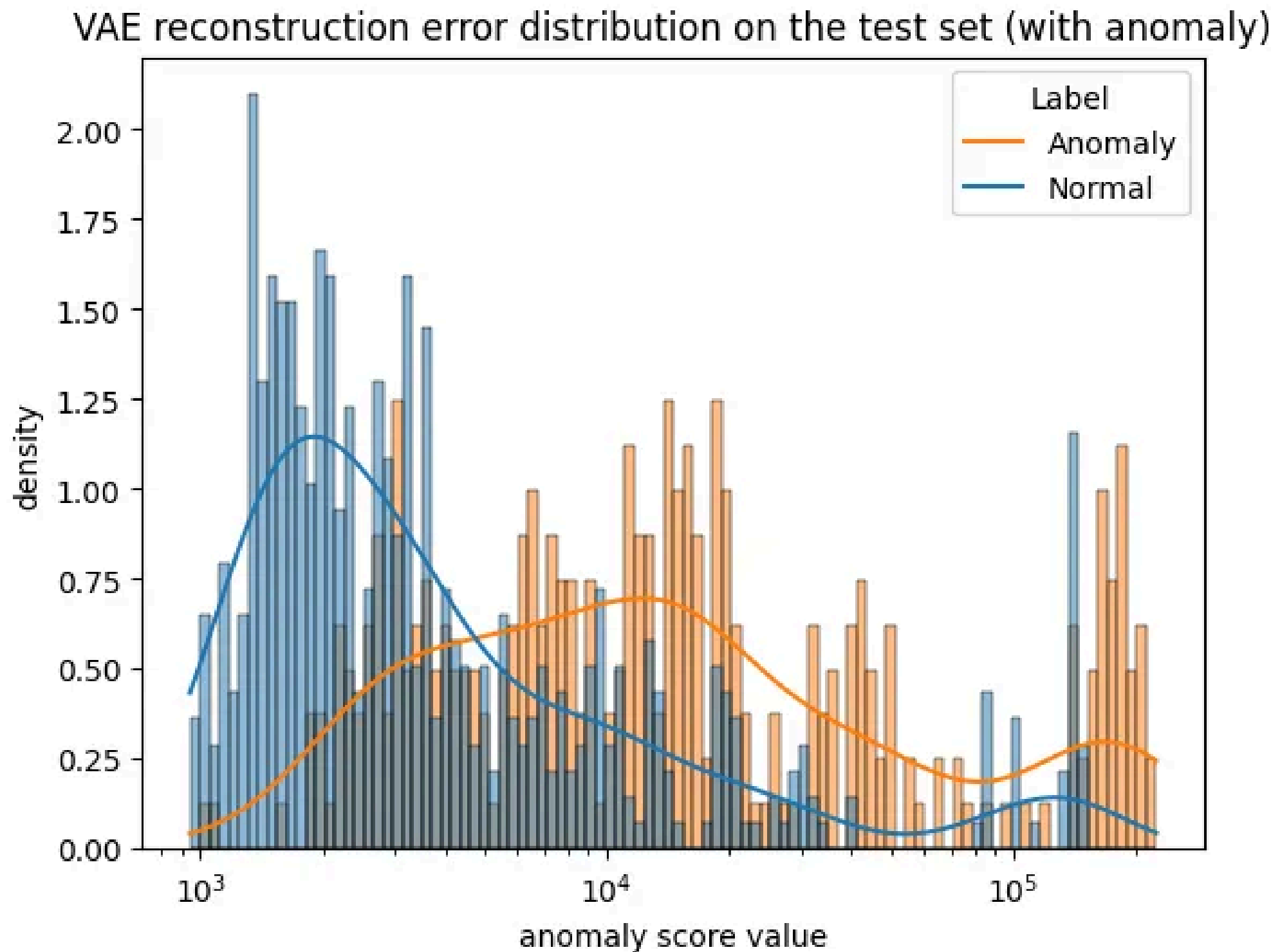
Experiment: MA-VAE

Precision	Recall	F1	AUROC
0.874	0.86	0.856	0.93



Experiment: MA-VAE

Normal vs Anomaly distributions

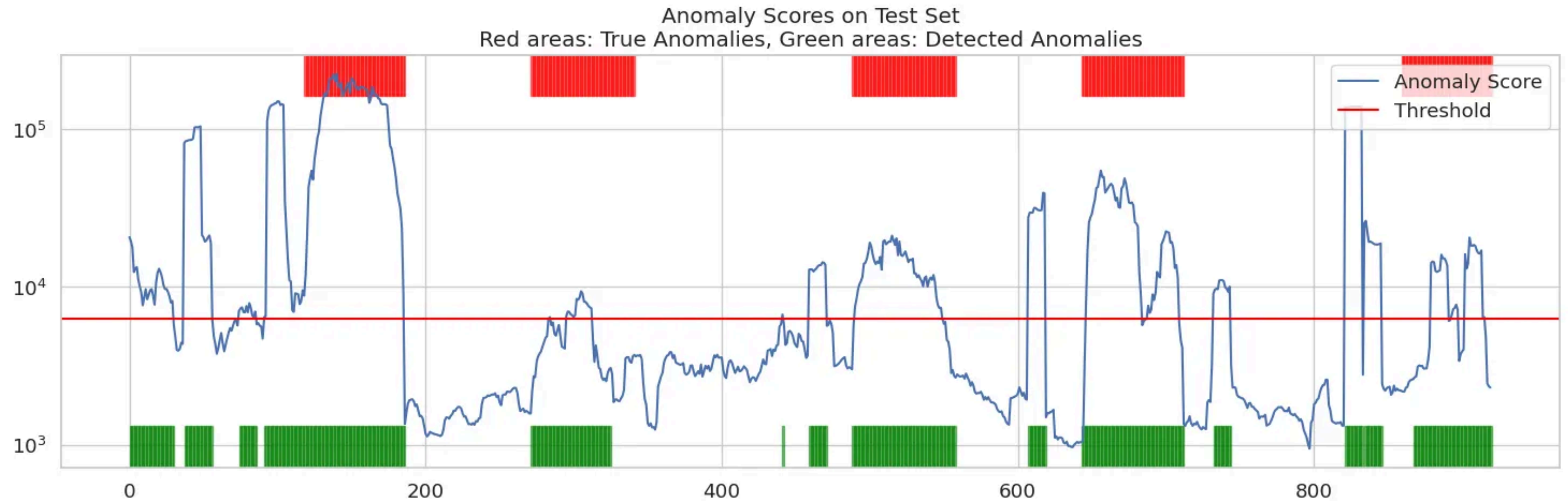


- **Strong overlap between Normal and Anomaly score distributions**

Example results on Fold 3 is used for qualitative analysis.

Experiment: MA-VAE

“How prediction actually behaves?”

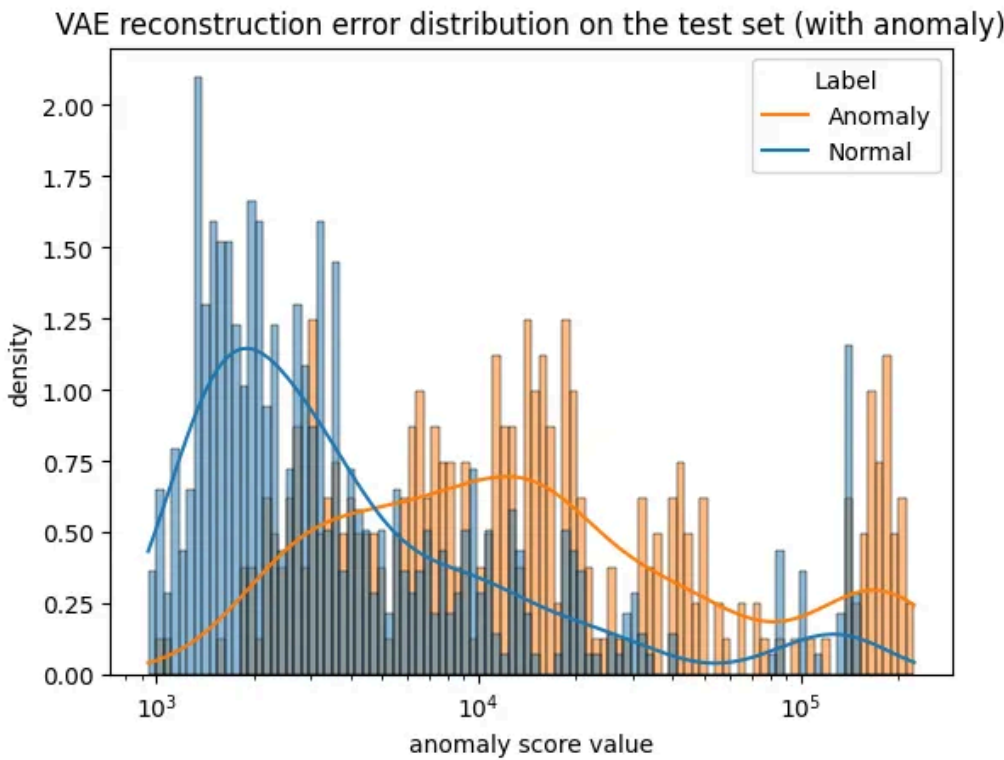
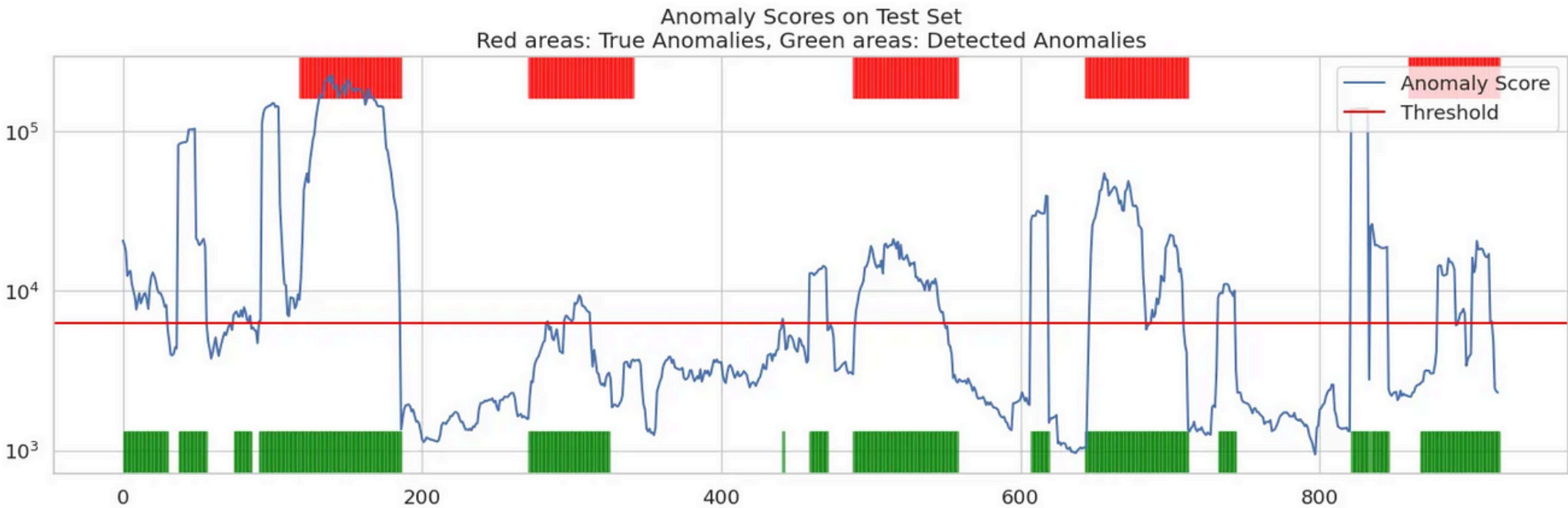


- Models only **reconstruction error** → Limitation of MA-VAE baseline
- Sensitive to any deviation, even normal variability

Example results on Fold 3 is used for qualitative analysis.

Experiment: MA-VAE

Result Analysis



Precision	Recall	F1	AUROC
0.874	0.86	0.856	0.93

STRENGTH:

- Learns **stable global reconstruction patterns**
- **Cyclical KL annealing** prevents KL-vanishing during training

LIMITATIONS:

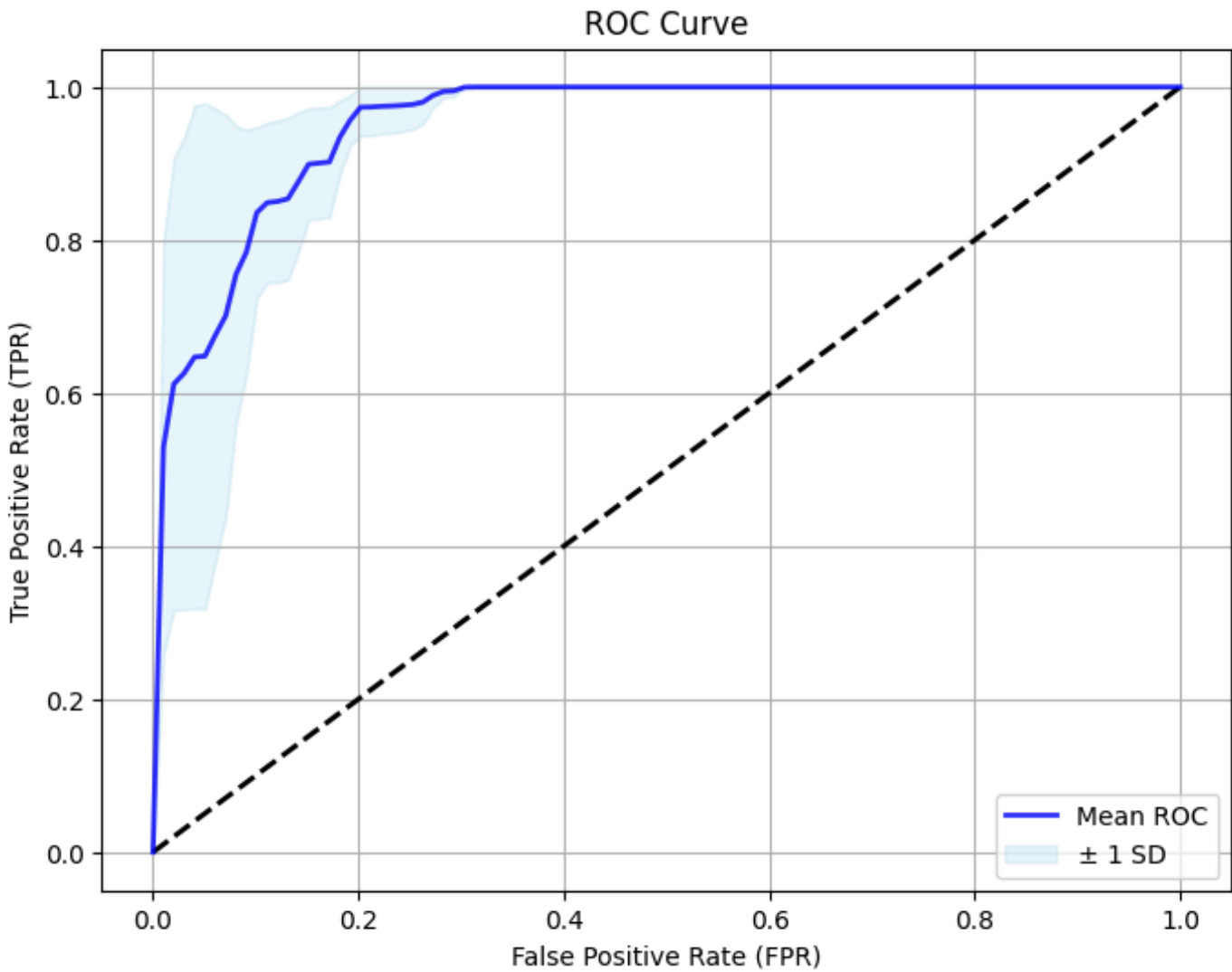
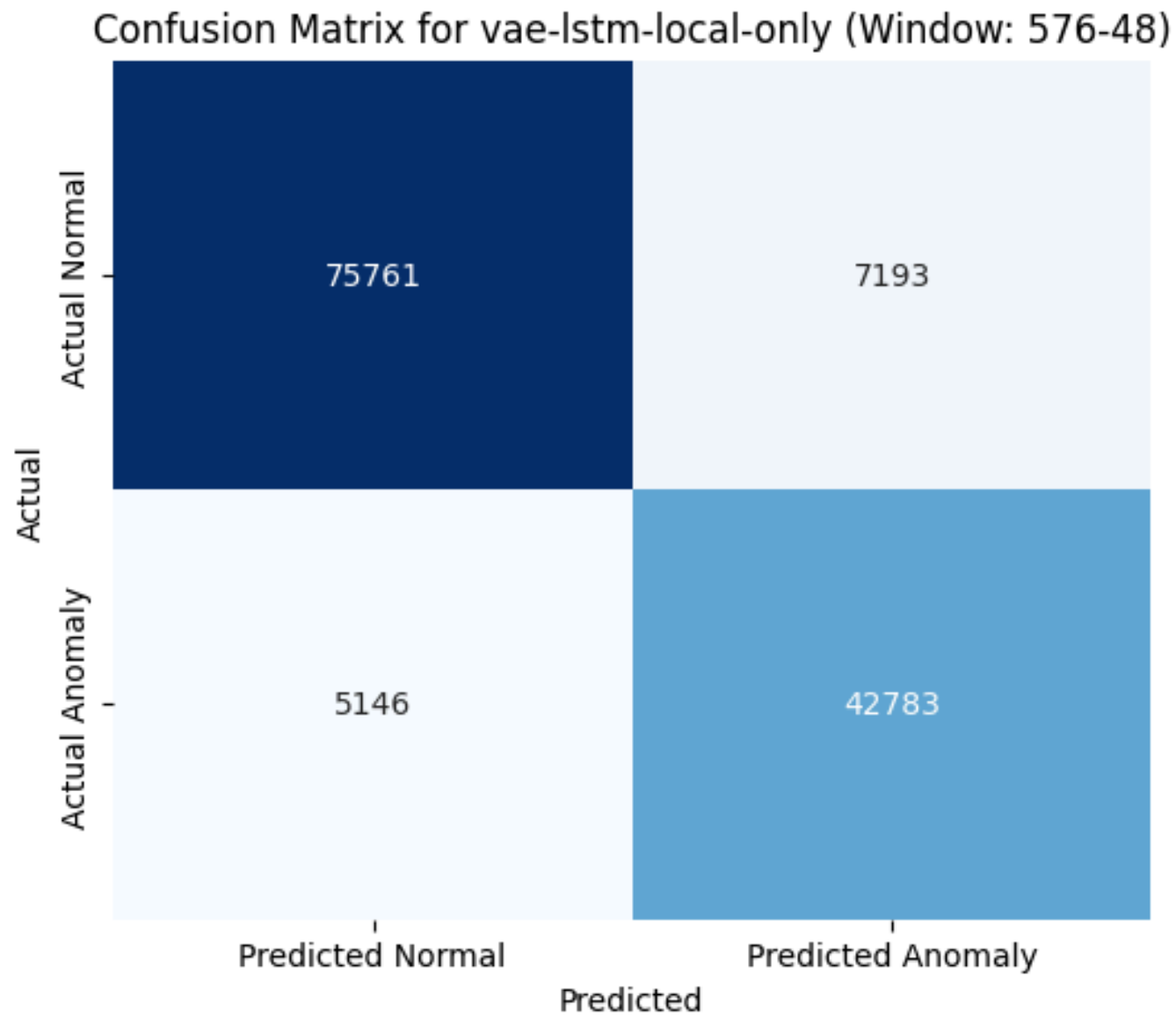
- **Uses only reconstruction error:** cannot capture temporal shifts
- Normal vs anomaly reconstruction error distributions **overlap** heavily: unstable boundaries

Example results on Fold 3 is used for qualitative analysis.

VAE-LSTM

Experiment: VAE-LSTM

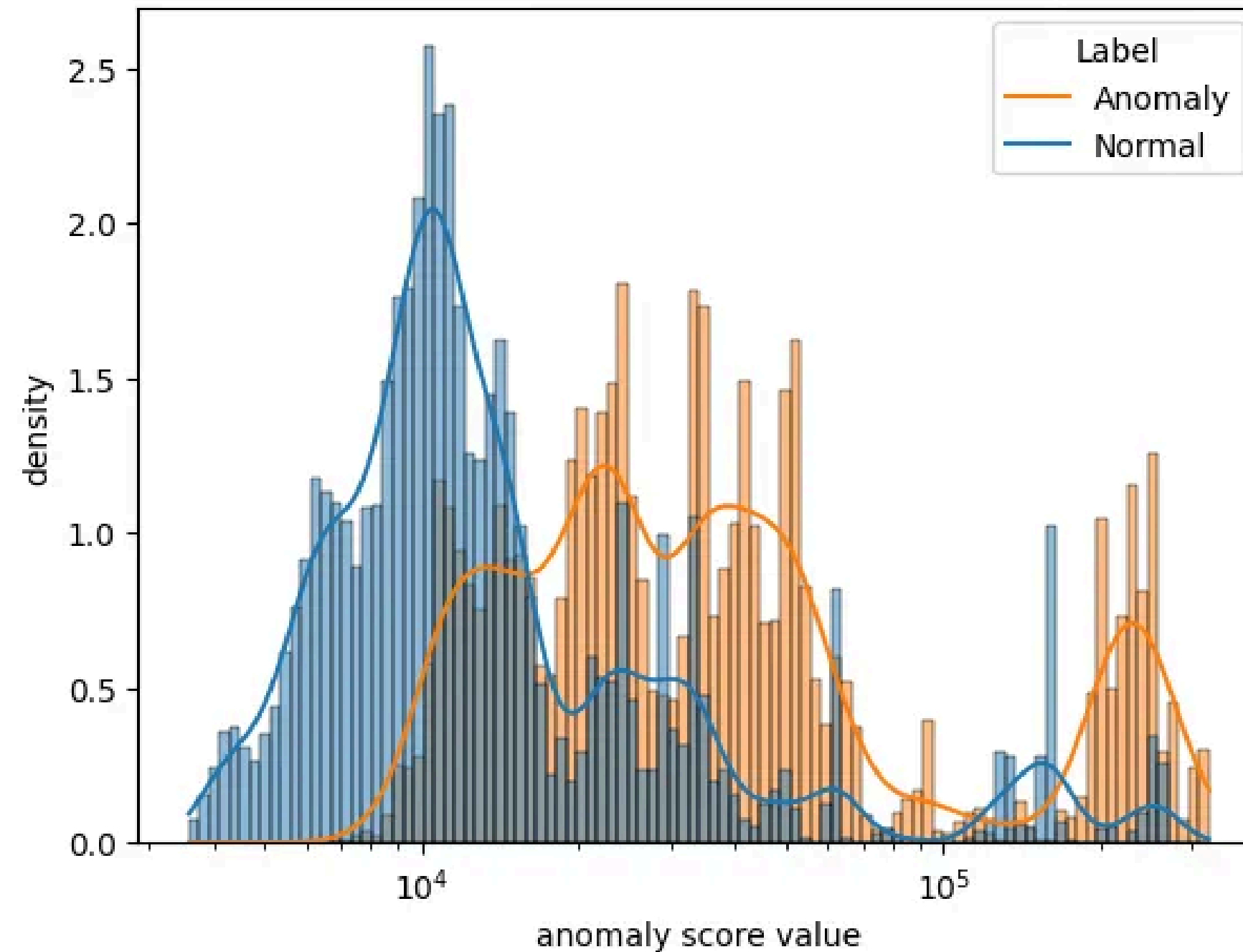
Precision	Recall	F1	AUROC
0.862	0.893	0.875	0.956



Experiment: VAE-LSTM

Normal vs Anomaly distributions

VAE reconstruction error distribution on the test set (with anomaly)

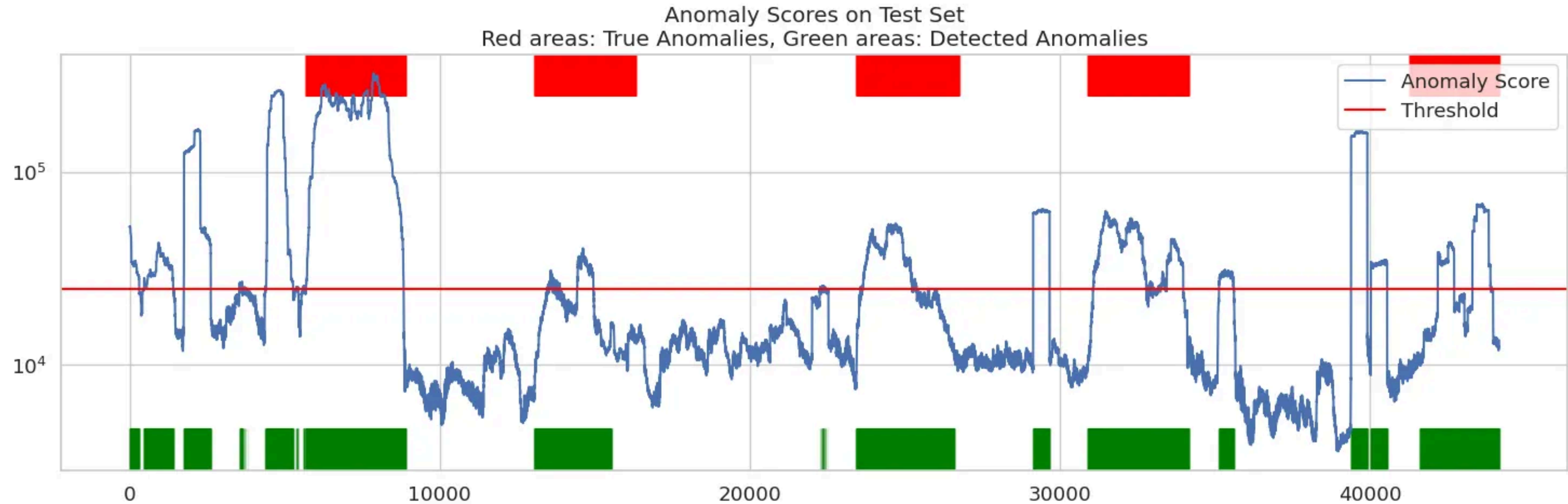


- Better separation than MA-VAE
- **Normal and Anomaly error still overlap because reconstruction error spikes even for normal data**

Example results on Fold 3 is used for qualitative analysis.

Experiment: VAE-LSTM

“How prediction actually behaves?”

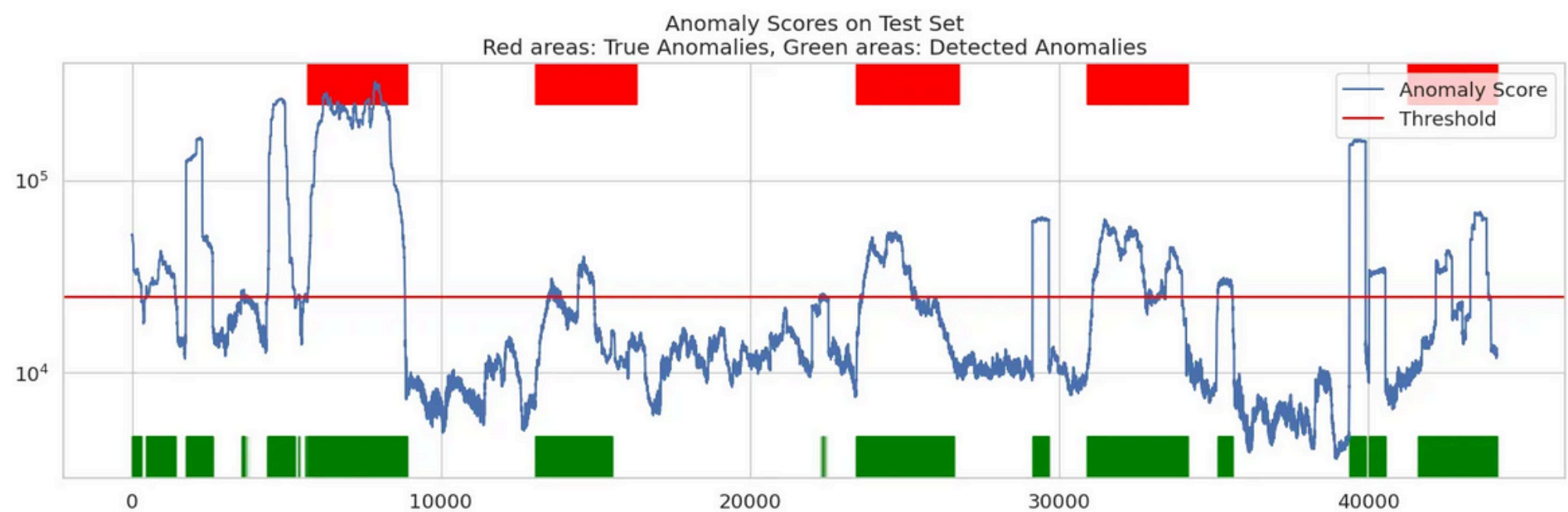


- **VAE-LSTM** successfully identifies most anomaly segments (**green areas** aligning with **red**)
- However, we see several **false alarms** in regions where the signal **changes rapidly** but is not truly anomalous

Example results on Fold 3 is used for qualitative analysis.

Experiment: VAE-LSTM

Result Analysis

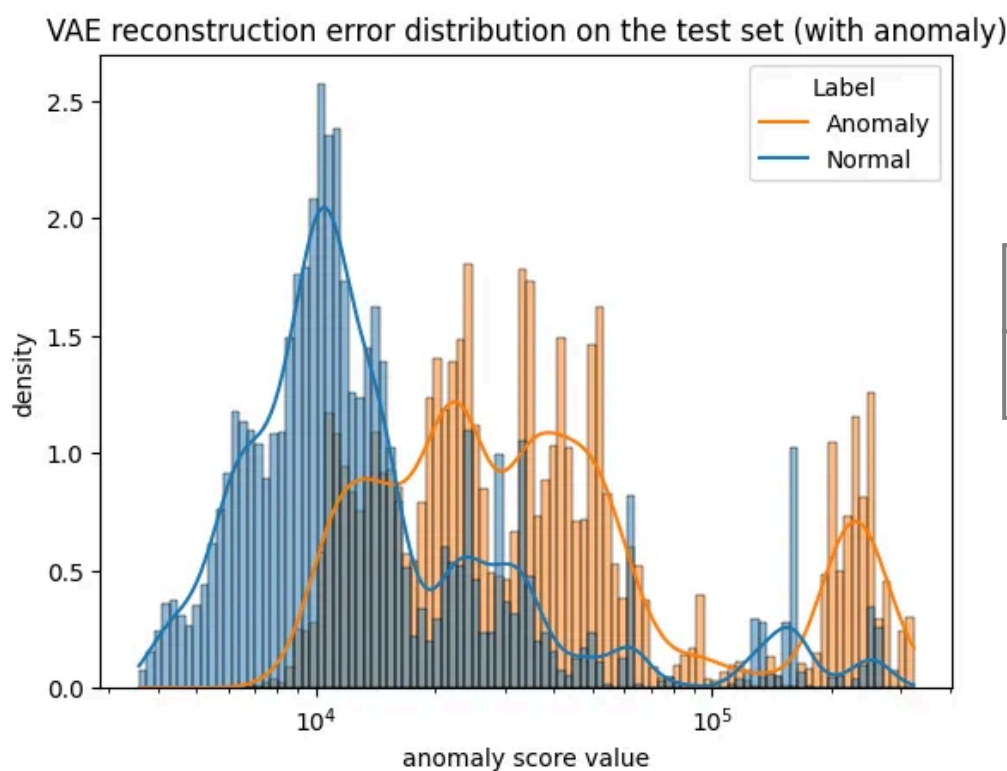


STRENGTH:

- **Good at detecting true anomalies**
- Learns temporal patterns well

LIMITATIONS:

- **Lack of global context:** sensitive to deviations
- Normal vs anomaly reconstruction error distributions overlap: unstable boundary



Precision	Recall	F1	AUROC
0.862	0.893	0.875	0.956

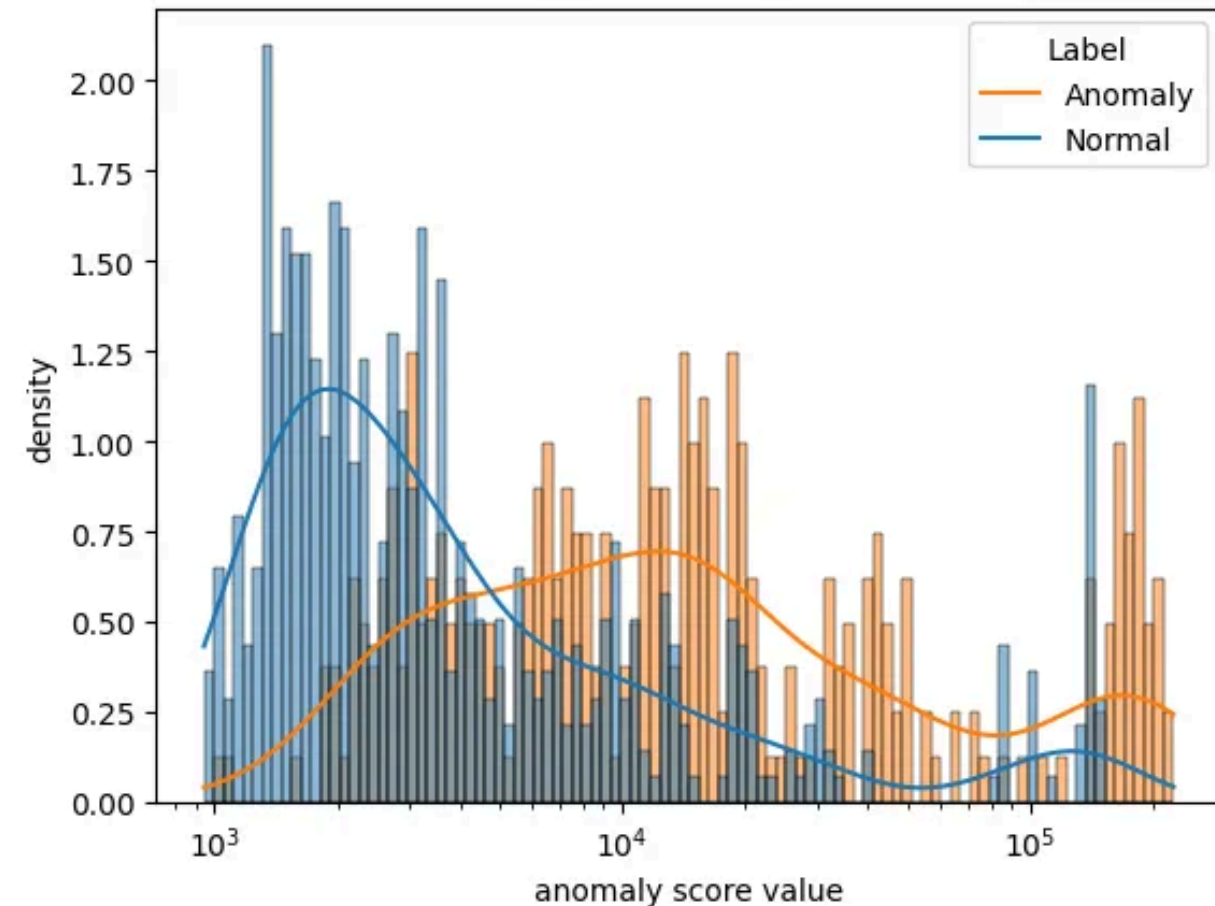
Example results on Fold 3 is used for qualitative analysis.

Limitations of Baseline Models

“Reconstruction-based vs Prediction-Based”

MA-VAE (Reconstruction Based)

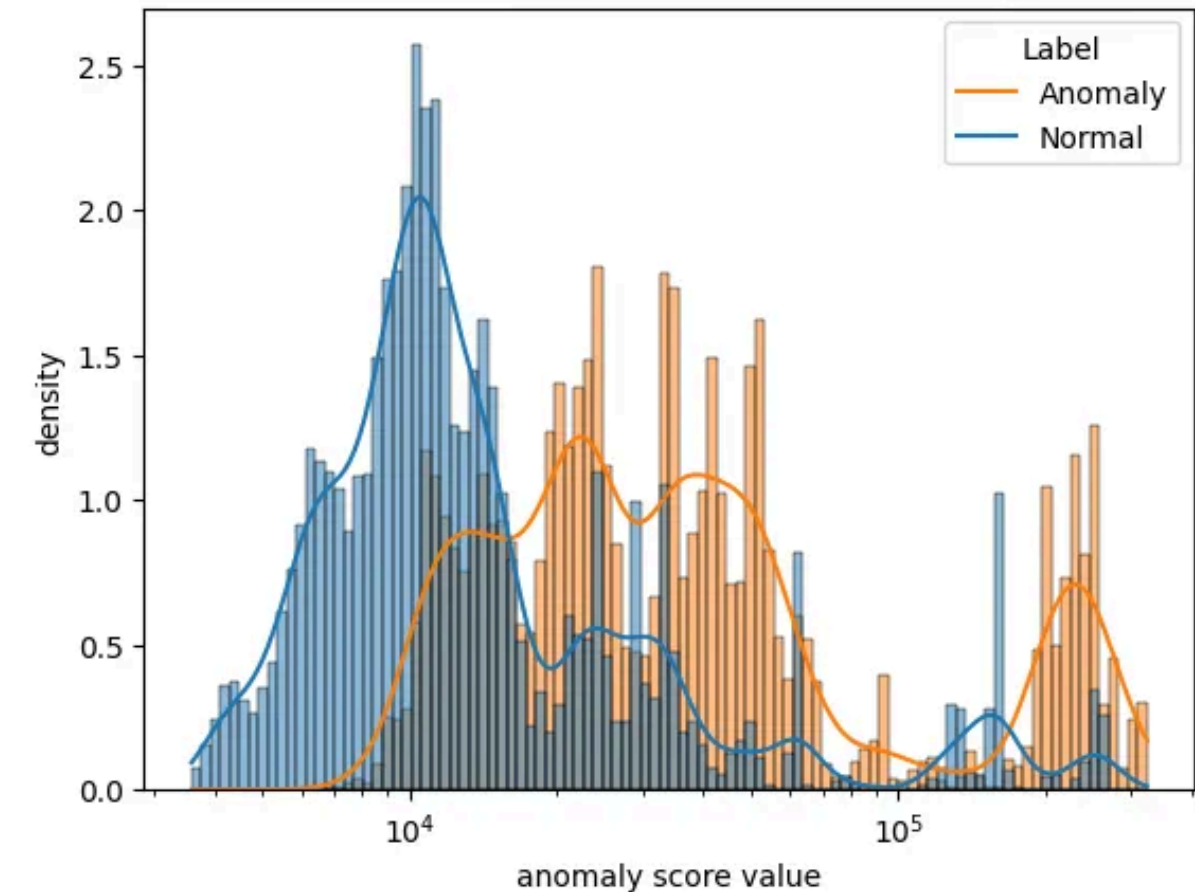
VAE reconstruction error distribution on the test set (with anomaly)



- Very large overlap (blue/orange mixed heavily)
- Not enough temporal sensitivity → misses subtle changes

VAE-LSTM (Prediction Based)

VAE reconstruction error distribution on the test set (with anomaly)



- Normal & anomaly overlap moderately
- Reconstruction errors spike even when data is normal
- Highlights lack of global context

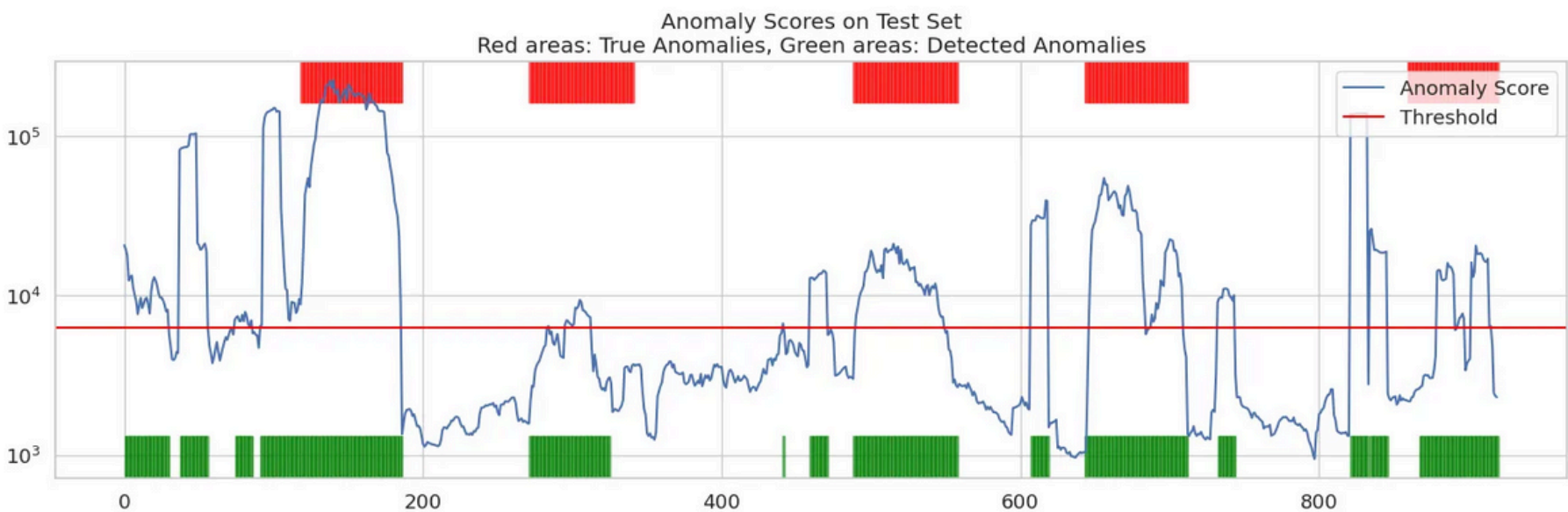
We need both global & temporal features

Example results on Fold 3 is used for qualitative analysis.

Limitations of Baseline Models

“Reconstruction-based vs Prediction-Based”

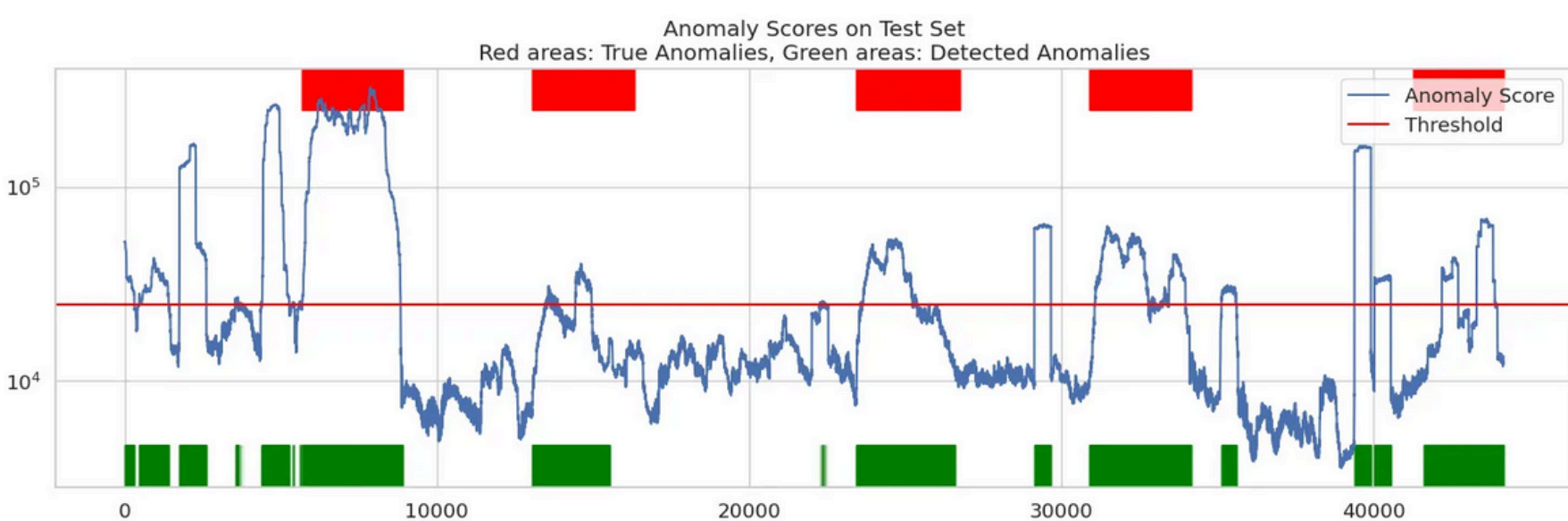
MA-VAE (Reconstruction Based)



Precision	Recall	F1	AUROC
0.874	0.86	0.856	0.93

- Misses anomalies due to lack of temporal information

VAE-LSTM (Prediction Based)



Precision	Recall	F1	AUROC
0.862	0.893	0.875	0.956

- Catches anomalies well but fires false positives on rapid normal changes

To overcome the weaknesses of both baselines,
we need a model that combines **local temporal patterns** (VAE-LSTM)
with **global contextual understanding** (inspired by MA-VAE).

This motivates our proposed Dual-branch Network!!

Dual-branch VAE-LSTM

(Linear Concat)

Dual-branch VAE-LSTM (Linear Concat)

Deep Convolutional Clustering-Based Time Series Anomaly Detection

by Gavneet Singh Chadha ^{1,*} , Intekhab Islam ¹ , Andreas Schwung ¹  and Steven X. Ding ² 

- ¹ Department of Automation Technology, South Westphalia University of Applied Sciences, 59494 Soest, Germany
² Department of Automatic Control and Complex Systems, University of Duisburg-Essen, 47057 Duisburg, Germany
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Sensors **2021**, *21*(16), 5488; <https://doi.org/10.3390/s21165488>

Submission received: 23 June 2021 / Revised: 9 August 2021 / Accepted: 10 August 2021 /
Published: 15 August 2021

MST-VAE: Multi-Scale Temporal Variational Autoencoder for Anomaly Detection in Multivariate Time Series

by Tuan-Anh Pham ¹ , Jong-Hoon Lee ^{1,*}   and Choong-Shik Park ²

- ¹ Department of AI Laboratory, MOADATA, Global Convergence Center, Seongnam-si 13449, Korea
² Department of Smart IT, U1 Univerisity, Asan 31409, Korea
* Author to whom correspondence should be addressed.

Appl. Sci. **2022**, *12*(19), 10078; <https://doi.org/10.3390/app121910078>

Submission received: 18 August 2022 / Revised: 26 September 2022 / Accepted: 6 October 2022 /
Published: 7 October 2022

Multi-Channel Multi-Scale Convolution Attention Variational Autoencoder (MCA-VAE): An Interpretable Anomaly Detection Algorithm Based on Variational Autoencoder

by Jingwen Liu ^{1,†}  , Yuchen Huang ^{1,†}  , Dizhi Wu ¹ , Yuchen Yang ² , Yanru Chen ¹  ,
Liangyin Chen ^{1,3}   and Yuanyuan Zhang ^{1,*}  

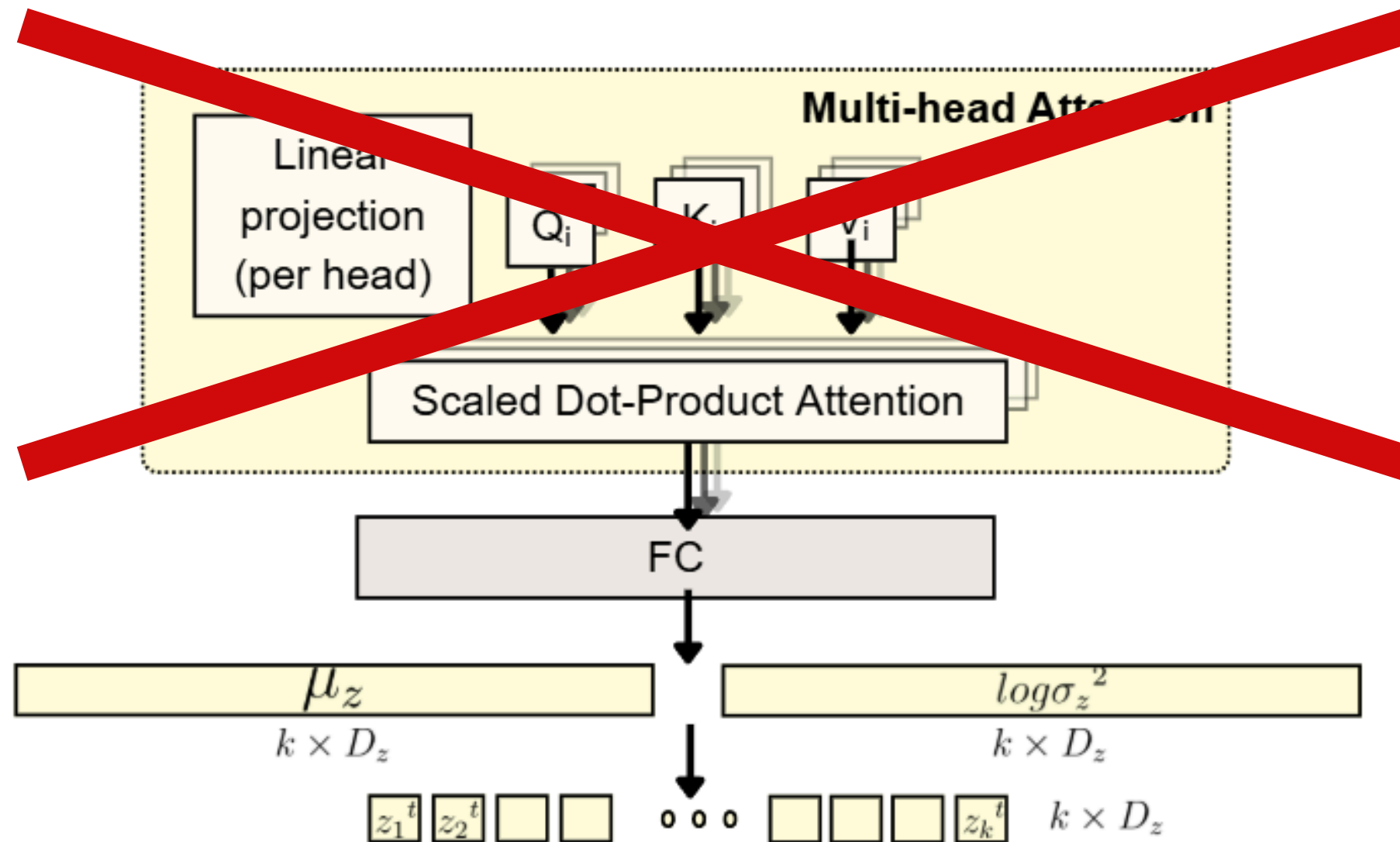
- ¹ College of Computer Science, Sichuan University, Chengdu 610065, China
² College of Computer Science, East China Normal University, Shanghai 200062, China
³ Institute for Industrial Internet Research, Sichuan University, Chengdu 610065, China
* Author to whom correspondence should be addressed.
† These authors contributed equally to this work.

Sensors **2024**, *24*(16), 5316; <https://doi.org/10.3390/s24165316>

Submission received: 23 July 2024 / Revised: 9 August 2024 / Accepted: 14 August 2024 /
Published: 16 August 2024

- Dual-branch encoder extracts **global and local latent features**.
- To validate the effectiveness of Multi-head attention, the latent representations were fused using a **simple linear concat** as an alternative.

Dual-branch VAE-LSTM (Linear Concat)

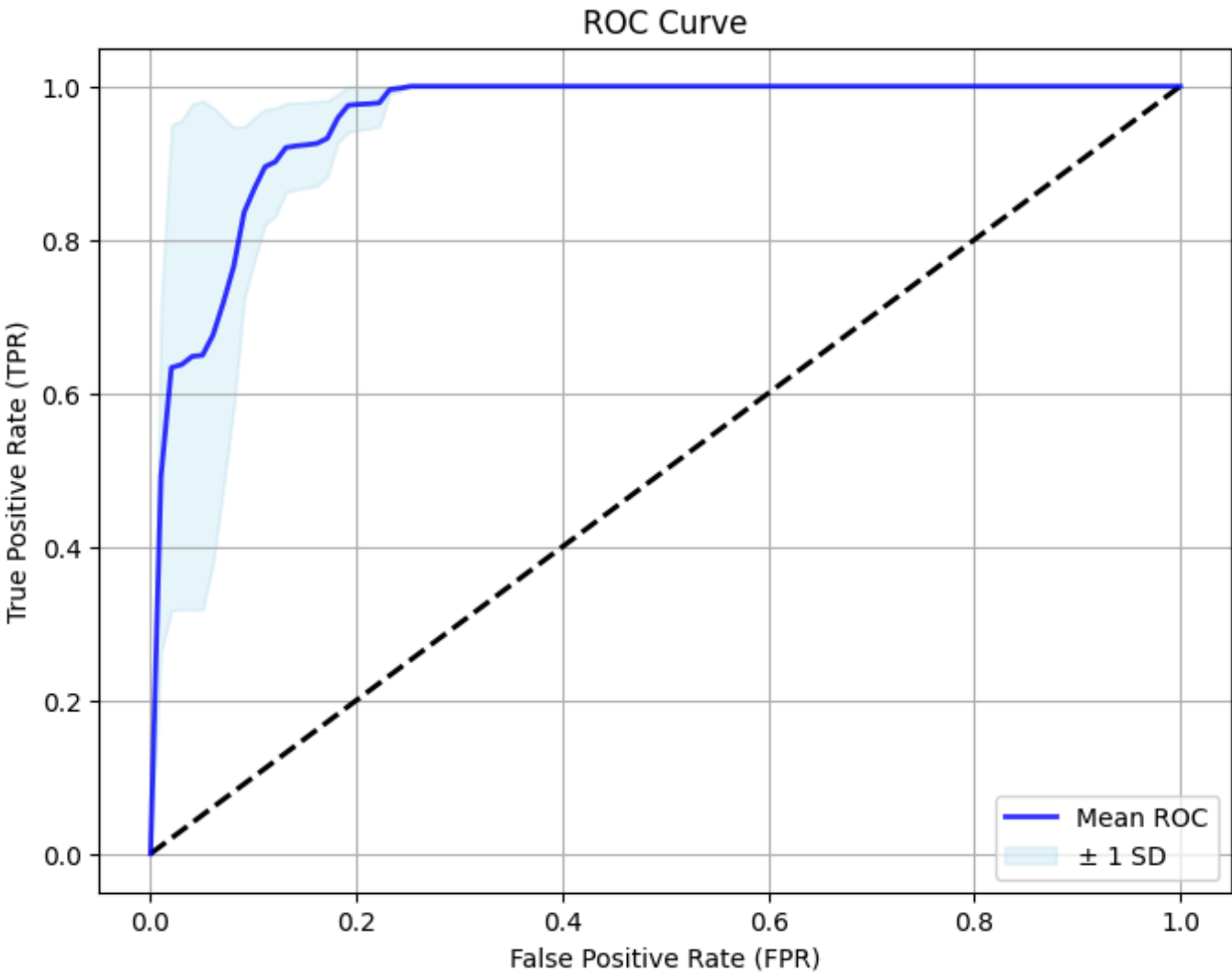
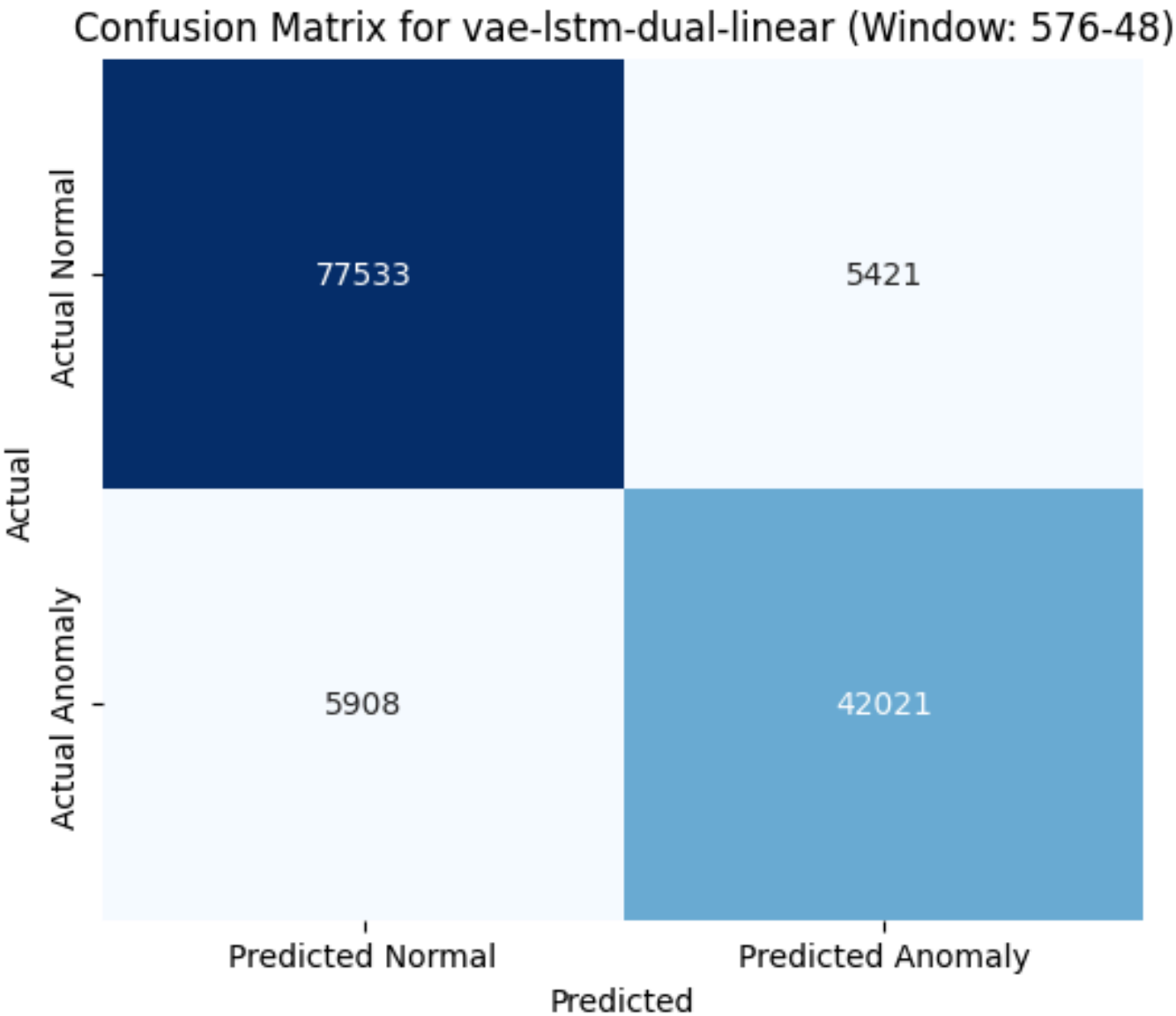


Ablation!

- Dual-branch encoder extracts **global and local latent features**.
- To validate the effectiveness of Multi-head attention, the latent representations were fused using a **simple linear concat** as an alternative.

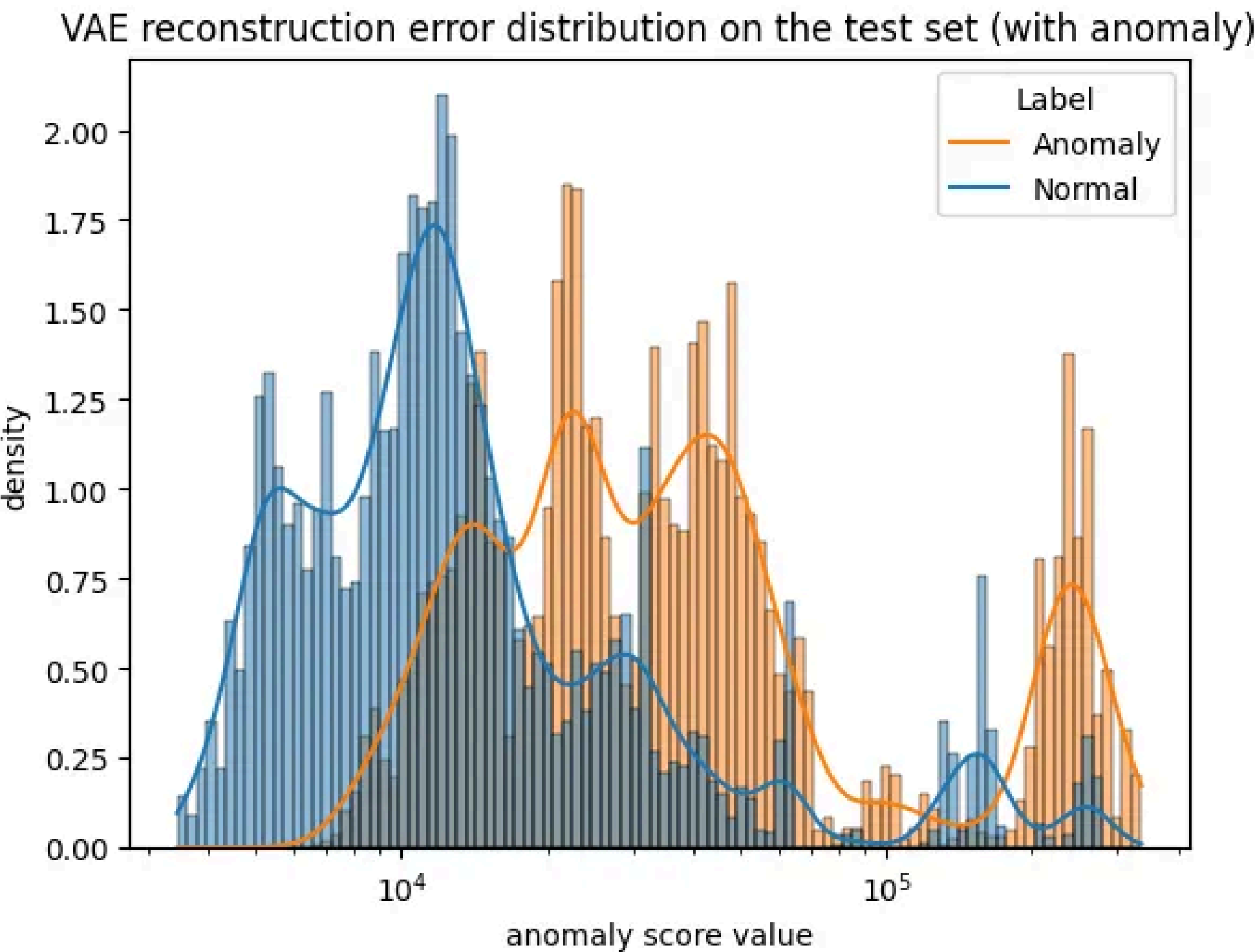
Experiment: Dual-branch VAE-LSTM (Linear Concat)

Precision	Recall	F1	AUROC
0.888	0.876	0.881	0.957



Experiment: Dual-branch VAE-LSTM (Linear Concat)

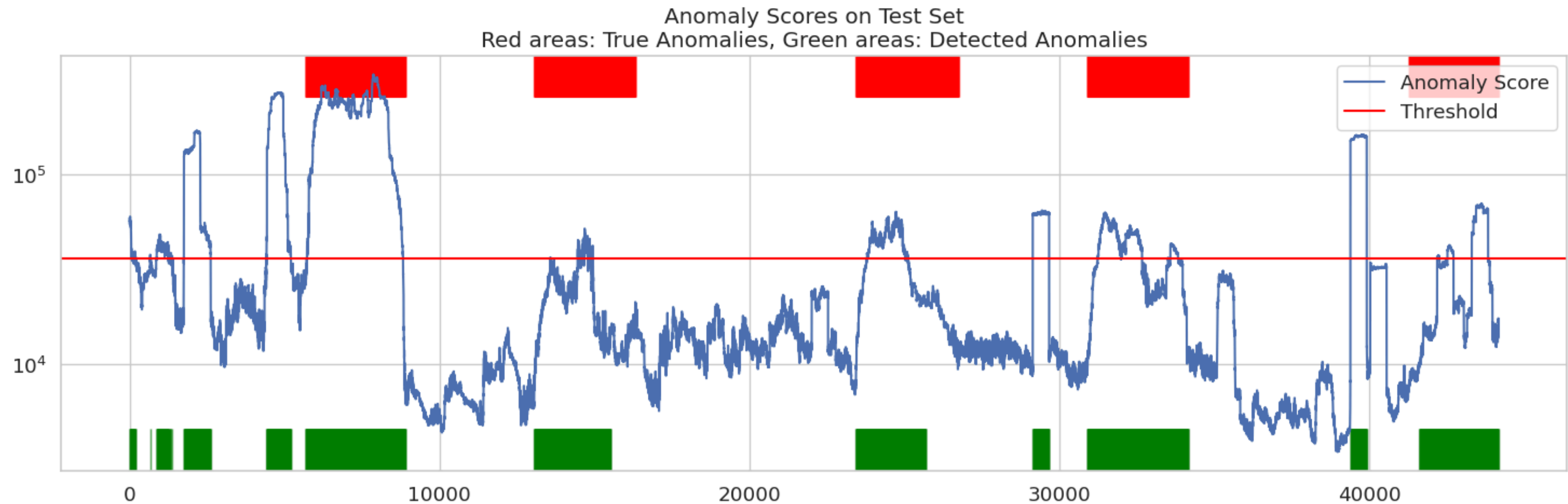
Normal vs Anomaly distributions



Example results on Fold 3 is used for qualitative analysis.

Experiment: Dual-branch VAE-LSTM (Linear Concat)

“How prediction actually behaves?”

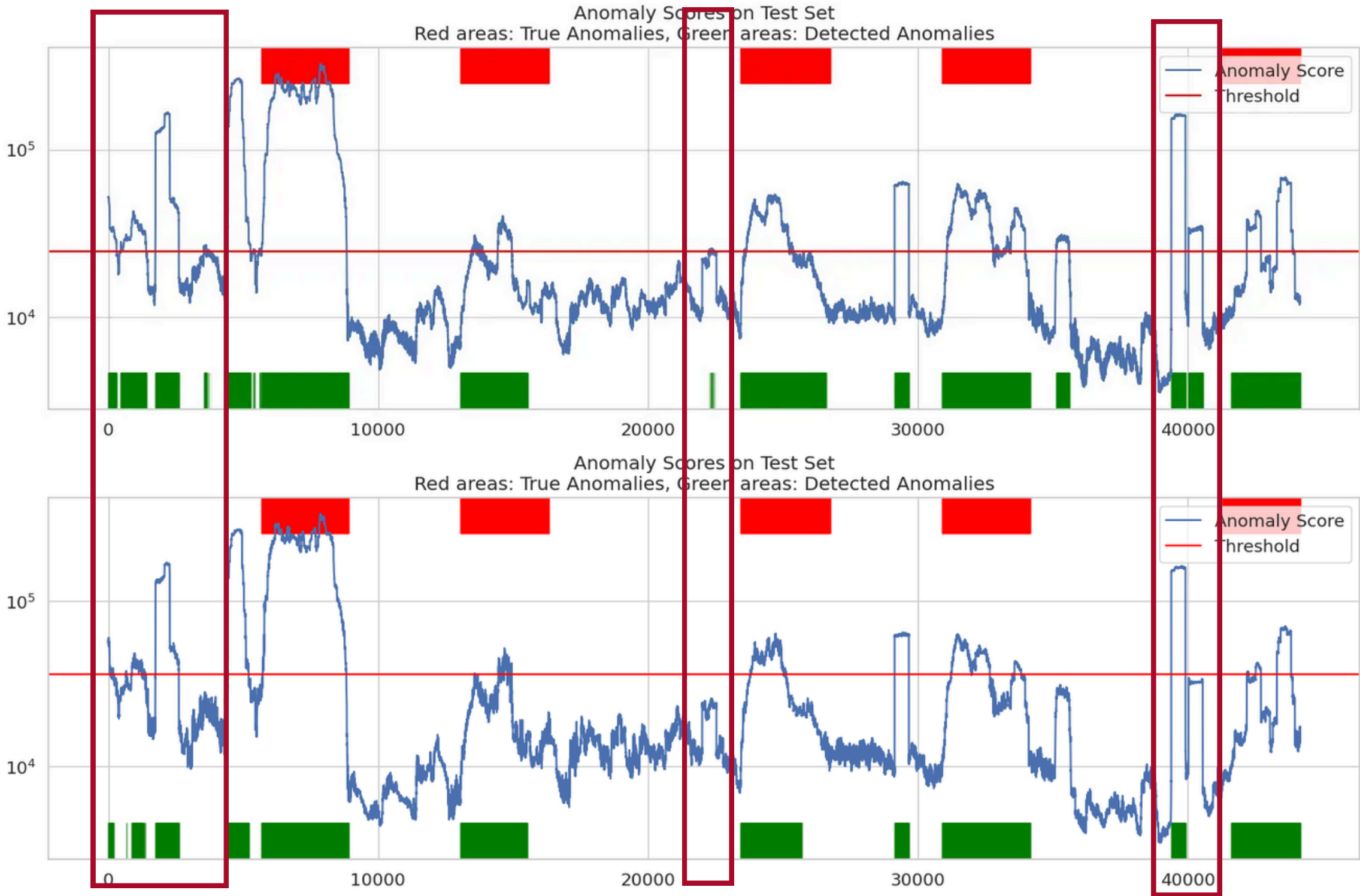


- **Captures anomaly patterns well** — green detections almost align with true anomaly segments (red).
- Injecting global latent feature **suppresses reconstruction error spikes**, reducing false positives.

Example results on Fold 3 is used for qualitative analysis.

Comparative Analysis

VAE-LSTM

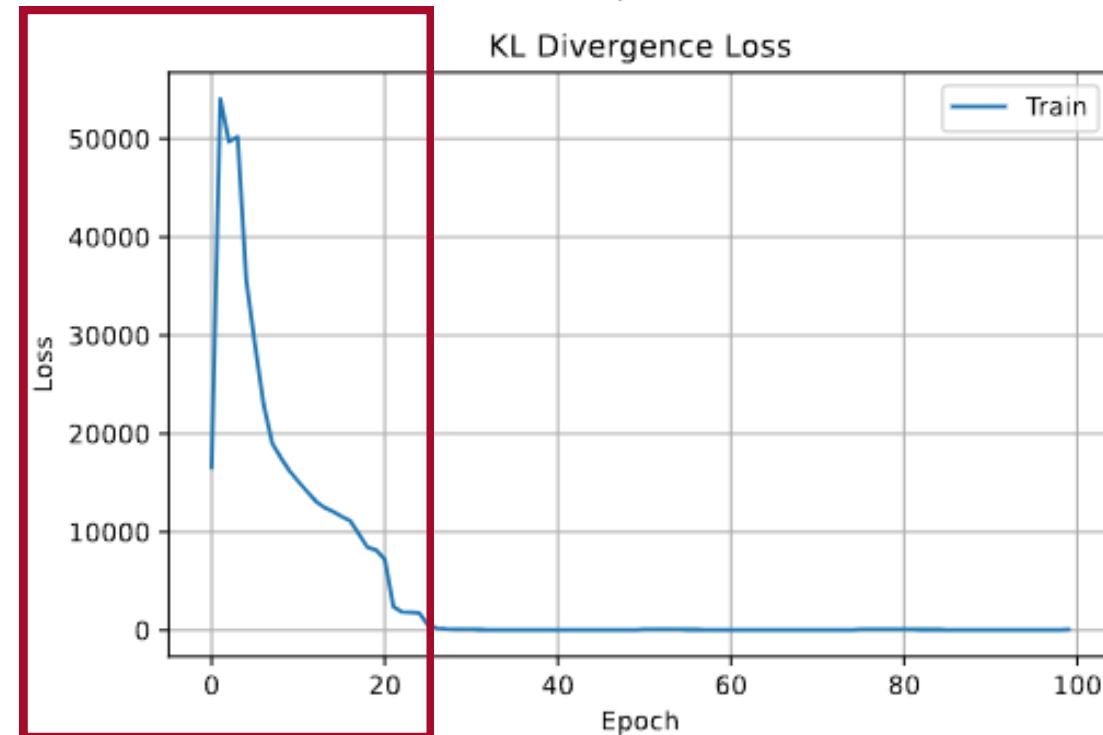
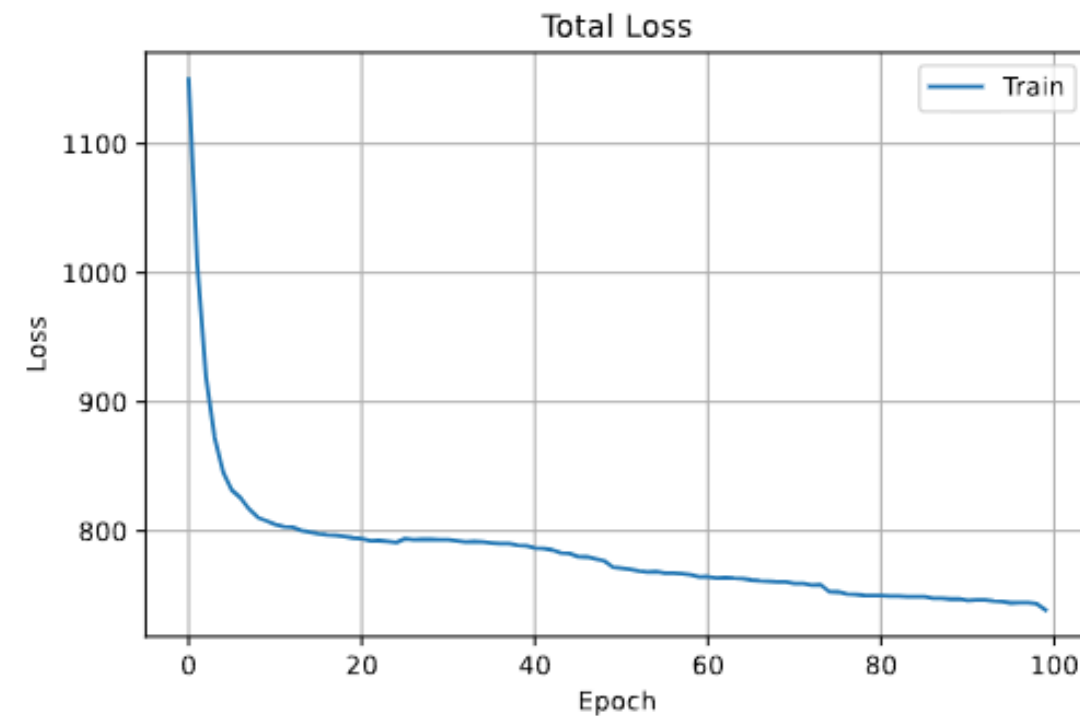


Dual-Branch VAE-LSTM (Linear Concat)

Example results on Fold 3 is used for qualitative analysis.

Experiment: Dual-branch VAE-LSTM (Linear Concat)

Limitation



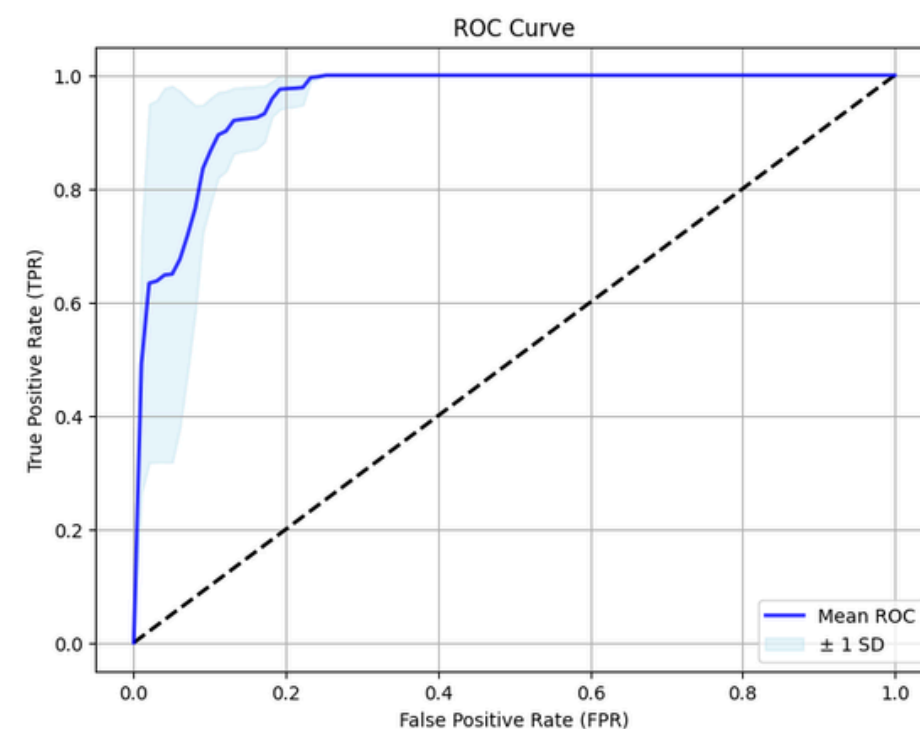
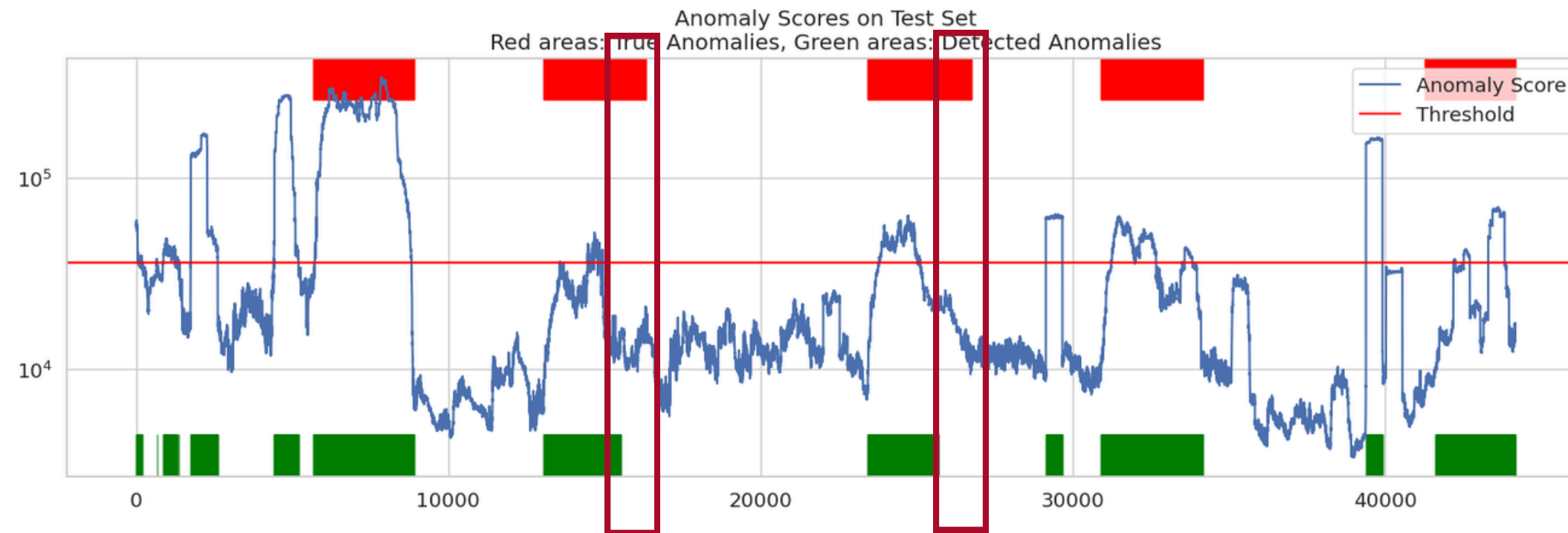
KL Divergence Explosion

- During training, the **KL divergence term suddenly blows up, destabilizing learning.**
- If we forcefully clamp the KL term, the model barely learns.
- If we do not clamp, KL term diverges and training fails.

“ This instability prevents Dual-branch VAE-LSTM (Linear Concat) from fully leveraging global–local latent features.”

Experiment: Dual-branch VAE-LSTM (Linear Concat)

Limitation

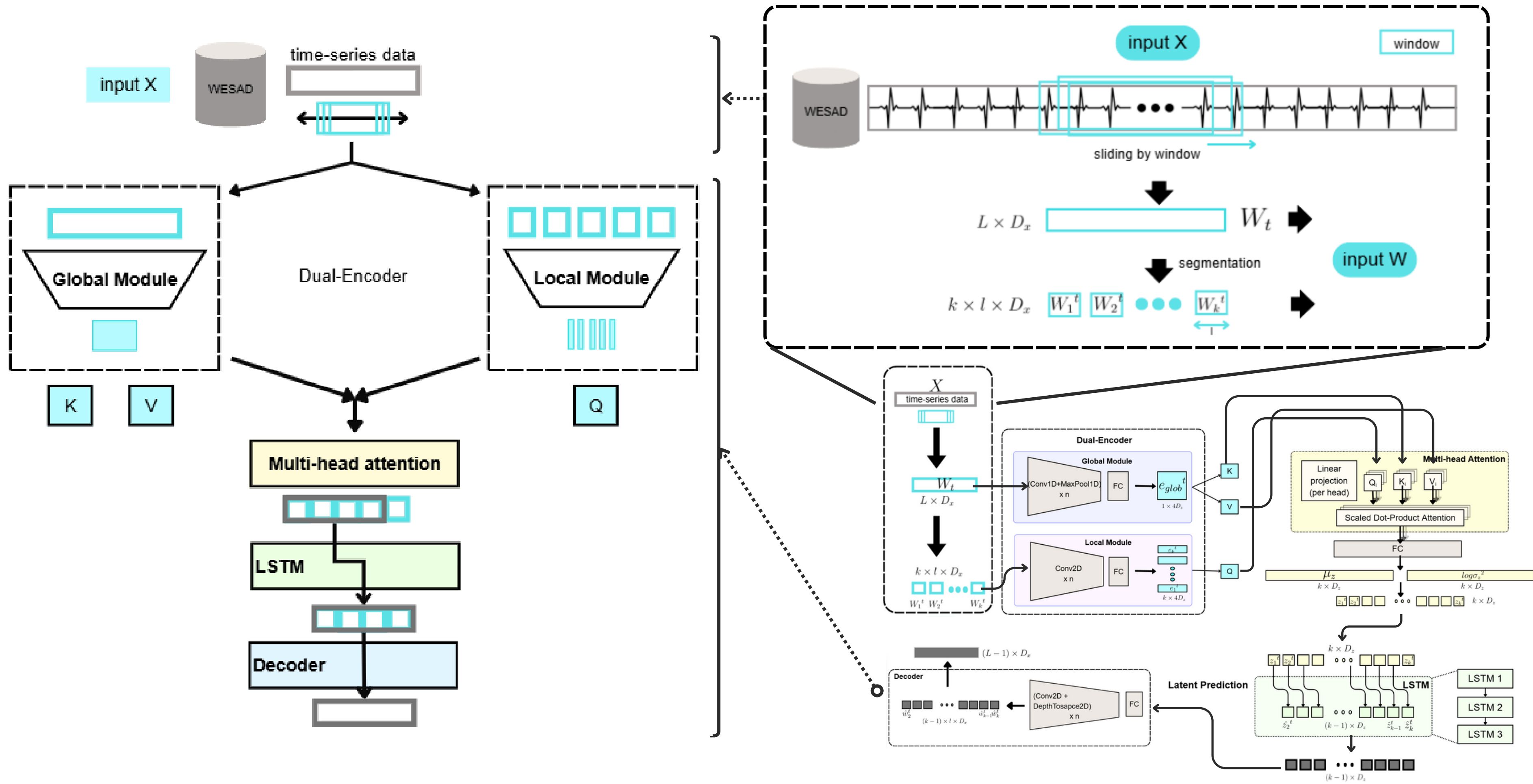


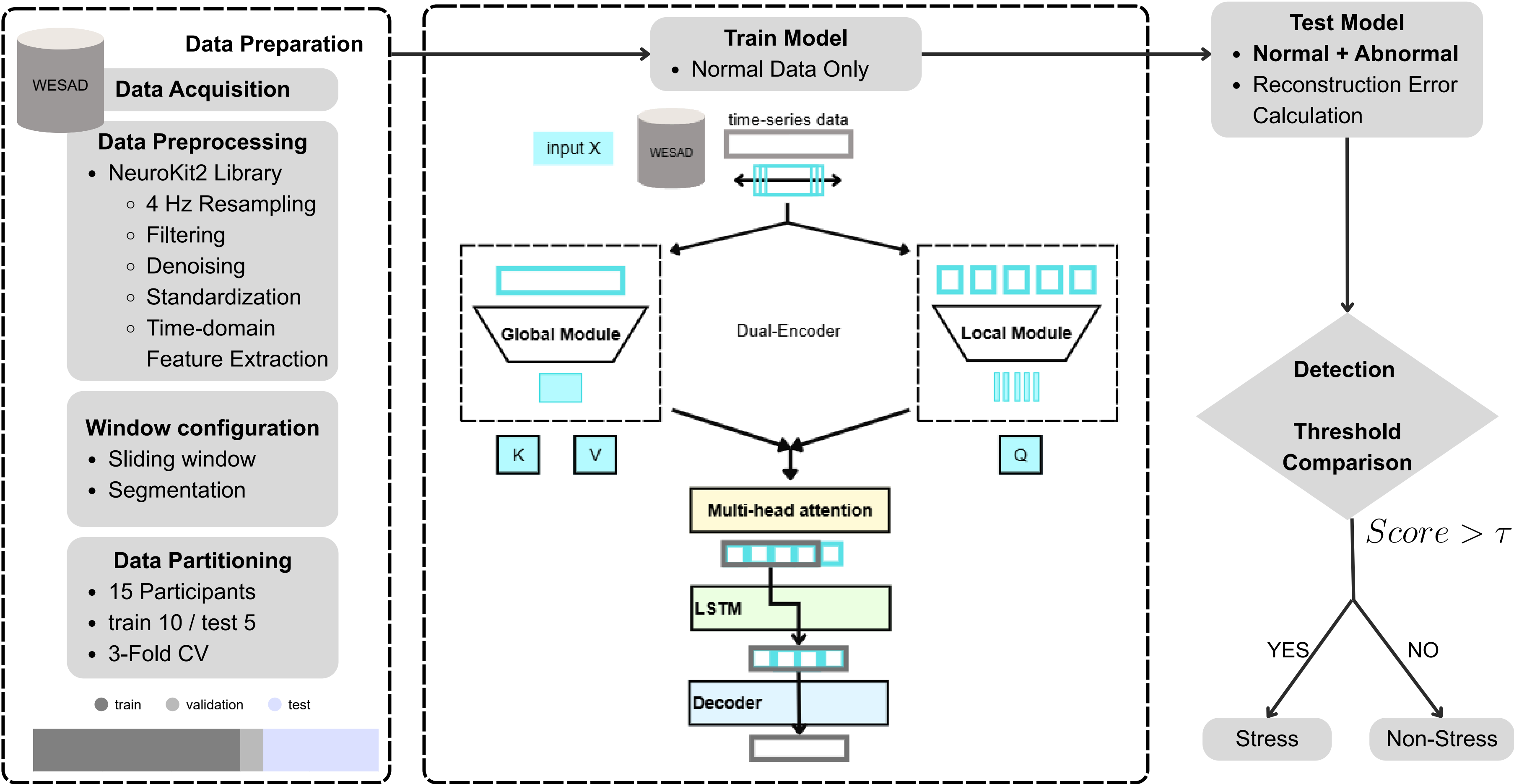
- Still miss some anomalies
- **Minimizing False Negatives in stress detection is more critical** than reducing false alarms
- Failing to identify true stress events poses a **direct risk to human health**.
- Further improvements remain achievable

Example results on Fold 3 is used for qualitative analysis.

Dual-branch VAE-LSTM

(Multi-head Attention)





Dual-branch VAE-LSTM (Multi-head Attention)

Attention Is All You Need

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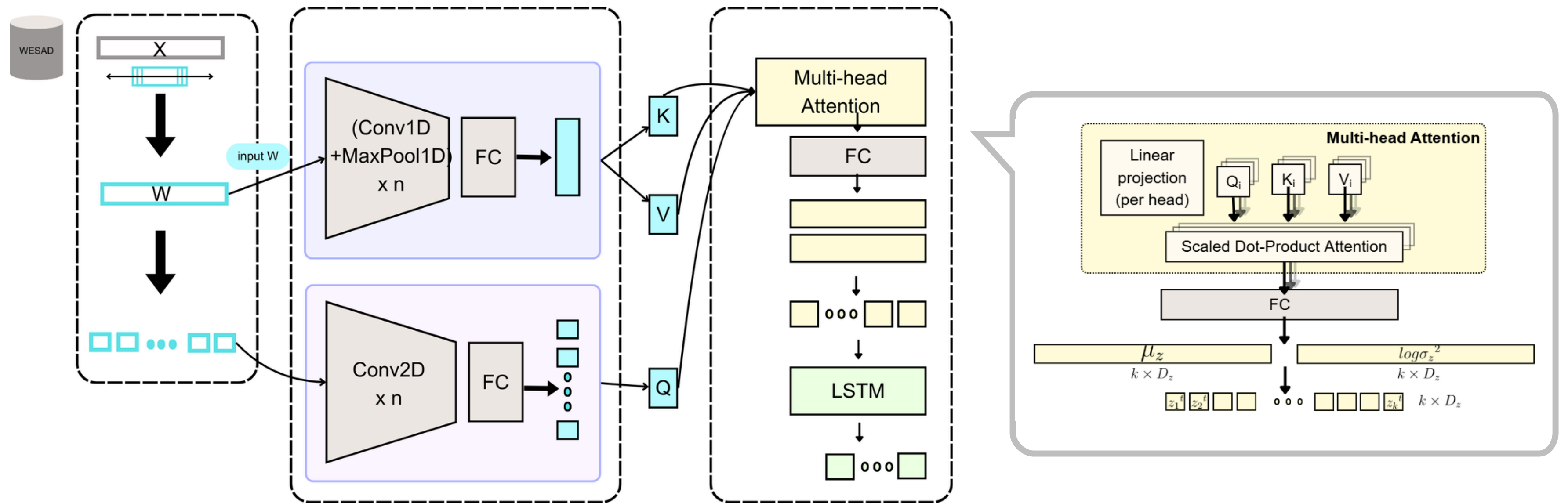
Illia Polosukhin* ‡
illia.polosukhin@gmail.com

- Attention allows models to **dynamically weight relevant information** while suppressing irrelevant parts
- This property addresses the problem we observed in Dual-branch VAE-LSTM with Local Concat.
- The fusion must **selectively weight** latent representations across time and features.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., N.Gomez, A., Kaiser, L., & Polosukhin, I. (2025b). Attention Is All You Need. <https://doi.org/10.65215/nxvz2v36>

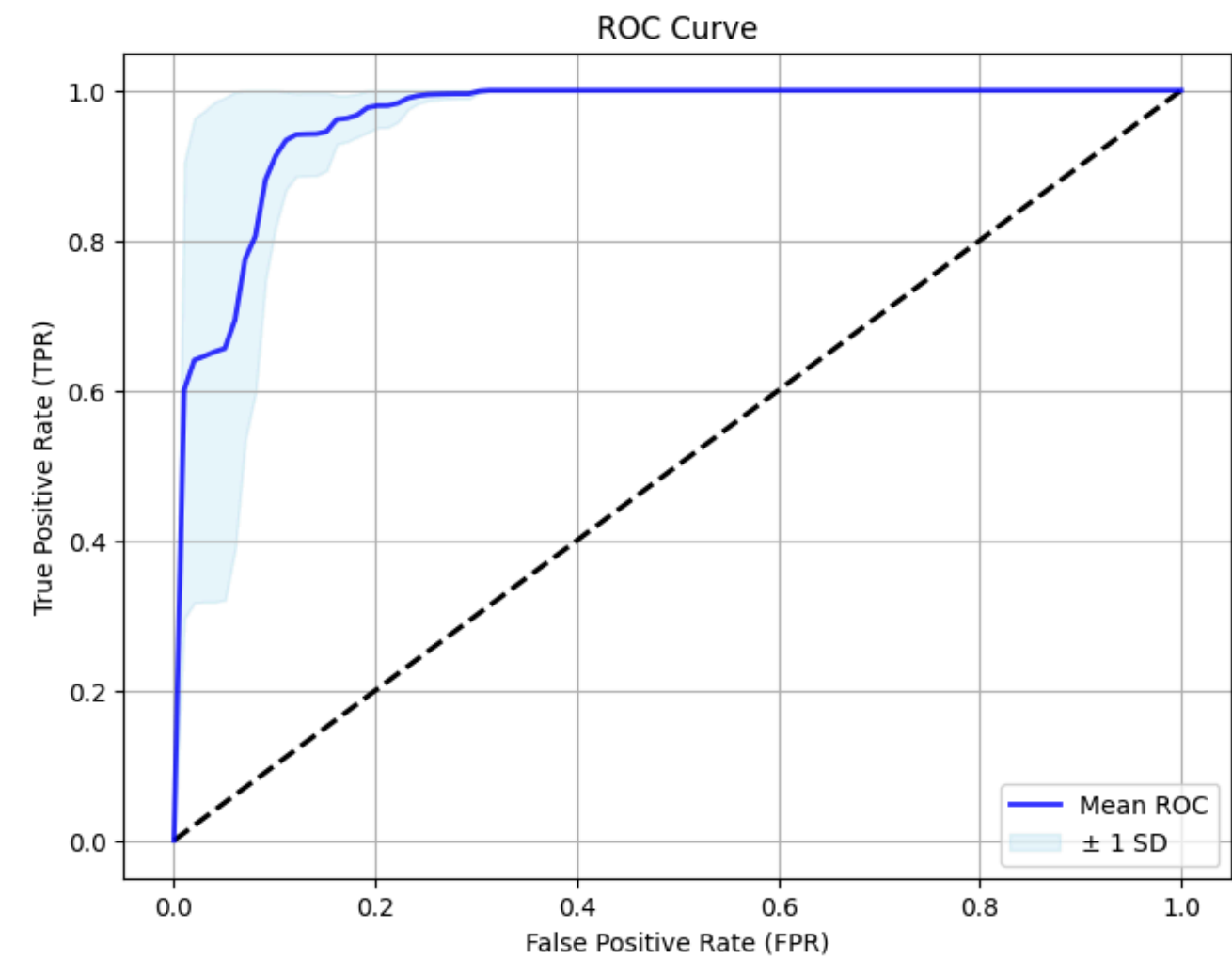
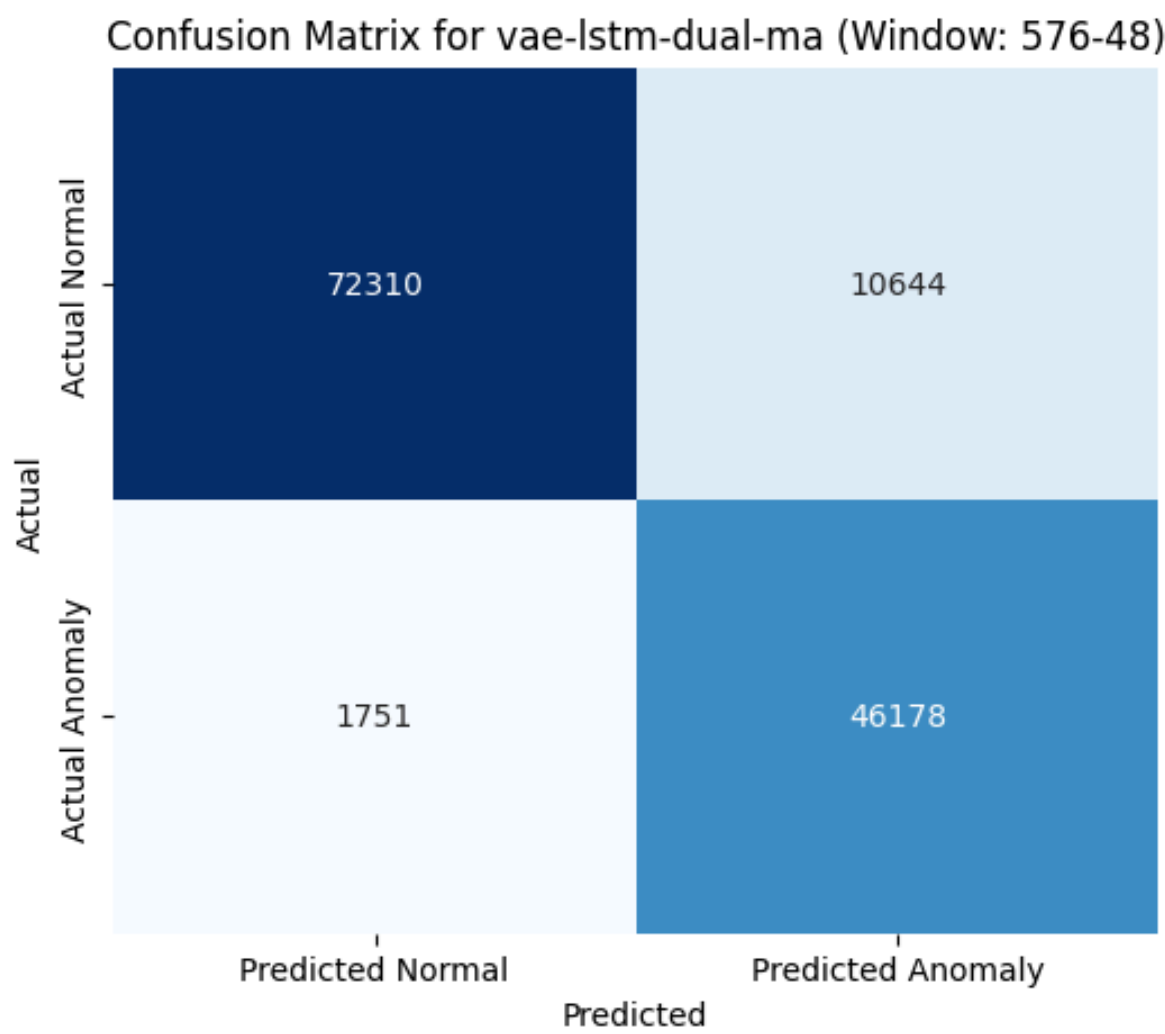
Dual-branch VAE-LSTM (Multi-head Attention)

Following the principle introduced in **Attention Is All You Need** and **baseline MA-VAE**, Adding Multi-Head Attention to the Model provides a learnable mechanism to fuse local (Q) and global (K, V) features, enabling the model to emphasize informative long-term context while suppressing irrelevant signals.



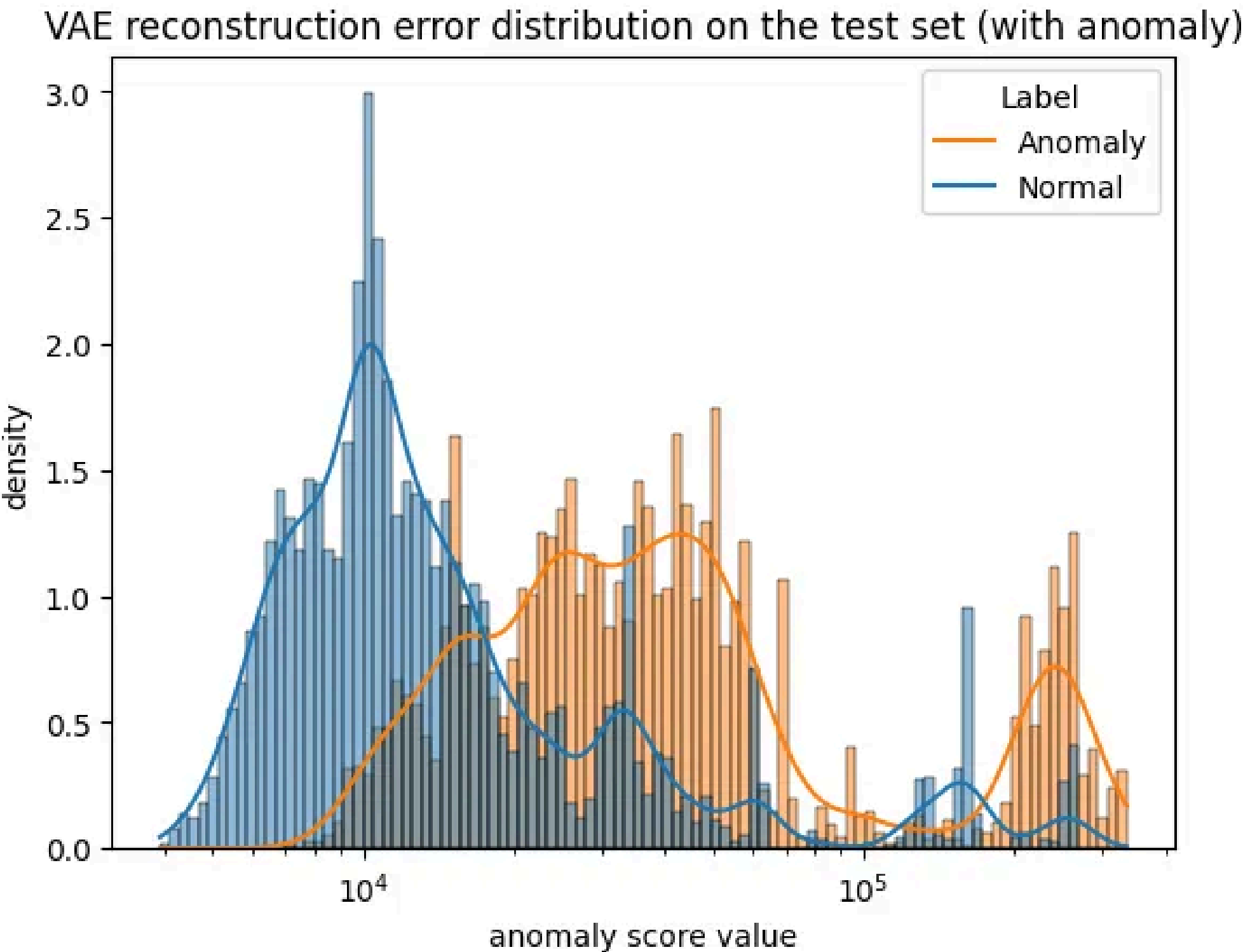
Experiment: Dual-branch VAE-LSTM (Multi-head Attention)

Precision	Recall	F1	AUROC
0.82	0.964	0.883	0.959



Experiment: Dual-branch VAE-LSTM (Multi-head Attention)

Normal vs Anomaly distributions

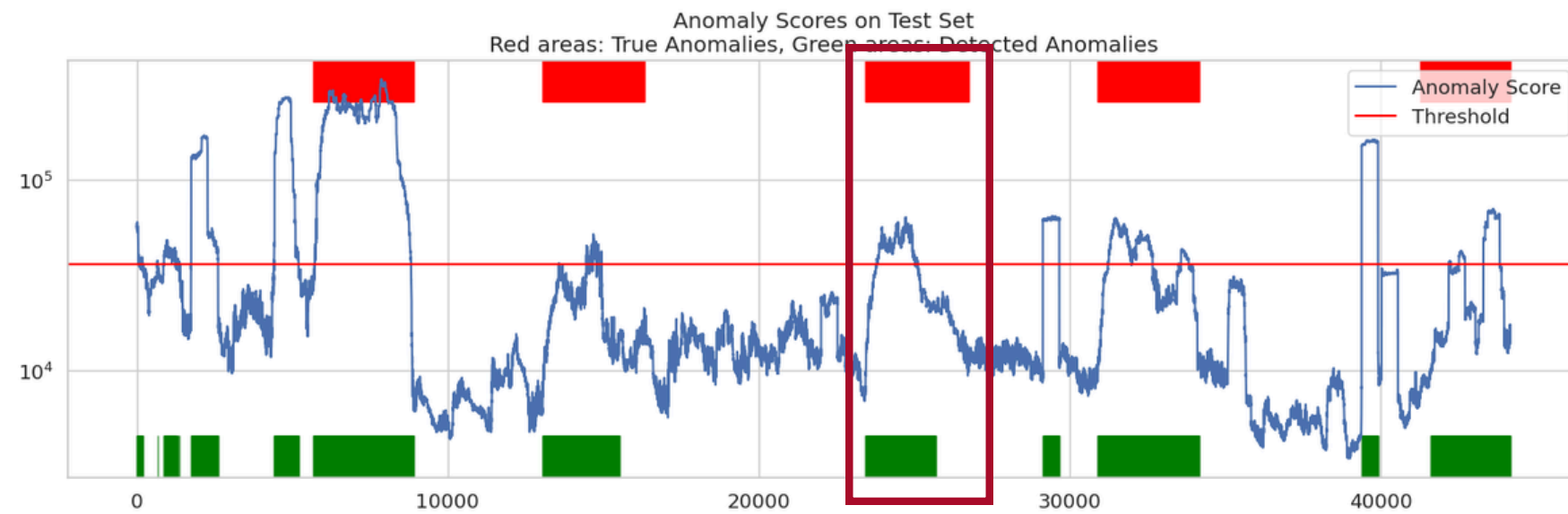


Example results on Fold 3 is used for qualitative analysis.

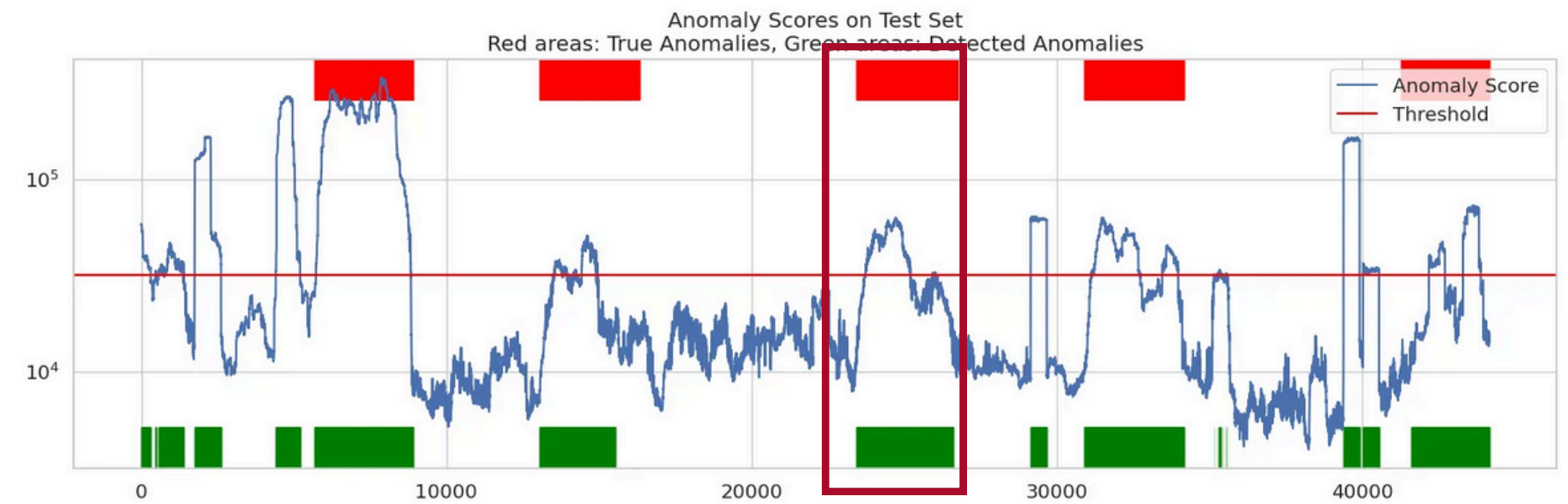
Comparative Analysis

Linear Concat, Multi-head Attention

Linear Concat



Multi-head Attention



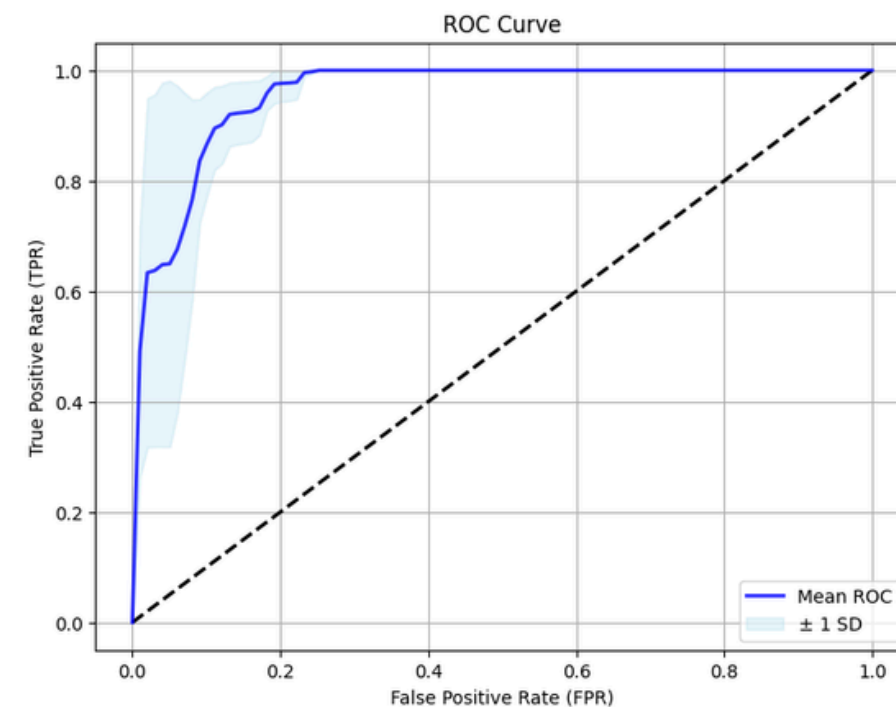
- Green detection blocks **align more cleanly** with red ground-truth anomalies, **showing improved boundary localization**.
- MA fusion **catches anomalies effectively**, showing fewer false negatives compared to linear concat.
- The model **maintains temporal consistency** while using global context to suppress noise-induced errors to a reasonable and acceptable level.

Example results on Fold 3 is used for qualitative analysis.

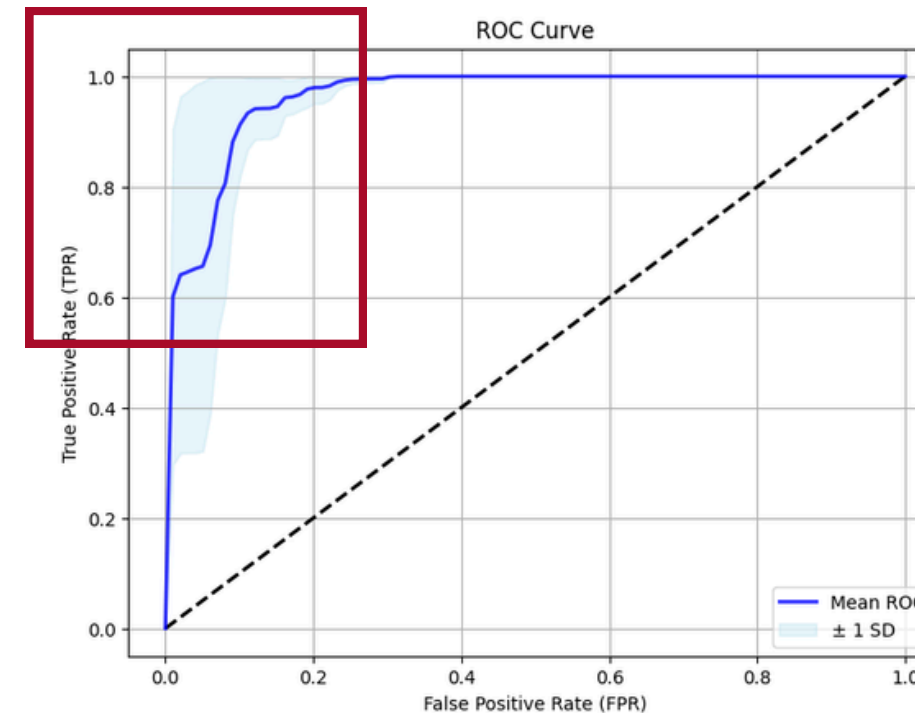
Comparative Analysis

Linear Concat, Multi-head Attention

Linear Concat



Multi-head Attention



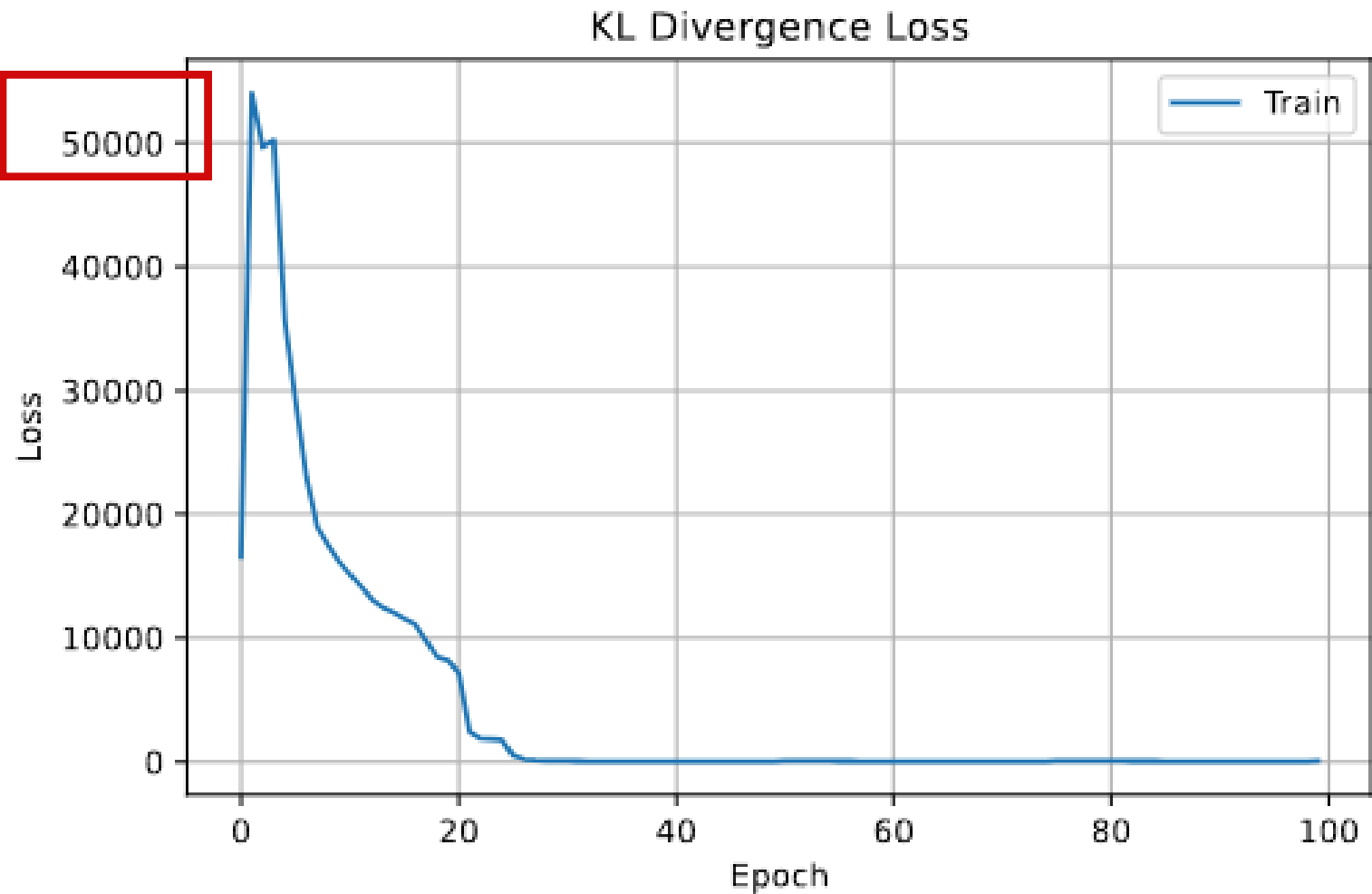
- The ROC Curve clearly indicates that the Multi-head Attention provides significantly **better discriminatory power** than Linear Concat.
- MA yielded a higher True Positive Rate (TPR), **reducing the risk of missed anomaly** (stress).

Example results on Fold 3 is used for qualitative analysis.

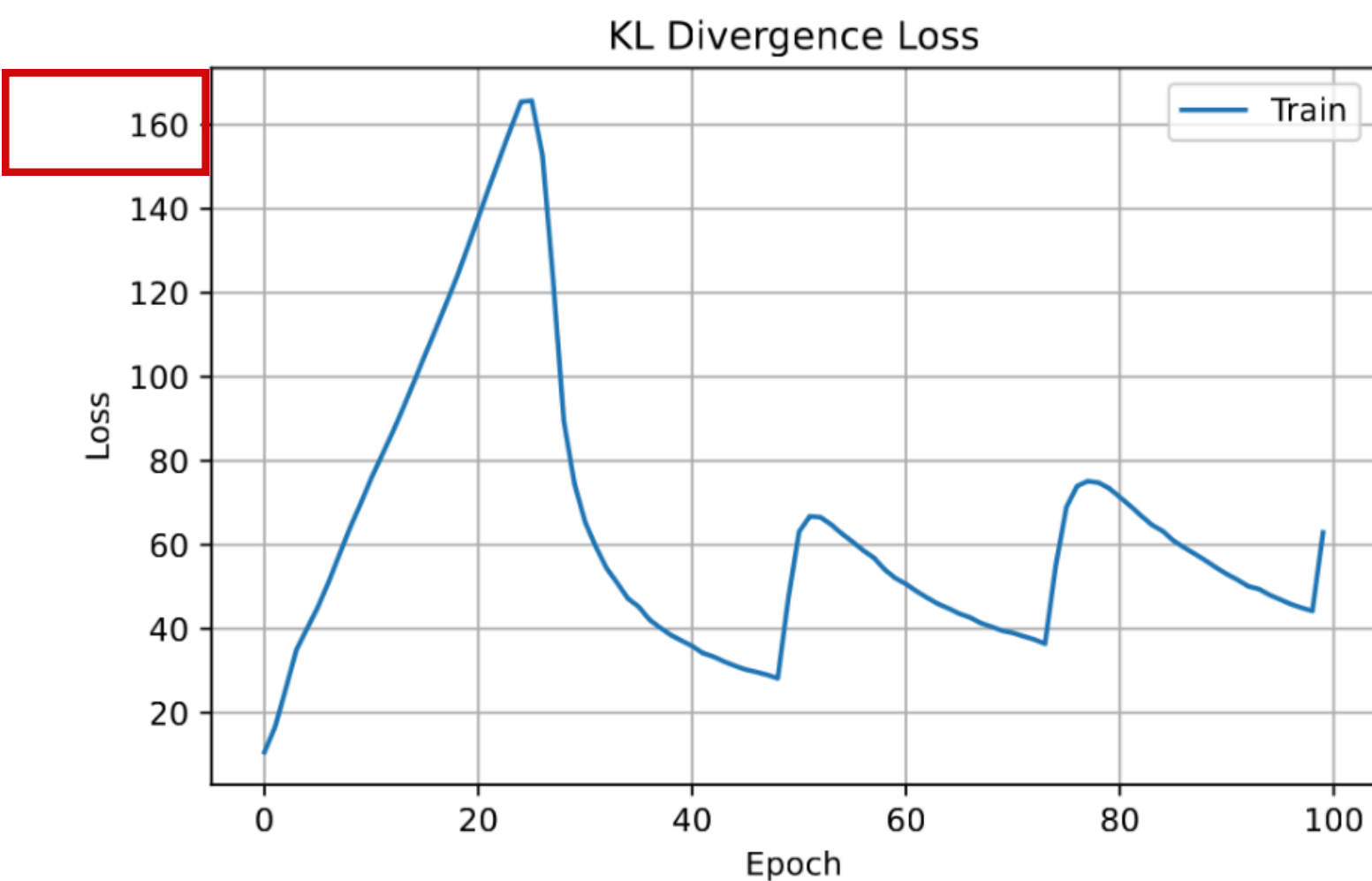
Comparative Analysis

Linear Concat, Multi-head Attention

Linear Concat



Multi-head Attention



Linear Concat suffers from KL divergence instability,
while **MA Fusion learns stably across epochs,** making it more robust.

05

Ablation Study Results

Ablation Table

Model	Encoder	Fusion Method	Temporal Learning
MA-VAE (Reference)	BiLSTM	Multi-Head Attention	None
VAE-LSTM (Baseline)	Local	None	LSTM
VAE-LSTM-Dual-Linear	Local + Global	Linear Concat	LSTM
VAE-LSTM-Dual-MA	Local + Global	Multi-Head Attention	LSTM

Fixed Window Size

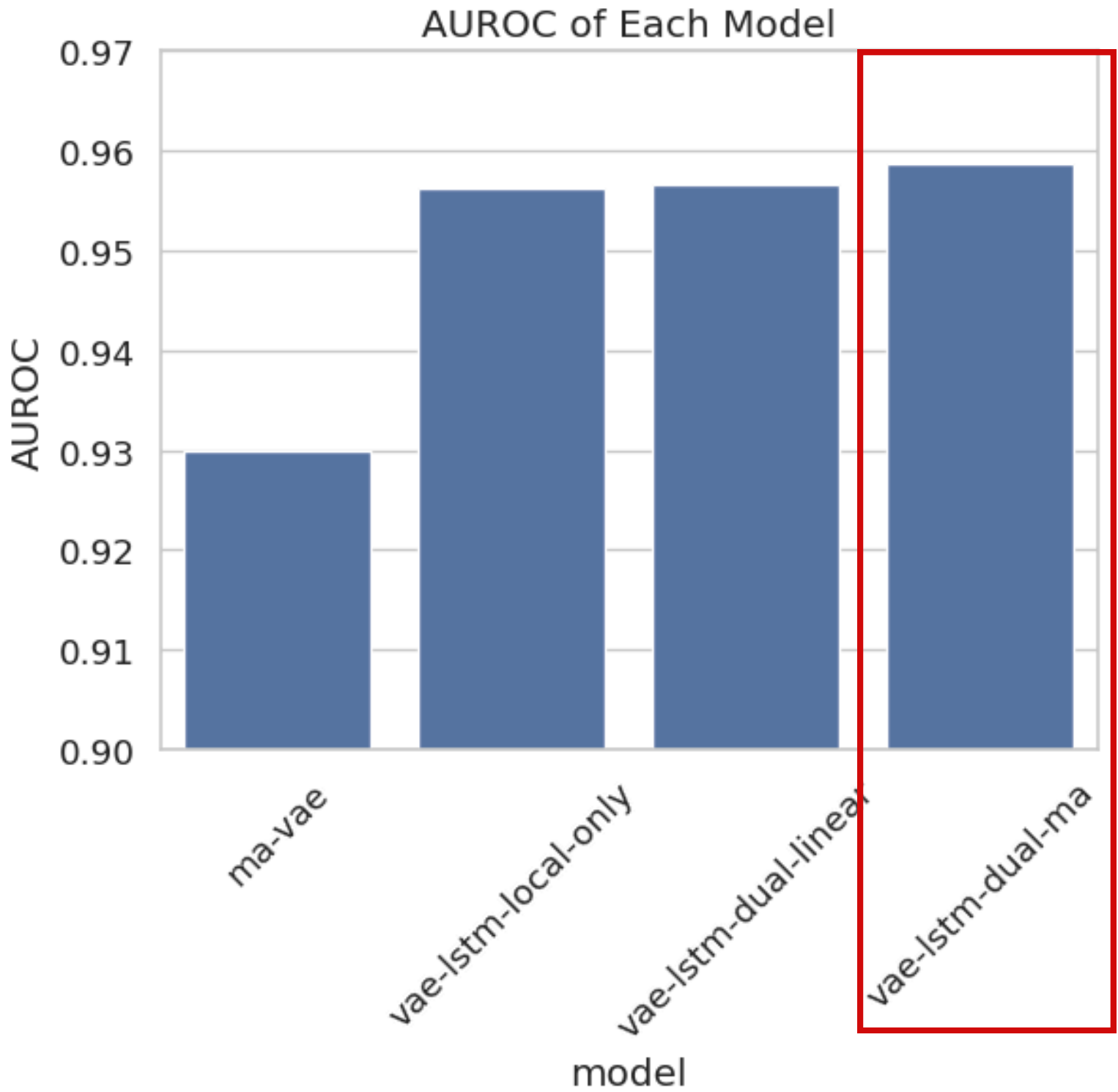
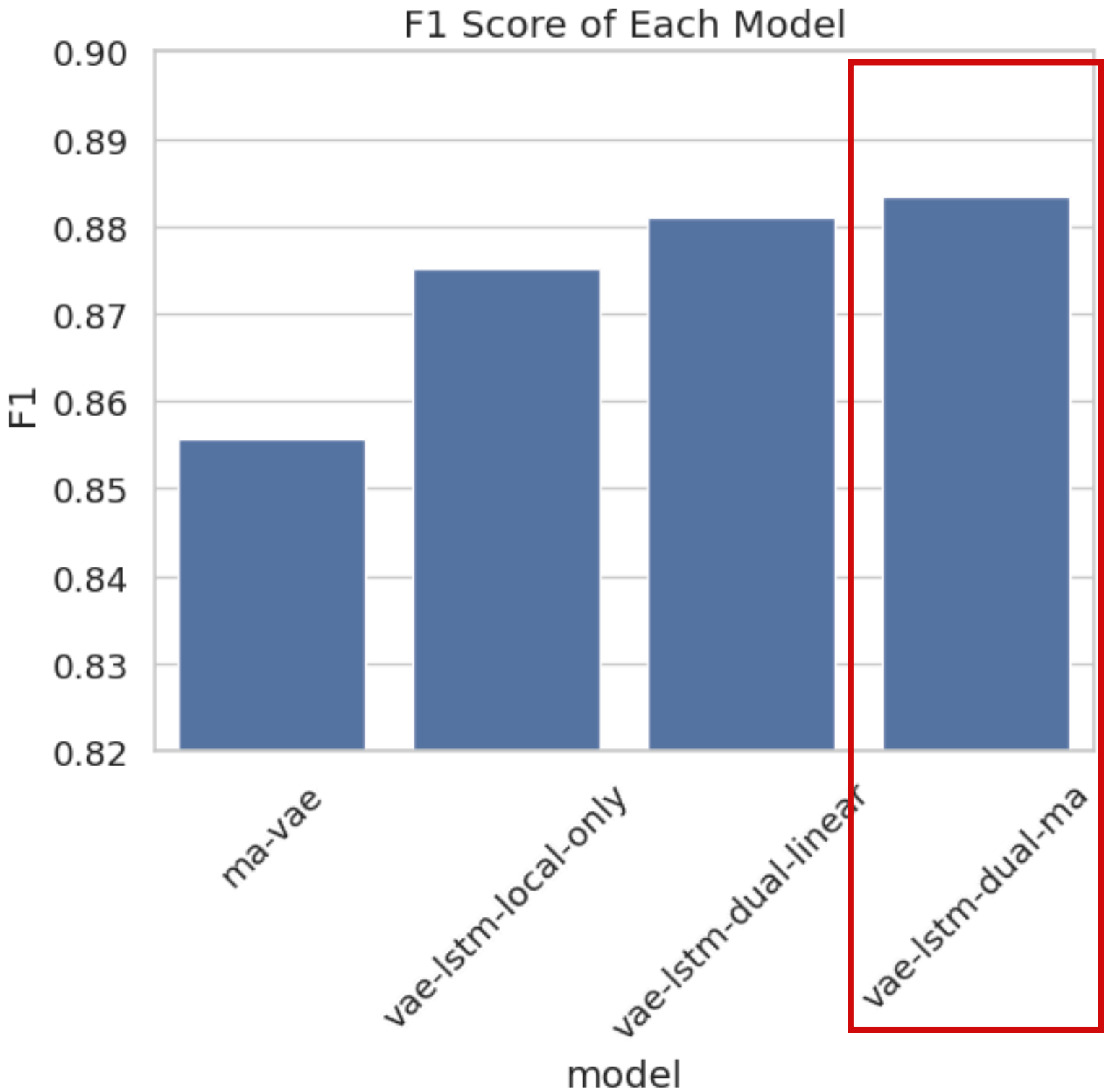
Results

(Window: 576–48)

Model	AUROC	F1	Precision	Recall
MA-VAE (Reference)	0.930	0.856	0.874	0.860
VAE-LSTM (Baseline)	0.956	0.875	0.862	0.893
VAE-LSTM-Dual-Linear	0.957	0.881	0.888	0.876
VAE-LSTM-Dual-MA	0.959	0.883	0.820	0.964

Results

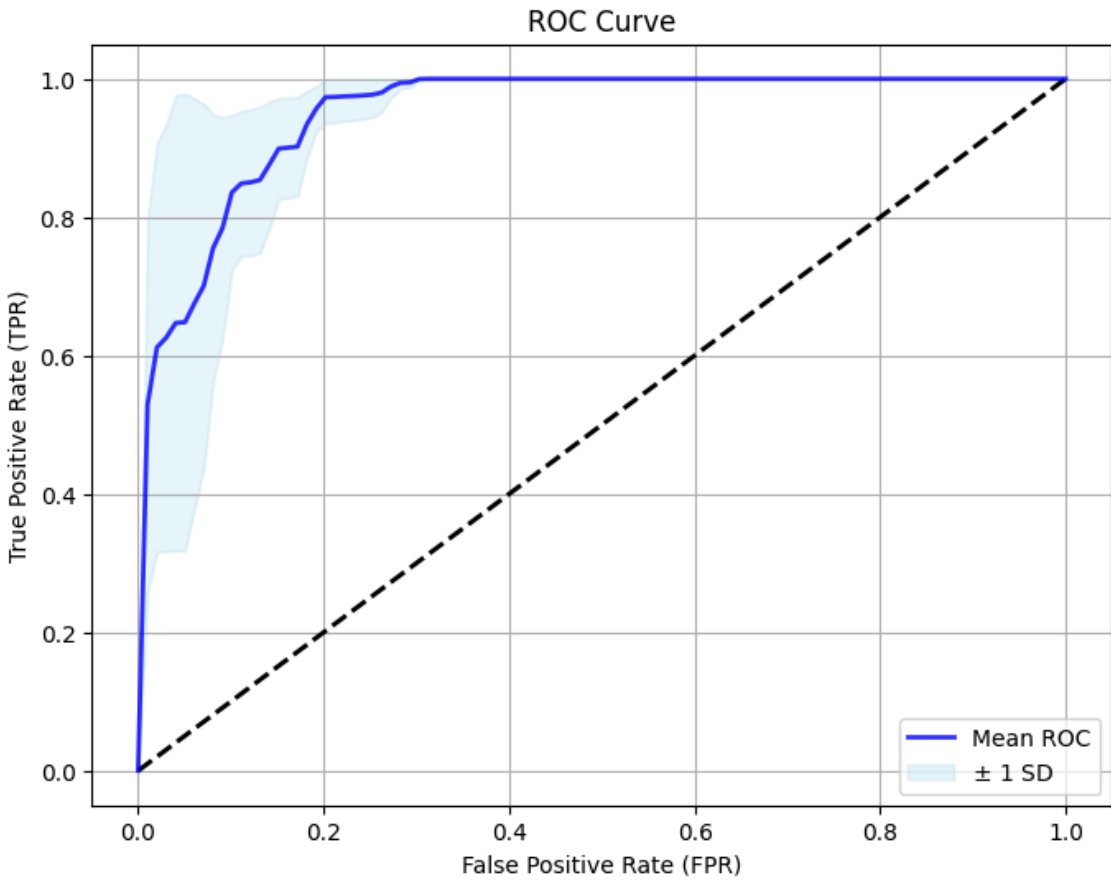
F1 Score & AUROC Comparison



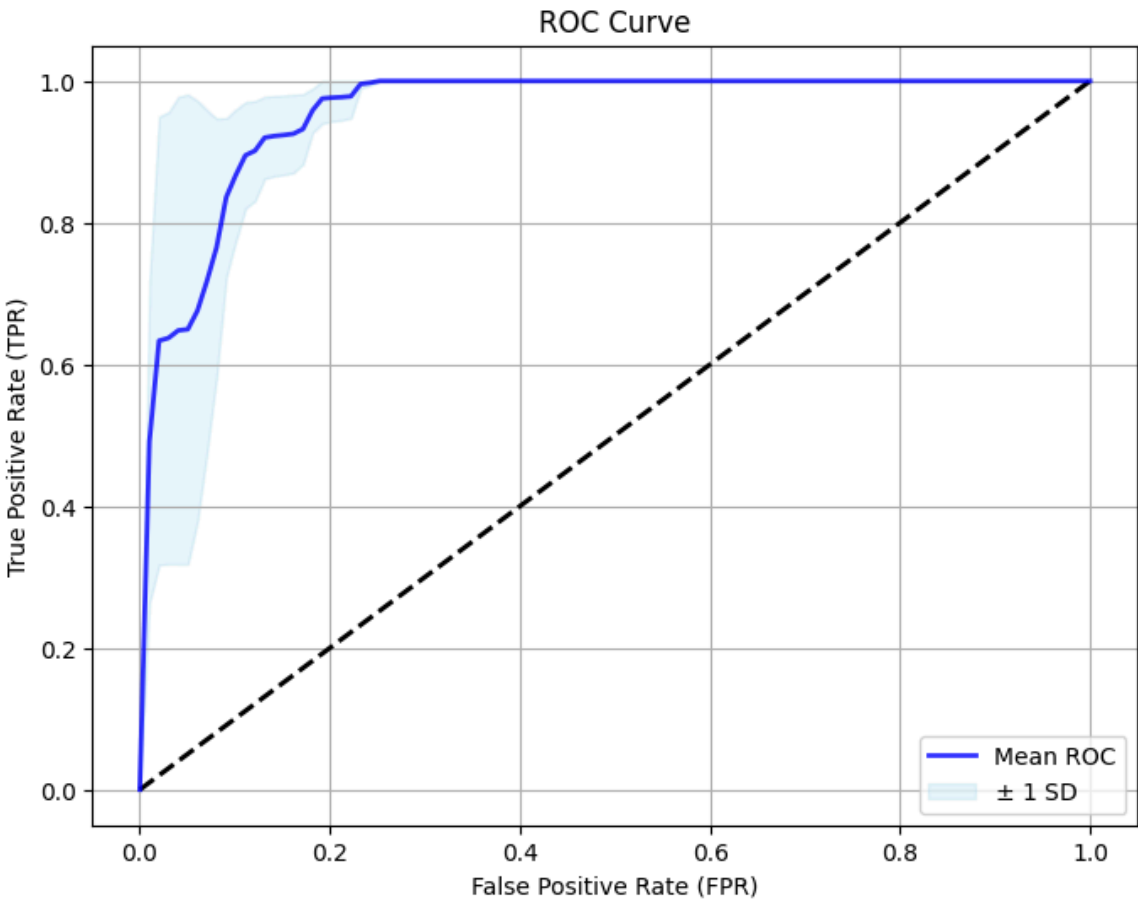
Results

ROC Curve Comparison

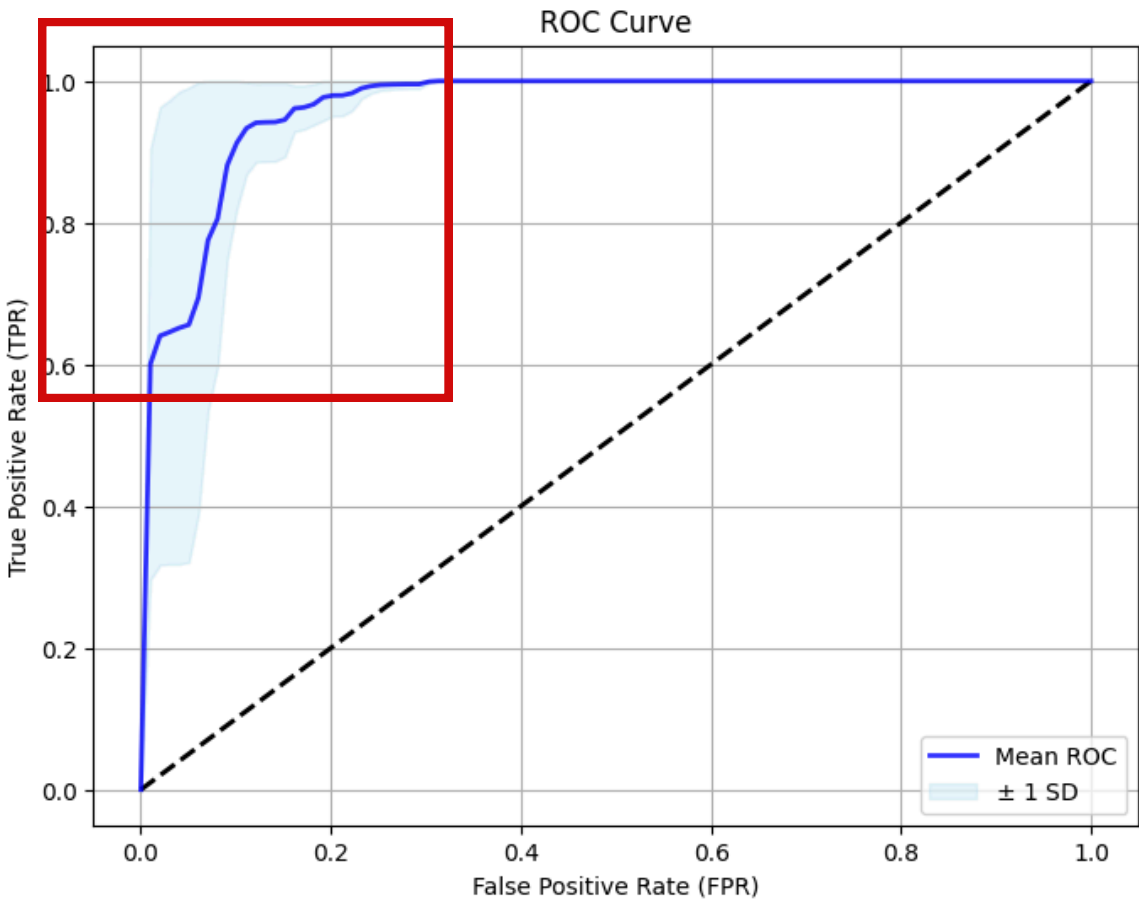
VAE-LSTM (Baseline)



Dual-branch VAE-LSTM
(Linear Concat)



Dual-branch VAE-LSTM
(Multi-head Attention)



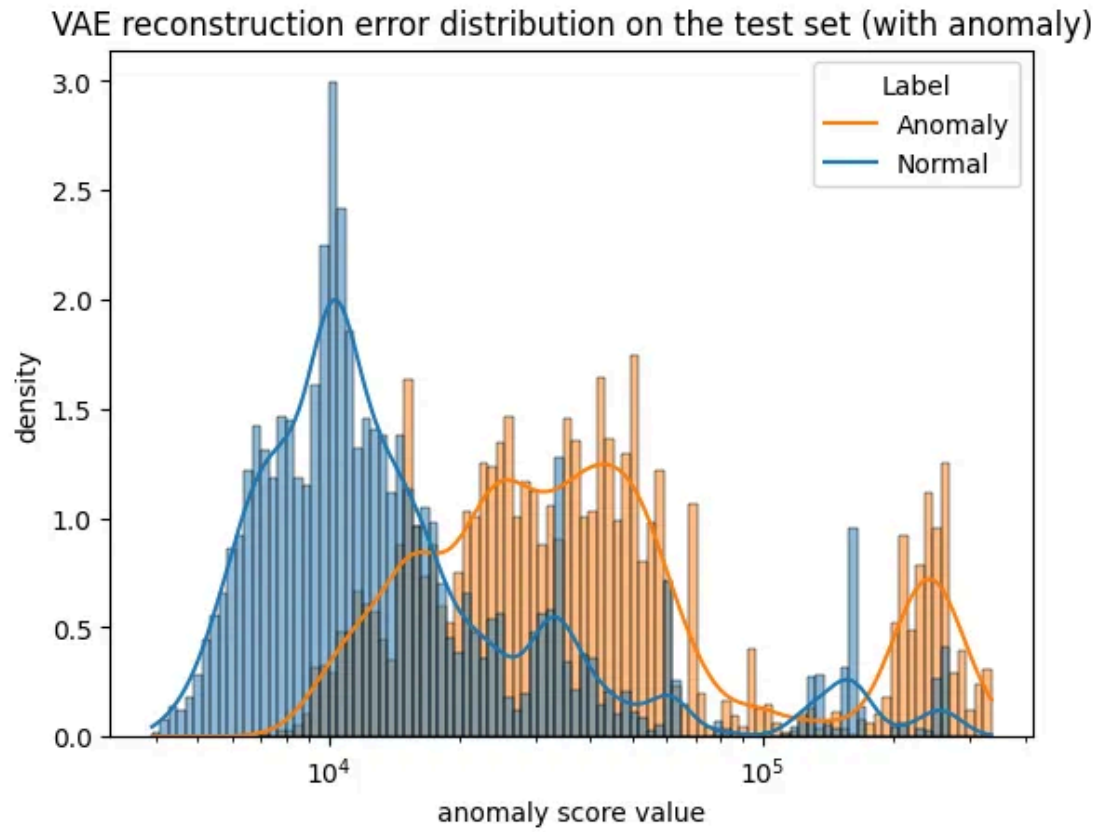
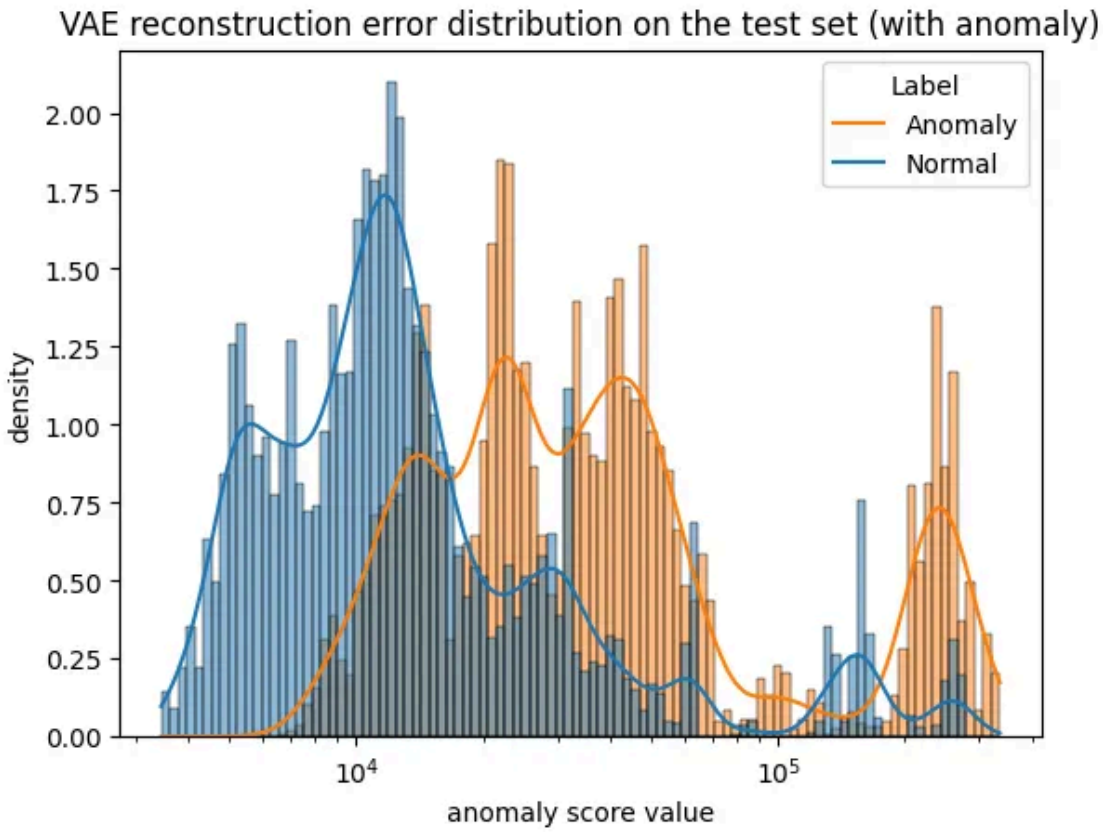
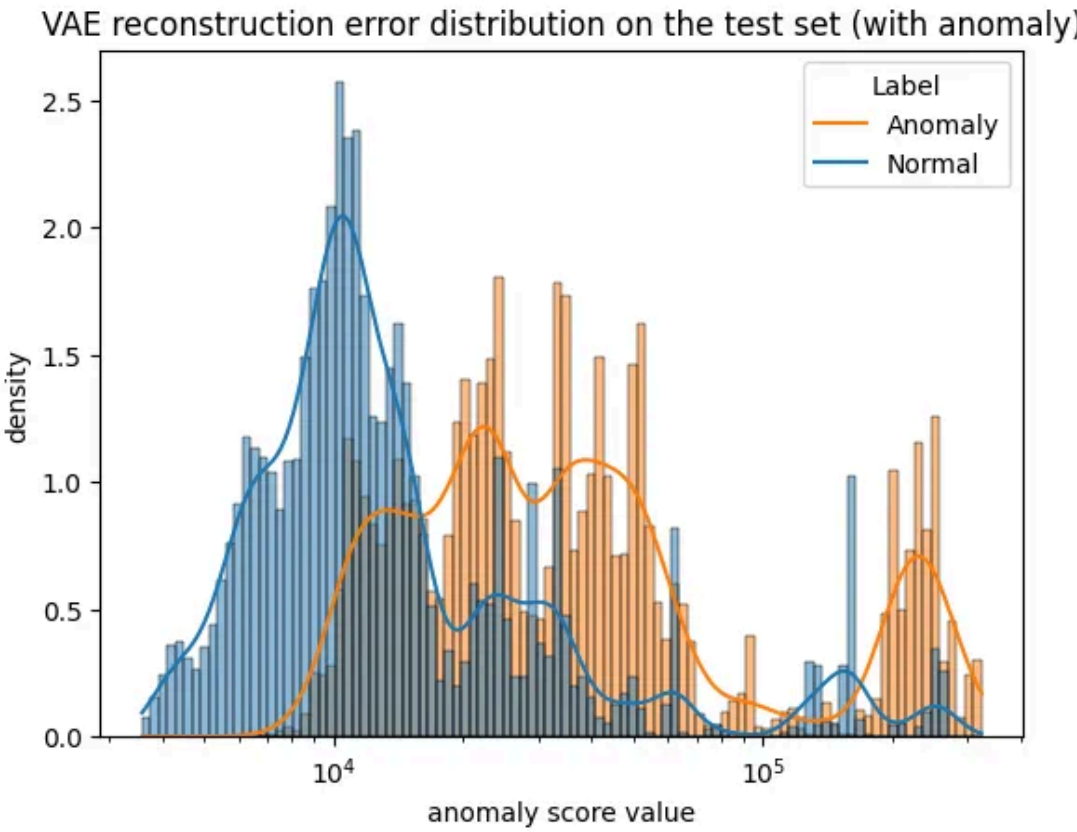
Results

Reconstruction Error Distribution Comparison

VAE-LSTM (Baseline)

Dual-branch VAE-LSTM
(Linear Concat)

Dual-branch VAE-LSTM
(Multi-head Attention)



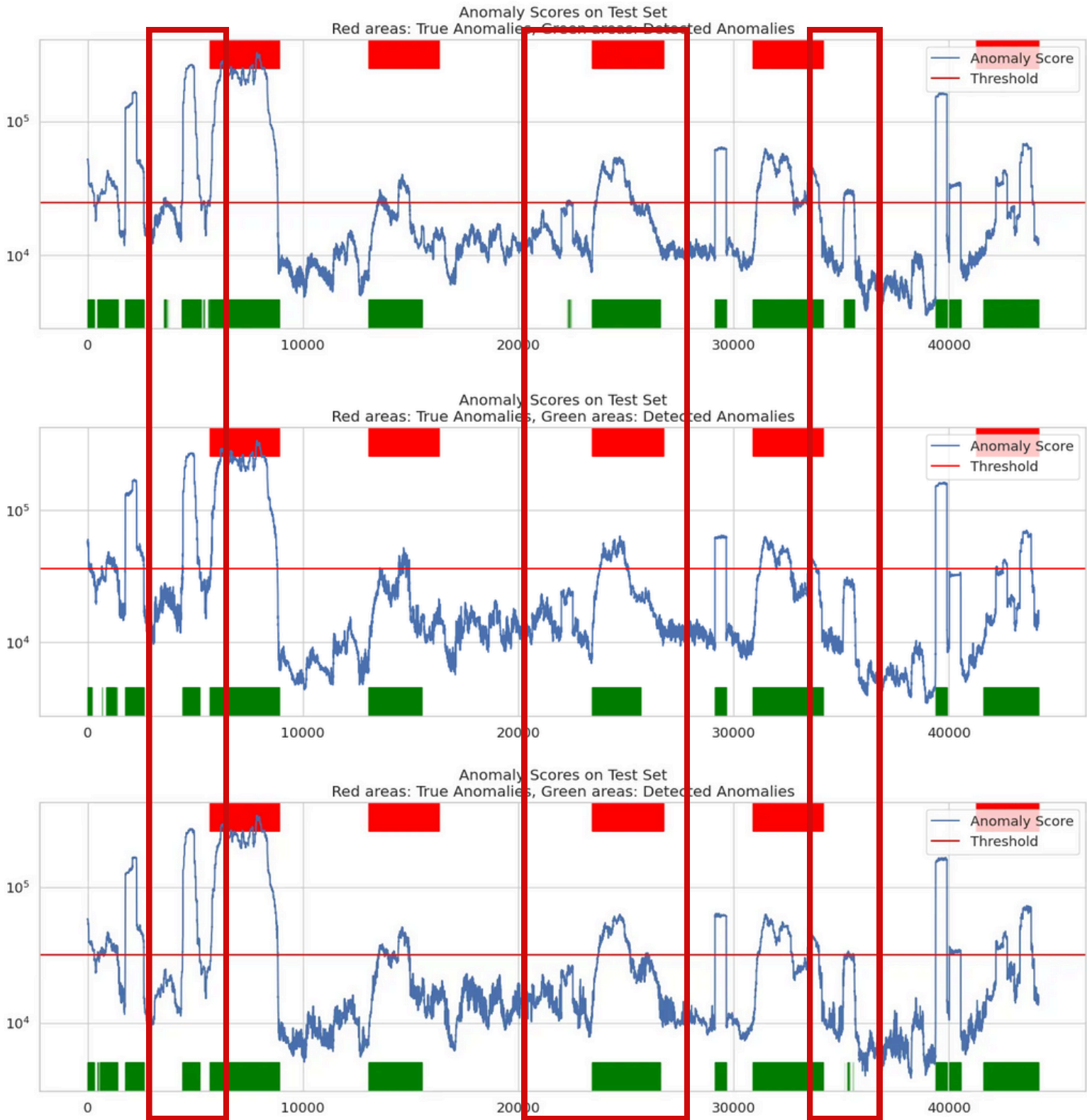
Results

Detection Result Comparison

VAE-LSTM (Baseline)

Dual-branch VAE-LSTM
(Linear Concat)

Dual-branch VAE-LSTM
(Multi-head Attention)



Variable Window Size

Ablation Table

Model	Encoder	Fusion Method	Temporal Learning
MA-VAE (Reference)	BiLSTM	Multi-Head Attention	None
VAE-LSTM (Baseline)	Local	None	LSTM
VAE-LSTM-Dual-Linear	Local + Global	Linear Concat	LSTM
VAE-LSTM-Dual-MA	Local + Global	Multi-Head Attention	LSTM

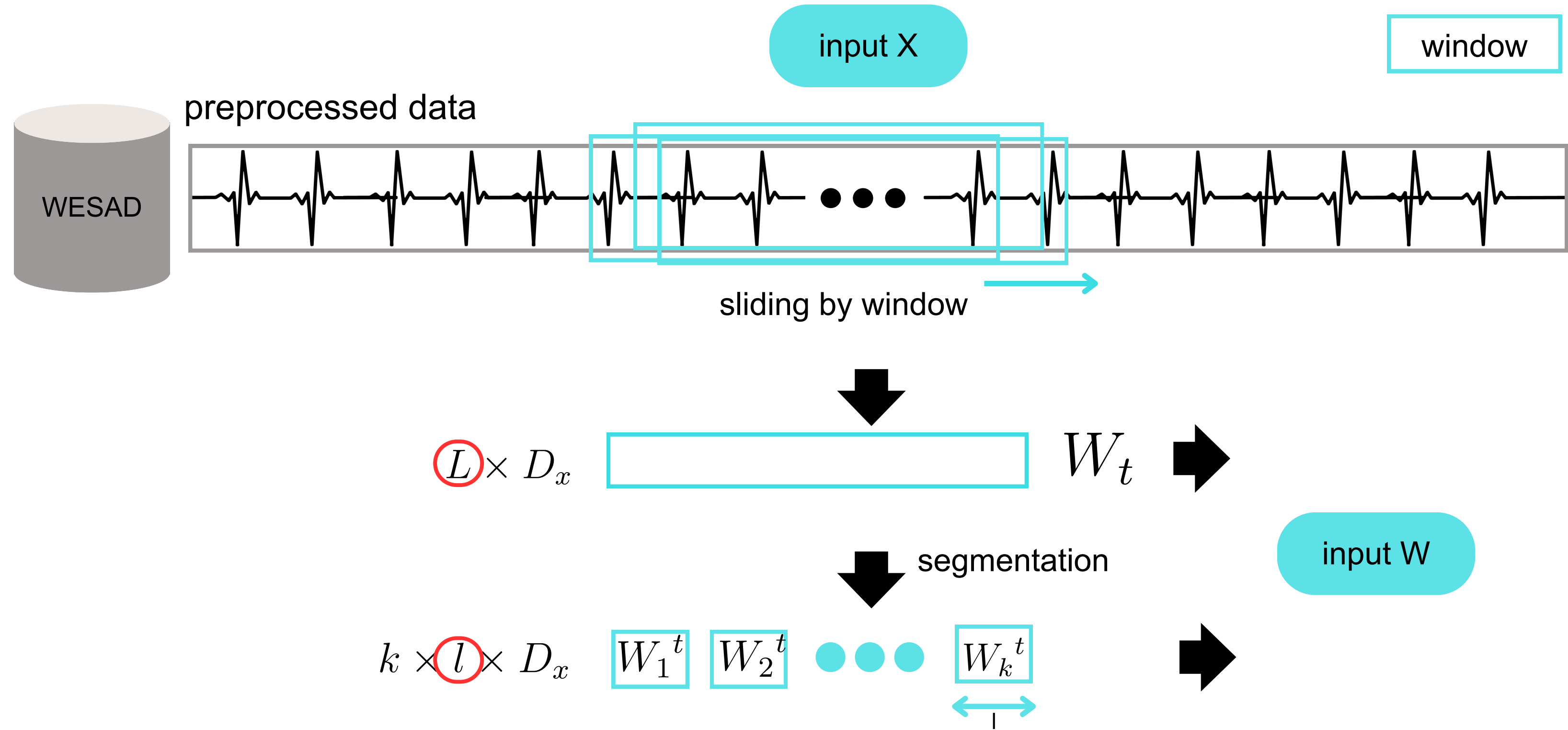
Window Setup	Window Length (L)	Segment Length (l)
576-48	576	48
288-48	288	48
288-24	288	24
144-24	144	24
72-24	72	24

4 models x 5 window setups x 3 fold = **60 experiments**

For MA-VAE, the window shift is variable instead of the segment length.

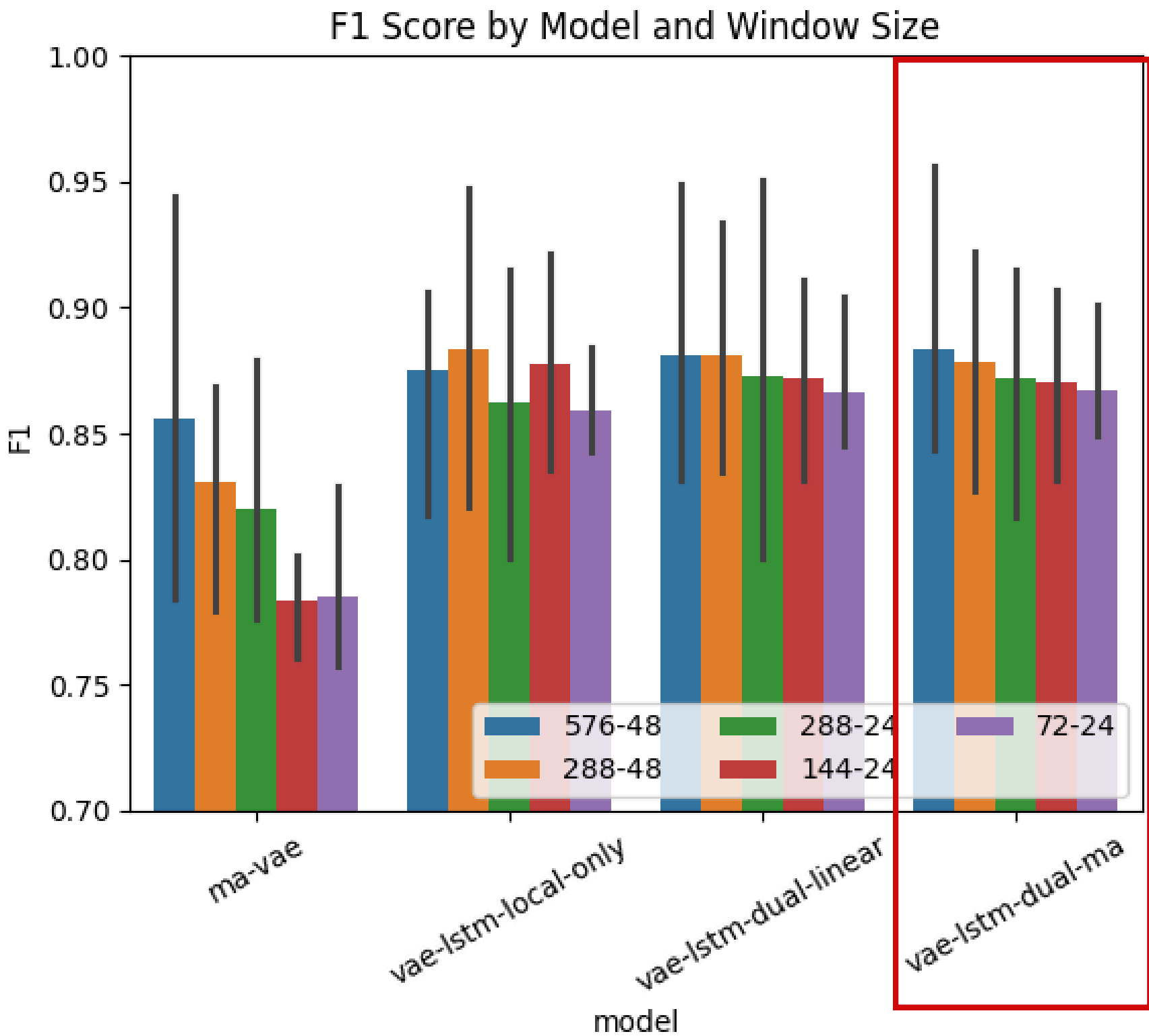
Window Size Robustness Analysis

“How stable are the models across different window sizes?”



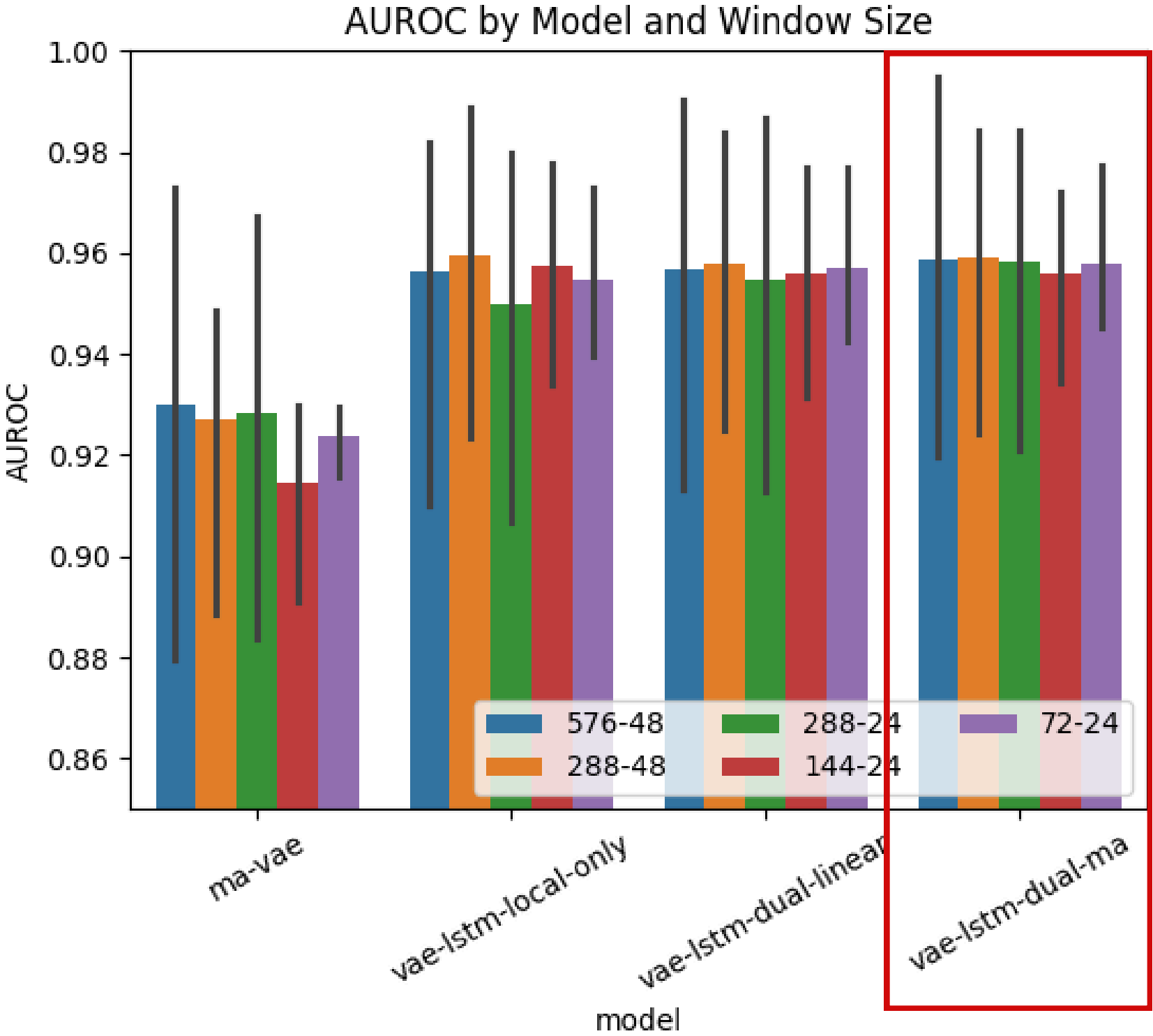
Window Size Robustness Analysis

“How stable are the models across different window sizes?”



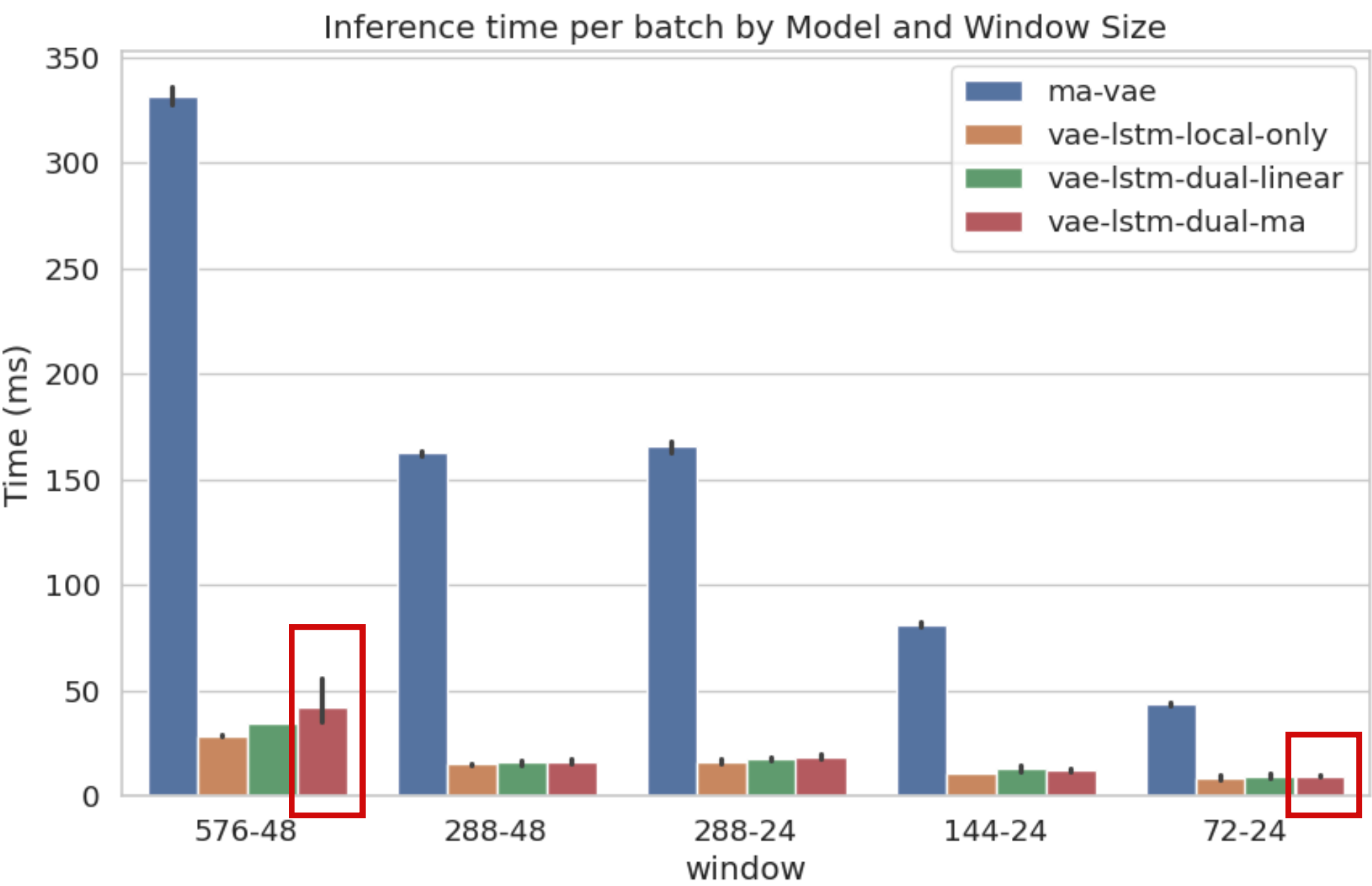
Window Size Robustness Analysis

“How stable are the models across different window sizes?”



Inference Time Analysis

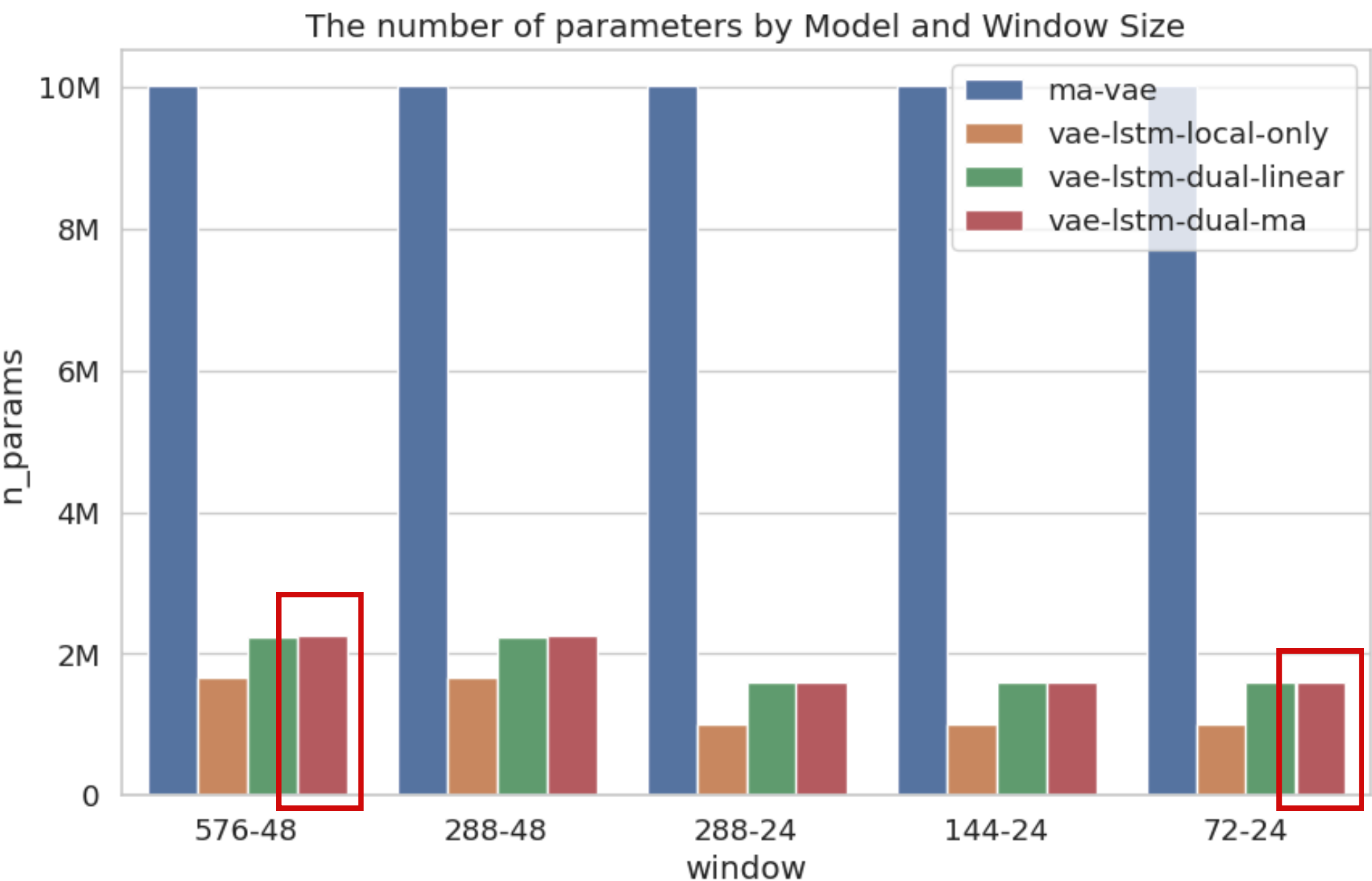
“How fast are the models to make output?”



Super Fast! (less than 50ms)

The Number of Parameters Analysis

“How light are the models?”



Super Light! (2M Parameters)

Variable Window Size Analysis (Full Data)

model	window	n_params	inference_time	average_latenc	AUROC	precision	recall	accuracy	F1
ma-vae	144-24	10020910	2387.597561	81.40396101	0.914652587	0.733646732	0.856450822	0.84874678	0.783742677
ma-vae	288-24	10020910	4801.935165	165.5839712	0.928421722	0.745034985	0.920787632	0.862945888	0.820087599
ma-vae	288-48	10020910	2439.228602	162.6152401	0.926931801	0.861597438	0.801502677	0.888357053	0.83022139
ma-vae	576-48	10020910	4855.86113	331.1880409	0.929922779	0.874179409	0.860085362	0.888403713	0.855822845
ma-vae	72-24	10020910	1271.899474	43.3651273	0.923627851	0.741301074	0.850110433	0.85284936	0.785041973
vae-lstm-dual-linear	144-24	1601679	1116.974057	12.87098711	0.955686642	0.87215763	0.876666668	0.917186875	0.871894553
vae-lstm-dual-linear	288-24	1601679	1517.822046	17.58781789	0.954654436	0.836169123	0.922234385	0.907395352	0.872755641
vae-lstm-dual-linear	288-48	2245327	1387.804459	16.07992023	0.958045008	0.863732726	0.905231252	0.916977911	0.880610126
vae-lstm-dual-linear	576-48	2245327	2971.338553	34.68512354	0.956676701	0.887971265	0.875876617	0.913465039	0.881024395
vae-lstm-dual-linear	72-24	1601679	830.6892808	9.585444027	0.957069237	0.870350452	0.865453419	0.917906312	0.86663479
vae-lstm-dual-ma	144-24	1606527	1066.968859	12.31179142	0.955828484	0.863374306	0.888412748	0.914583093	0.87063415
vae-lstm-dual-ma	288-24	1606527	1605.628189	18.60963001	0.958150209	0.832166838	0.922409197	0.907898543	0.871692688
vae-lstm-dual-ma	288-48	2250175	1420.911351	16.45506851	0.95930576	0.854620591	0.908219384	0.915137219	0.878250891
vae-lstm-dual-ma	576-48	2250175	3621.505026	42.40552427	0.9586449	0.819801951	0.963697849	0.905169434	0.883402729
vae-lstm-dual-ma	72-24	1606527	840.9681678	9.702573123	0.957876381	0.870009756	0.86933965	0.917978986	0.867234241
vae-lstm-local-only	144-24	1014319	942.0351674	10.86751346	0.957369899	0.865634342	0.899365043	0.918745076	0.877477887
vae-lstm-local-only	288-24	1014319	1416.755848	16.42128399	0.949788902	0.854702379	0.873232257	0.905683826	0.862229332
vae-lstm-local-only	288-48	1657967	1332.355148	15.43323007	0.959542956	0.851697259	0.929562508	0.915818207	0.883461336
vae-lstm-local-only	576-48	1657967	2464.424791	28.76133928	0.956262862	0.861893284	0.893278369	0.90596888	0.875143631
vae-lstm-local-only	72-24	1014319	749.3432671	8.637486156	0.954783932	0.868954414	0.852759688	0.913249133	0.858685169

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Conclusion

Conclusion

1. Innovation: Overcoming Limitations with Dual-branch Architecture

2. Performance: Enhanced Stability via Multi-head Attention fusion

Conclusion

3. Robustness: Robust to Variable Window Size

4. Real-time: Fast Interence Time

5. Impact: Contribution to Preventive Medicine

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